

Yann Guittet

Réseau



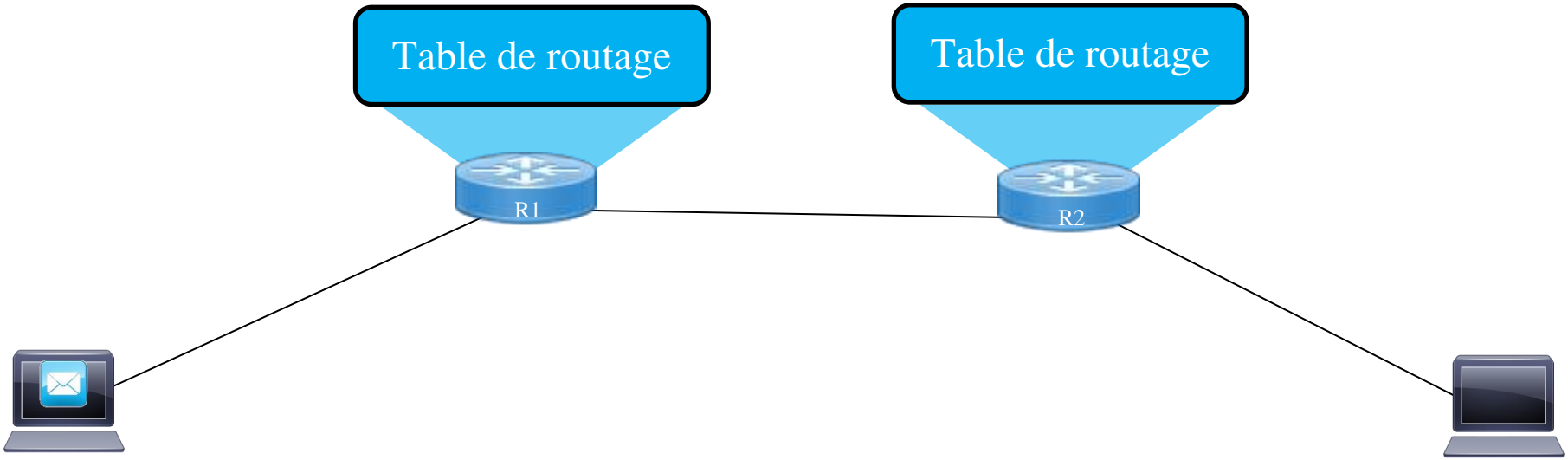
Routage



Principe du routage



Le routeur



Séquence de démarrage

Version de l'IOS

Version du bootstrap

Quantité de ram

Type d'UC

Nombre et type d'interfaces

Quantité de NVram

Quantité de mémoire Flash

```
Router#show version
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

ROM: System Bootstrap, Version 12.3(8r)T8, RELEASE SOFTWARE (fc1)

System returned to ROM by power-on
System image file is "flash:c1841-advipservicesk9-mz.124-15.T1.bin"


This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

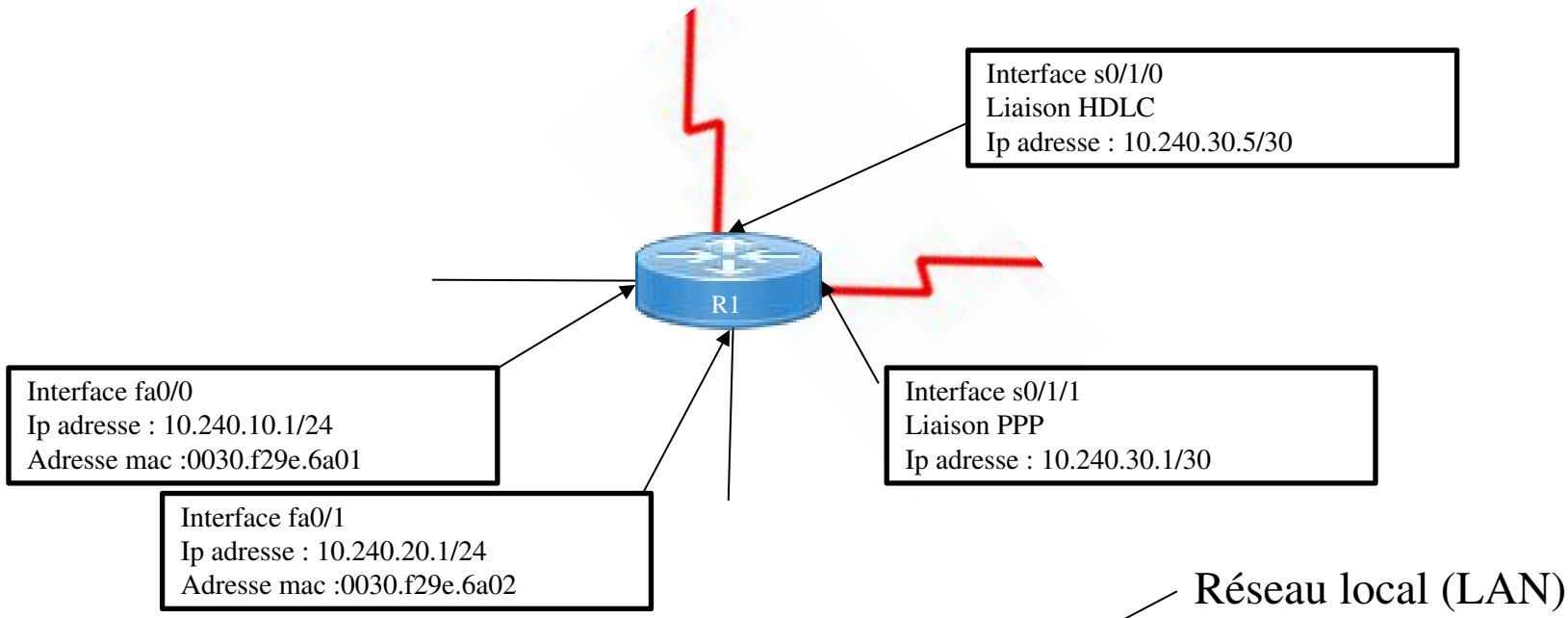
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco 1841 (revision 5.0) with 114688K/16384K bytes of memory.
Processor board ID FTX0947Z18E
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)

Configuration register is 0x2102
```

Les types d'interfaces

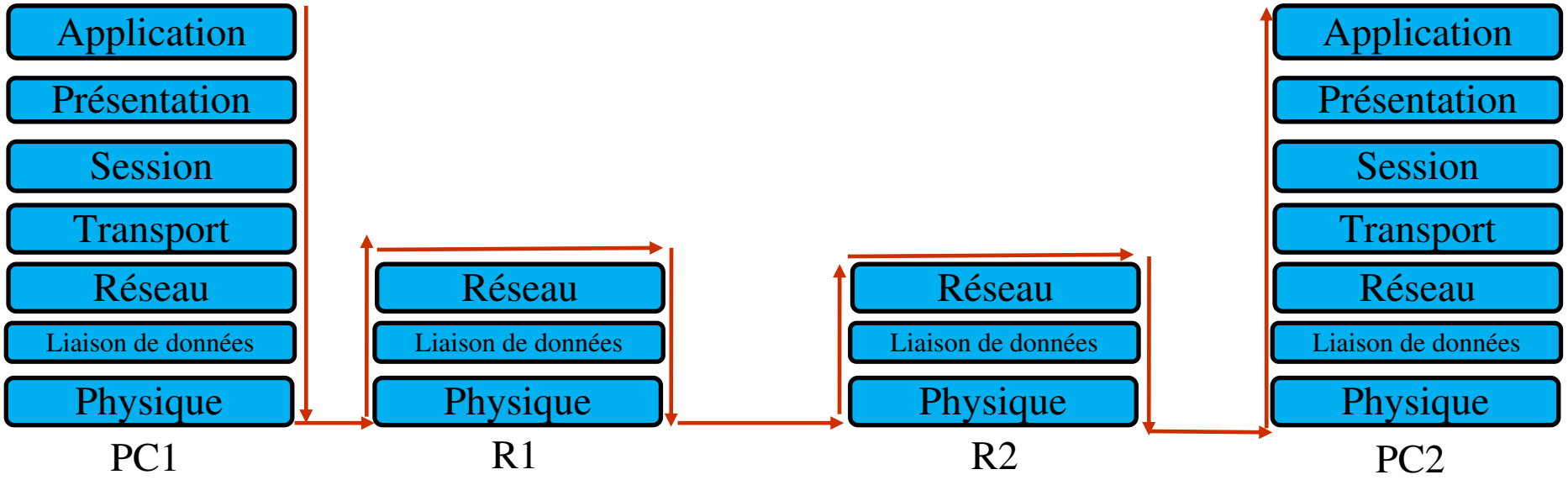


Le routage dans le modèle OSI



Adresse IP de destination	Adresse IP Source

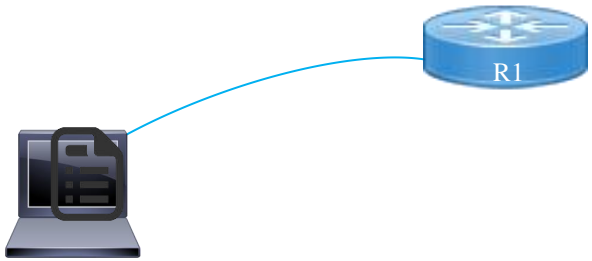
Adresse IP de destination	Adresse IP Source



Configuration du hostname

Commande : (config)*#hostname R1*

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```



Définition des mots de passe

Commande : (config)*#enable secret insta*

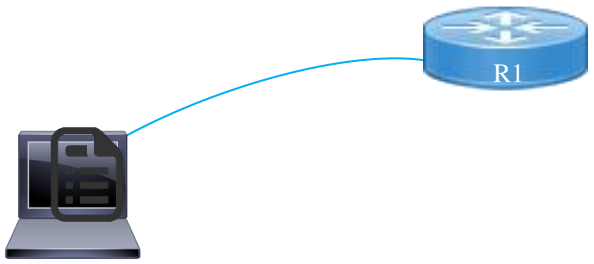
```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#enable secret insta
```

Commande : (config)*#Line console 0*
Commande : (config)*#password insta0*
Commande : (config)*#login*

```
R1#configure terminal
Enter configuration commands, one per
R1(config)#line console 0
R1(config-line)#password insta0
R1(config-line)#login
```

Commande : (config)*#Line VTY 0 4*
Commande : (config)*#password instavty*
Commande : (config)*#login*

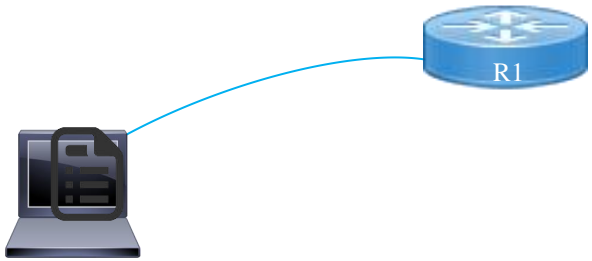
```
R1#configure terminal
Enter configuration commands, one per
R1(config)#line vty 0 4
R1(config-line)#password instavty
R1(config-line)#login
```



Configuration du message du jour

Commande : (config)*#banner motd # bienvenue sur le routeur de l'insta #*

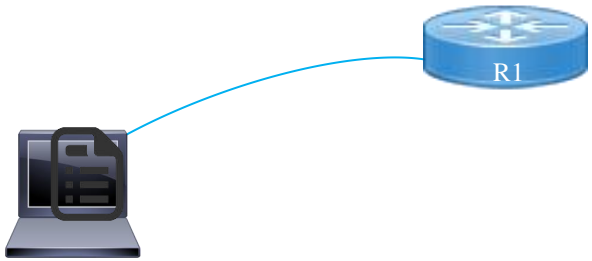
```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#banner motd # bienvenue sur le routeur de linsta #
```



Configuration d'interface

Commande : (config)*#interface Fa0/0*
Commande : (config-if)*#ip addresse 10.240.30.1 255.255.255.0*
Commande : (config-if)*#description « gateway du lan de presentation »*
Commande : (config-if)*#no shutdown*

```
R1(config)#interface fa0/0
R1(config-if)#ip address 10.240.30.1 255.255.255.0
R1(config-if)#description "gateway du lan de prsentation"
R1(config-if)#no shutdown
```



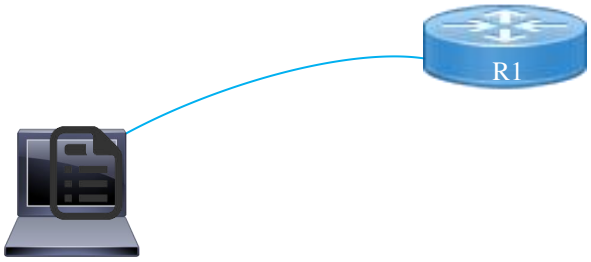
Sauvegarde de la configuration

Commande : *#copy running-config startup-config*

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Commande : *#copy running-config flash:*

```
R1#copy running-config flash:
Destination filename [running-config]? R1.cfg
Building configuration...
[OK]
```



Vérification de la configuration

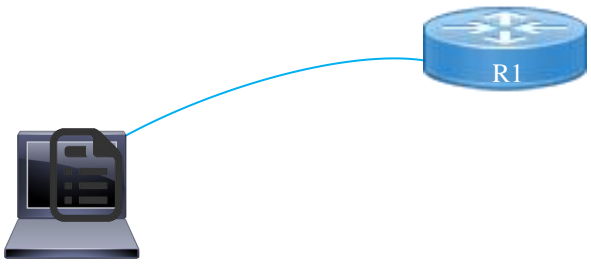
Commande : *#show running-config*

```
R1#show running-config
Building configuration...

Current configuration : 748 bytes
!
version 12.4
no service timestamps log datetime msec
```

Commande : *#show startup-config:*

```
R1#show startup-config
Using 748 bytes
!
version 12.4
no service timestamps log datetime msec
```



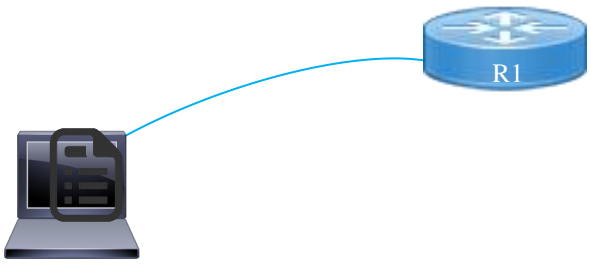
Affichage de la table de routage

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.240.30.0 is directly connected, FastEthernet0/0
```



Affichage de la configuration des interfaces

Commande : *#show ip interface brief*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

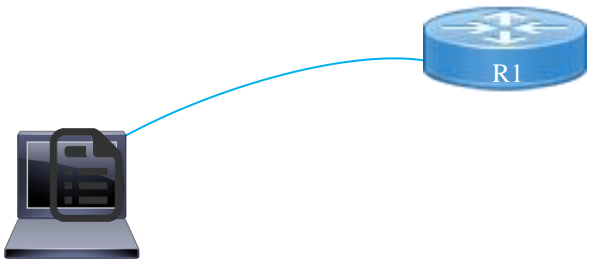
FastEthernet0/0        10.240.30.1     YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Vlan1                  unassigned      YES unset  administratively down down
```

Commande : *#show interface*

```
R1#show interfaces
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0003.e470.8c01 (bia 0003.e470.8c01)
  Description: "gateway du lan de prsentation"
  Internet address is 10.240.30.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```



La Table de routage

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 2 subnets
C       10.240.0.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
```



Routage statique

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.240.0.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 [1/0] via 10.240.0.2
```



Routage dynamique

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.240.0.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
R       10.240.20.0 [120/1] via 10.240.0.2, 00:00:15, Serial0/1/0
```



Principe du routage

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

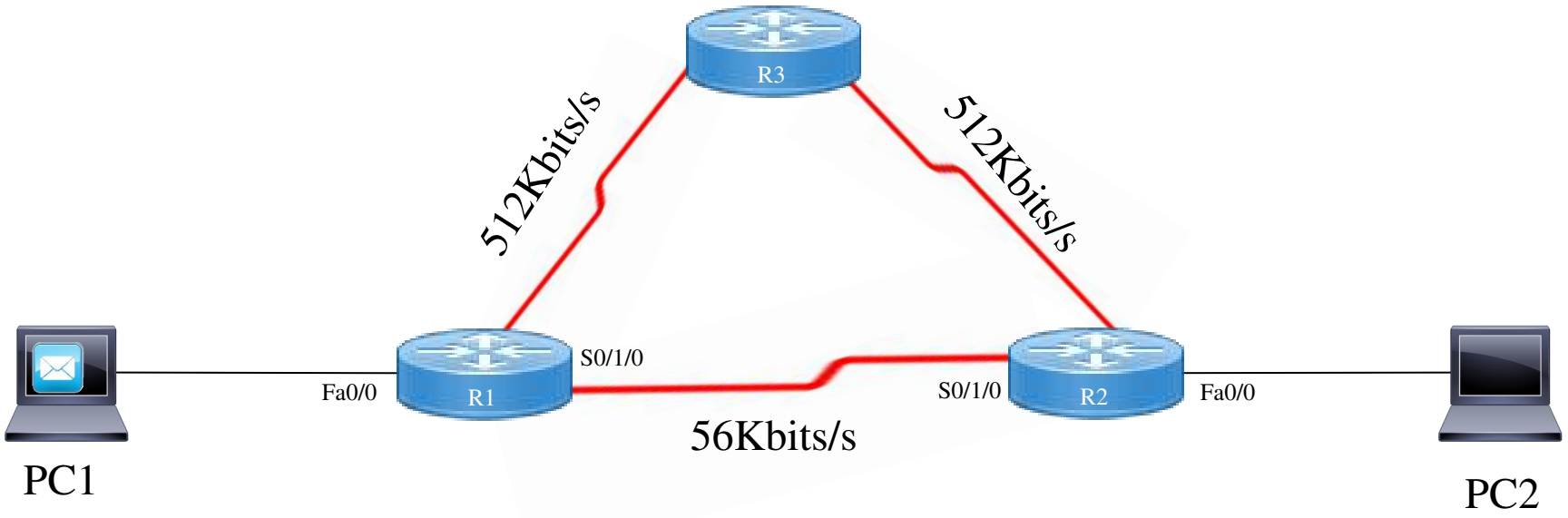
Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.240.0.0 is directly connected, Serial0/1/0
R       10.240.10.0 [120/1] via 10.240.0.1, 00:00:25, Serial0/1/0
C       10.240.20.0 is directly connected, FastEthernet0/0
```

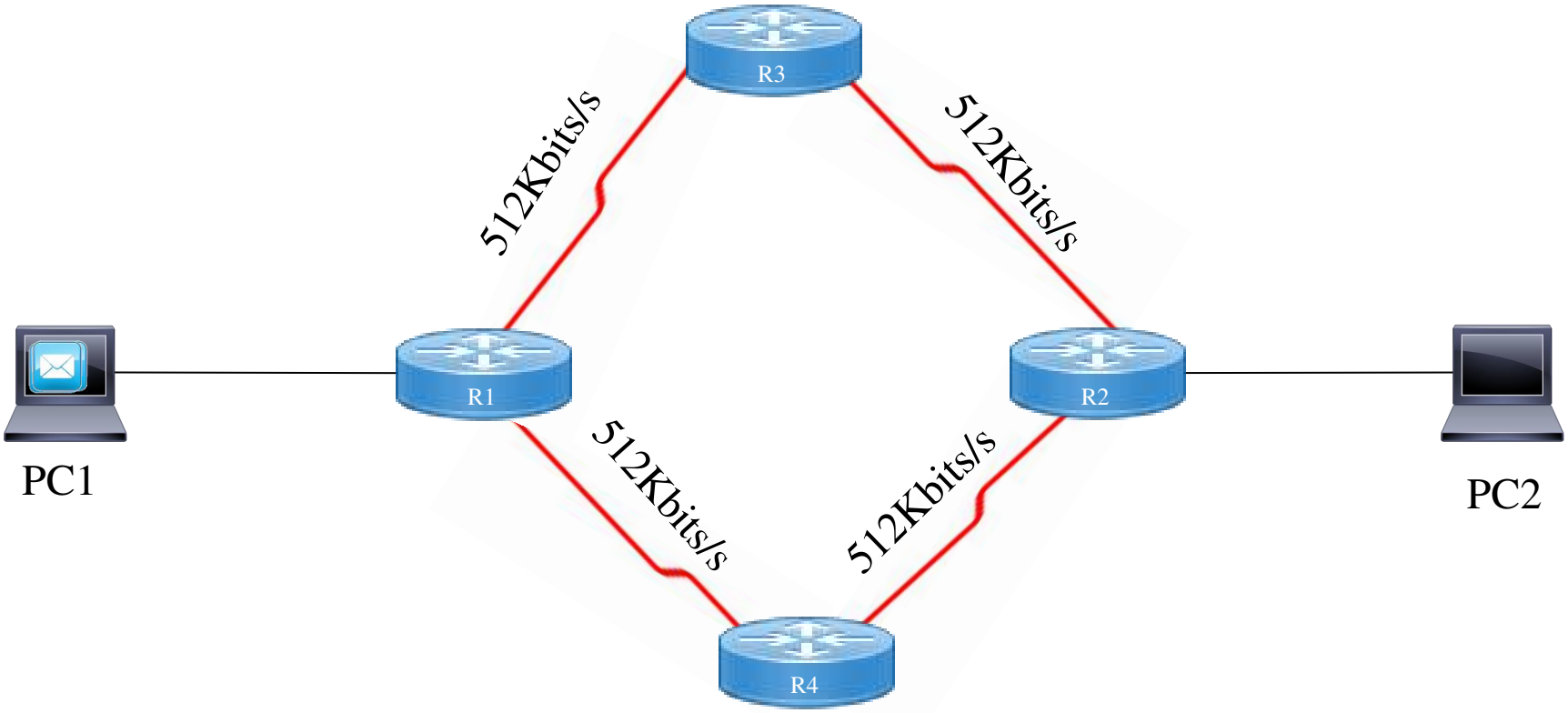


Détermination meilleur chemin

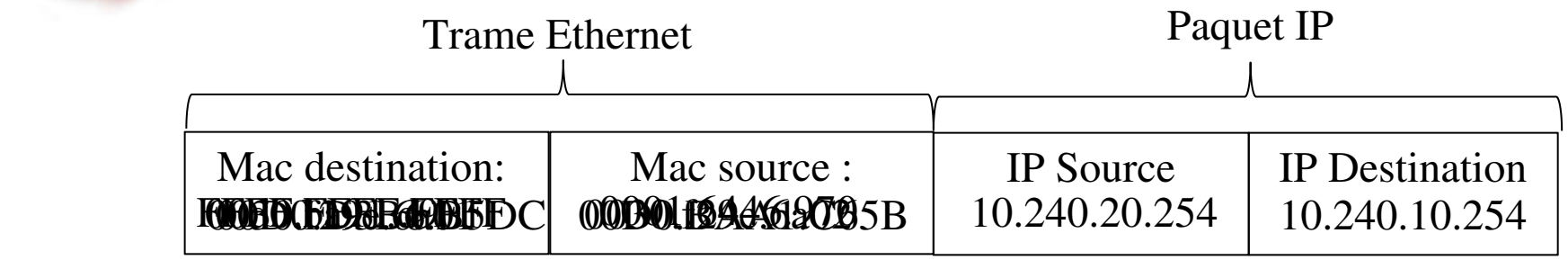
Bande passante

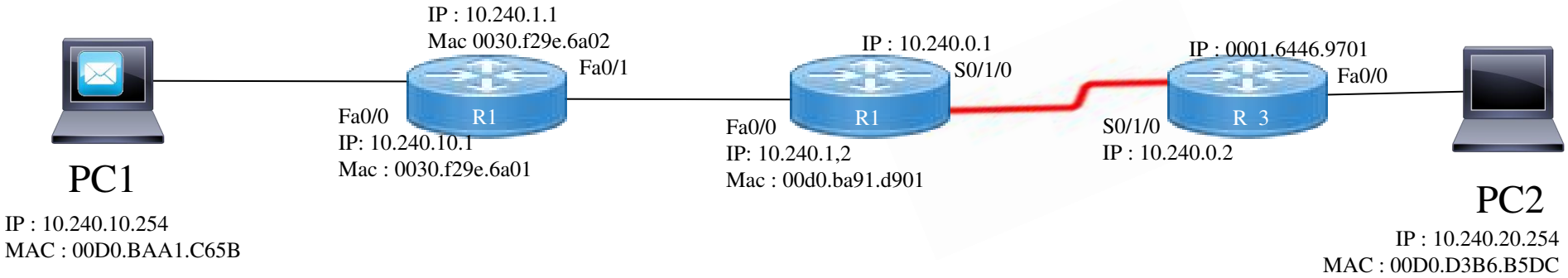


Equilibrage de charge



Principe du routage






```
R3#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.240.20.1             -   0001.6446.9701  ARPA   FastEthernet0/0
Internet  10.240.20.254           46  00D0.D3B6.B5DC  ARPA   FastEthernet0/0
C        10.240.20.0 is directly connected, FastEthernet0/0
```

Routage Statique



Topologie

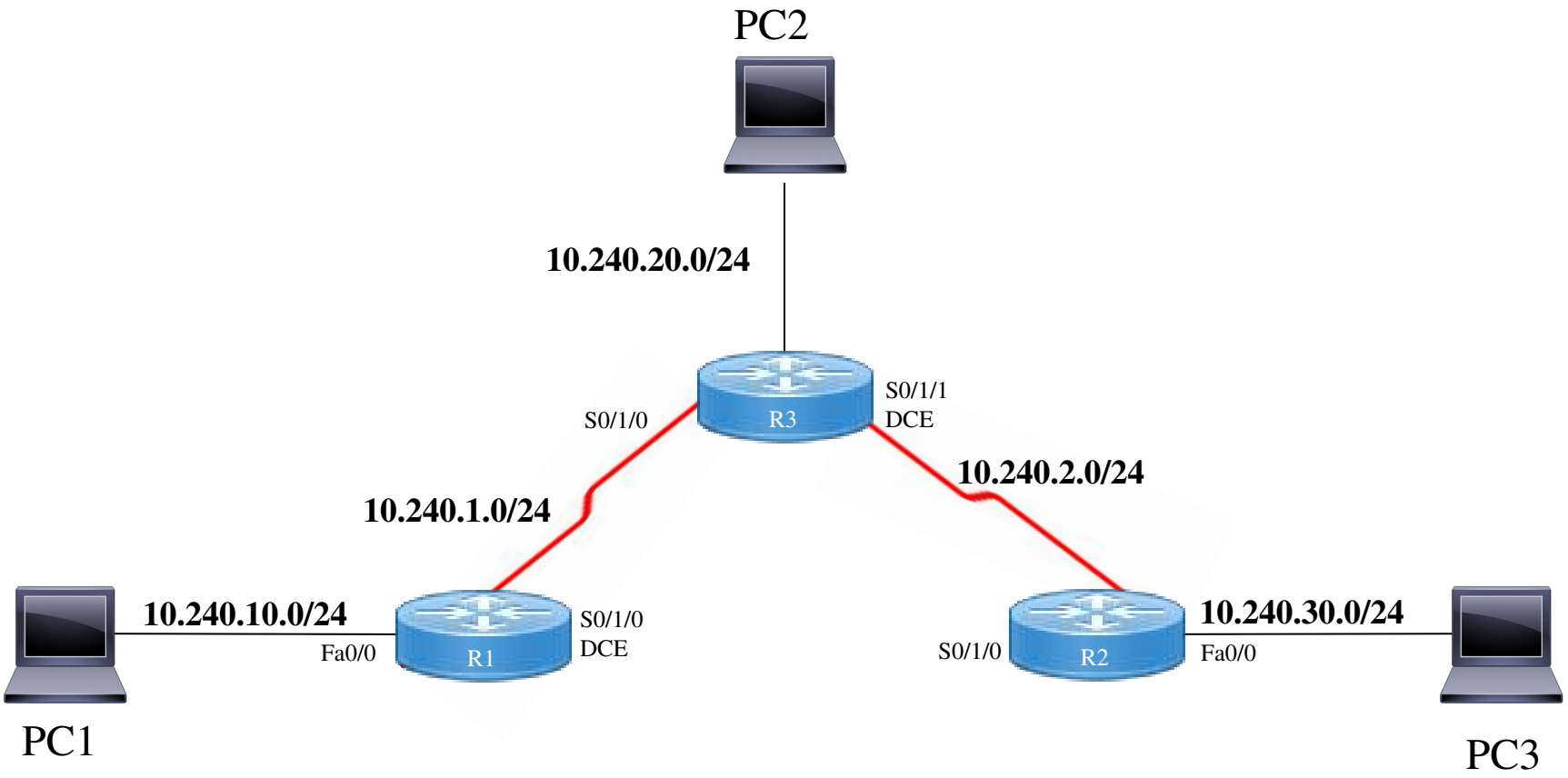

PC2
10.240.20.0/24
S0/1/0
R3
S0/1/1
DCE
10.240.2.0/24
10.240.1.0/24
10.240.10.0/24
Fa0/0
R1
S0/1/0
DCE
S0/1/0
R2
10.240.30.0/24
Fa0/0
PC1
PC3



Table d'adressage

Host	Interface	Adresse IP	Masque	Gateway
R1	Fa0/0	10.240.10.1	255.255.255.0	N/A
	S0/1/0	10.240.1.1	255.255.255.0	N/A
R2	Fa0/0	10.240.20.1	255.255.255.0	N/A
	S0/1/0	10.240.1.2	255.255.255.0	N/A
	S0/1/1	10.240.2.1	255.255.255.0	N/A
R3	.Fa0/0	10.240.30.1	255.255.255.0	N/A
	S0/1/0	10.240.2.2	255.255.255.0	N/A
PC1	NIC	10.240.10.254	255.255.255.0	10.20.240.10.1
PC2	NIC	10.240.20.254	255.255.255.0	10.20.240.20.1
PC3	NIC	10.240.30.254	255.255.255.0	10.20.240.30.1

Examen des interfaces routeur

Commande : *#show ip route*

```
R1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

Commande : *#show interface*

```
R1#show interfaces
FastEthernet0/0 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0001.9613.2e01 (bia 0001.9613.2e01)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
```

Commande : *#show ip interface brief*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Commande : *#show running-config*

```
R1#show running-config
Building configuration...

!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/1/0
 no ip address
 shutdown
```



Configuration des interfaces

Commande : (config)*#interface fa0/0*
Commande : (config-if)*#ip address 10.240.20.1 255.255.255.0*
Commande : (config)*#no shutdown*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fa 0/0
R1(config-if)#ip address 10.240.10.1 255.255.255.0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

Session synchrone

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fa 0/0
R1(config-if)#ip address 10.240.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#descrip
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to uption
R1(config-if)#
```

Commande : (config)*#line console 0*
Commande : (config-line)*#logging synchronous*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fa 0/0
R1(config-if)#ip address 10.240.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#description
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#description
```

Vérification de la table de routage

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.240.10.0 is directly connected, FastEthernet0/0
```



Vérification de l'interface

Commande : *#show interface fa 0/0*

```
R1#show interfaces fa0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.9613.2e01 (bia 0001.9613.2e01)
  Internet address is 10.240.10.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Commande : *#show ip interface brief*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.240.10.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Configuration de l'interface série

Commande : (config)*#interface s0/1/0*
Commande : (config-if)*#ip address 10.240.1.1 255.255.255.0*
Commande : (config)*#ino shutdown*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/1/0
R1(config-if)#ip address 10.240.1.1 255.255.255.0
R1(config-if)#no shu
R1(config-if)#no shutdown
```

Commande : *#show ip interface brief*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        10.240.10.1     YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/1/0            10.240.1.1      YES manual down                  down

Serial0/1/1            unassigned      YES unset  administratively down down
```

Configuration de l'interface série

Commande : (config)*#interface s0/1/0*
Commande : (config-if)*#ip address 10.240.1.1 255.255.255.0*
Commande : (config)*#ino shutdown*

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface s0/1/0
R2(config-if)#ip address 10.240.1.1 255.255.255.0
R2(config-if)#no shutdown
```

Commande : *#show ip interface brief*

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.240.10.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            10.240.1.1      YES manual up                    down
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Configuration de l'horloge

Commande : (config)*#show controllers s0/1/0*

```
R1#show controllers s0/1/0
Interface Serial0/1/0
Hardware is PowerQUICC MPC860
DCE V.35, no clock
idb at 0x81081AC4, driver data structure at 0x81084AC0
SCC Registers:
```

Commande : (config-if)*#clock rate 4000000*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/1/0
R1(config-if)#clock rate 4000000
```

Vérification de la configuration

Commande : *#show ip interface brief*

```
R1#show ip inter brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.240.10.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            10.240.1.1      YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 2 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
```

Commande : *#ping 10.240.1.2*

```
R1#ping 10.240.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.240.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/18 ms
```

débogage

Commande : *#debug ip routing*

```
R2#debug ip routing
IP routing debugging is on
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface fa0/0
R2(config-if)#ip address 10.240.20.1 255.255.255.0
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

RT: interface FastEthernet0/0 added to routing table
RT: SET_LAST_RDB for 10.240.20.0/24

    NEW rdb: is directly connected


RT: add 10.240.20.0/24 via 0.0.0.0, connected metric [0/0]

RT: NET-RED 10.240.20.0/24

R2#undebug all
All possible debugging has been turned off
```



CDP

Commande : *#show cdp neighbors*

```
R2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
R1               Ser 0/1/0         179           R       C1841     Ser 0/1/0
R3               Ser 0/1/1         164           R       C1841     Ser 0/1/0
R2#
```

Commande : *#sho cdp neighbors detail*

```
R2#show cdp neighbors detail

....

Device ID: R3
Entry address(es):
  IP address : 10.240.2.2
Platform: cisco C1841, Capabilities: Router
Interface: Serial0/1/1, Port ID (outgoing port): Serial0/1/0

....
```



Désactiver CDP

Commande (config) *#no cdp run*

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no cdp run
```

Commande : (config-if)*#no cdp run enable*

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface fa 0/0
R2(config-if)#no cdp enable
```

Identification des réseaux non connu

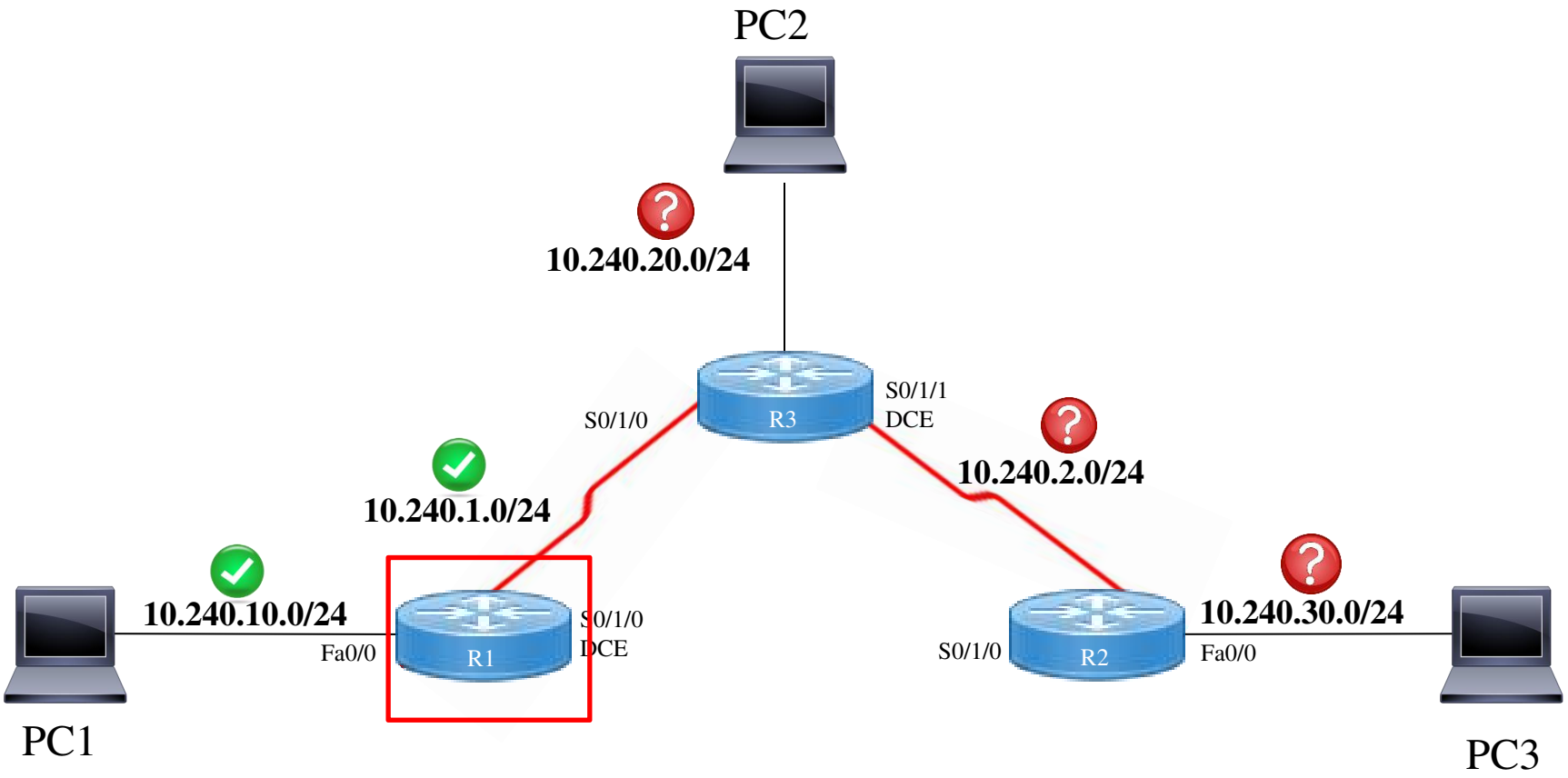




Table de routage

Host	Réseau	Masque	Interface de sortie	Gateway	Distance
R1	10.240.20.0	255.255.255.0	S0/1/0	10.240.1.2	1
	10.240.30.0	255.255.255.0	S0/1/0	10.240.1.2	2
	10.240.2.0	255.255.255.0	S0/1/0	10.240.1.2	1
R2	10.240.10.0	255.255.255.0	S0/1/1	10.240.1.1	1
	10.240.30.0	255.255.255.0	S0/1/0	10.240.2.1	1
R3	10.240.10.1	255.255.255.0	S0/1/0	10.240.2.1	2
	10.240.20.0	255.255.255.0	S0/1/0	10.240.2.1	1
	10.240.1.0	255.255.255.0	S0/1/0	10.240.2.1	1



Ajout de route statique

Commande : (config)*#ip route 10.240.20.0 255.255.255.0 10.240.1.2*

```
R1#configure terminal
R1(config)#ip route 10.240.20.0 255.255.255.0 10.240.1.2
|
...
RT: add 10.240.20.0/24 via 10.240.1.2, static metric [1/0]

...

R1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 [1/0] via 10.240.1.2
```

Vérifier la configuration des routes statiques

Commande : *# show running-config*

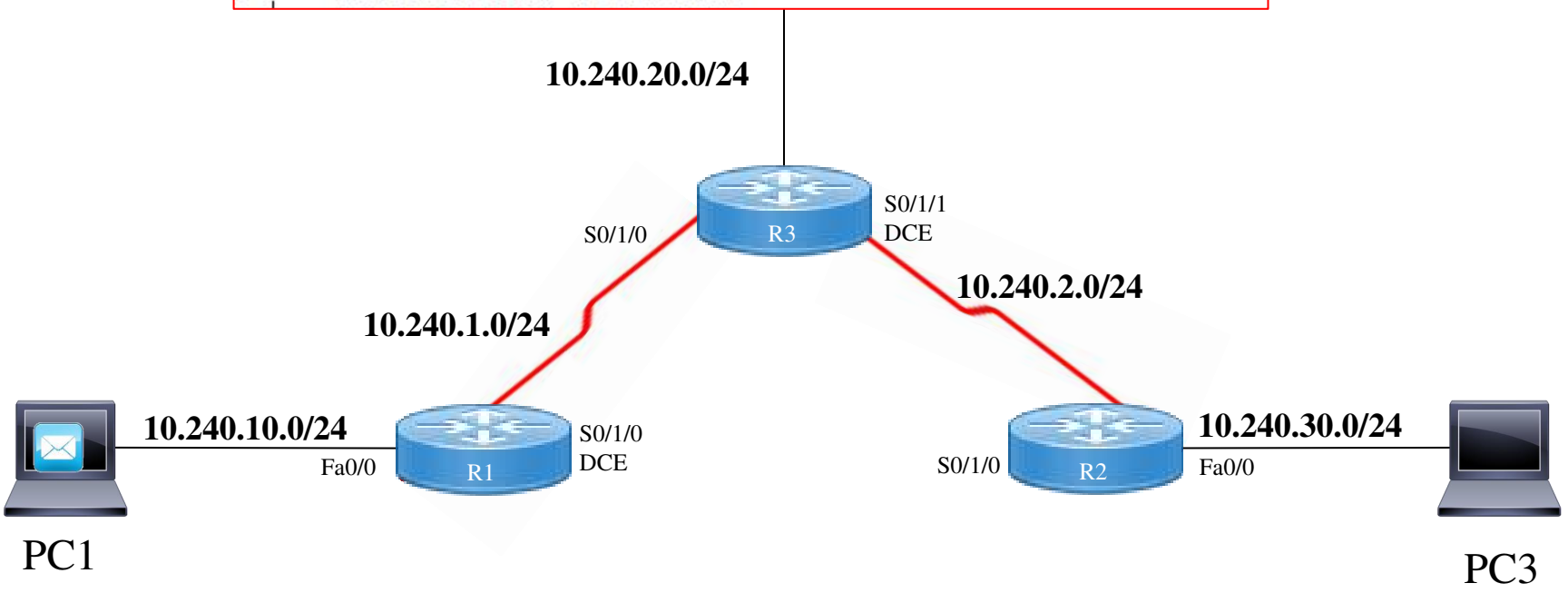
```
R1#show running-config
Building configuration...
...
!
ip classless
ip route 10.240.20.0 255.255.255.0 10.240.1.2
ip route 10.240.2.0 255.255.255.0 10.240.1.2
ip route 10.240.30.0 255.255.255.0 10.240.1.2
!
...
!
```

Lecture de la table de routage

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 [1/0] via 10.240.1.2
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 [1/0] via 10.240.1.2
S       10.240.30.0 [1/0] via 10.240.1.2
```



Interface de sortie down

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config-if)#interface s0/1/0
R1(config-if)#shutdown

RT: interface Serial0/1/0 removed from routing table
RT: del 10.240.1.0 via 0.0.0.0, connected metric [0/0]
RT: delete network route to 10.240.1.0
RT: del 10.240.20.0 via 10.240.1.2, static metric [1/0]
RT: delete network route to 10.240.20.0
RT: del 10.240.2.0 via 10.240.1.2, static metric [1/0]
RT: delete network route to 10.240.2.0
RT: del 10.240.30.0 via 10.240.1.2, static metric [1/0]
RT: NET-RED 10.240.30.0/24
```

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.240.10.0 is directly connected, FastEthernet0/0
```

Route statique par interface de sortie

Commande : (config)*#no ip route 10.240.20.0 255.255.255.0 10.240.1.2*

```
R1#configure terminal
R1(config)#no ip route 10.240.20.0 255.255.255.0 10.240.1.2
R1(config)#no ip route 10.240.2.0 255.255.255.0 10.240.1.2
R1(config)#no ip route 10.240.30.0 255.255.255.0 10.240.1.2
```

Commande : (config)*#ip route 10.240.20.0 255.255.255.0 s0/1/0*

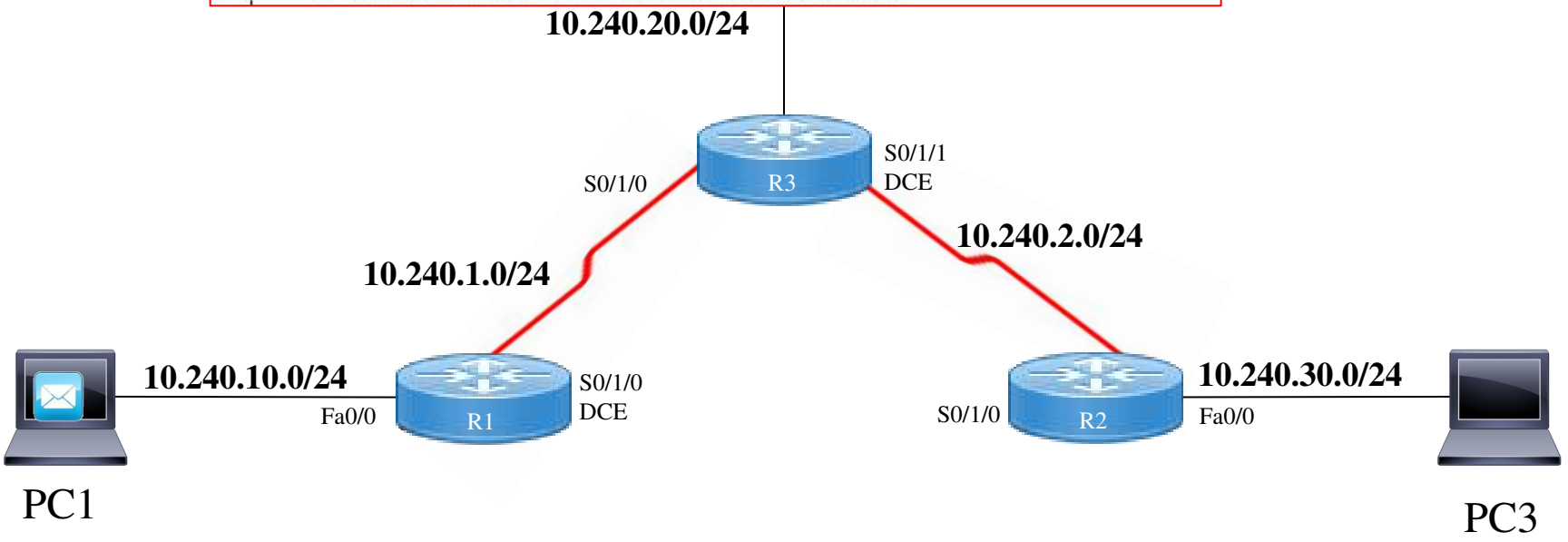
```
R1#configure terminal
R1(config)#ip route 10.240.20.0 255.255.255.0 s0/1/0
R1(config)#ip route 10.240.2.0 255.255.255.0 s0/1/0
R1(config)#ip route 10.240.30.0 255.255.255.0 s0/1/0
```

Lecture de la table de routage

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 is directly connected, Serial0/1/0
S       10.240.30.0 is directly connected, Serial0/1/0
```

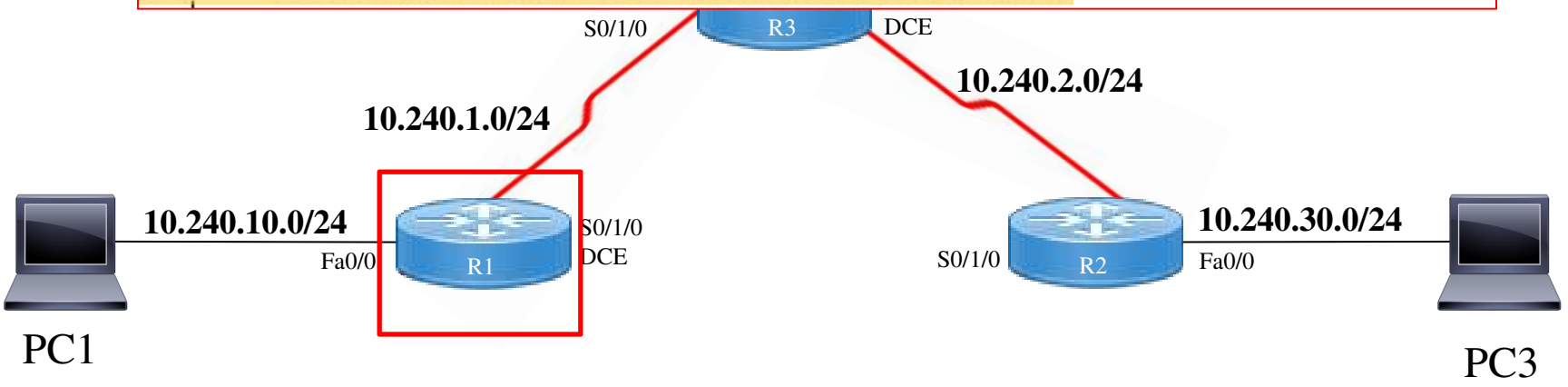


Route par default

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 is directly connected, Serial0/1/0
S       10.240.30.0 is directly connected, Serial0/1/0
```





Configuration de la route par default

Commande : (config)*#no ip route 10.240.20.0 255.255.255.0 s0/1/0*

```
R1#configure terminal
R1(config)#no ip route 10.240.20.0 255.255.255.0 s0/1/0
R1(config)#no ip route 10.240.2.0 255.255.255.0 s0/1/0
R1(config)#no ip route 10.240.30.0 255.255.255.0 s0/1/0
```

Commande : (config)*#ip route 0.0.0.0 0.0.0.0 s0/1/0*

```
R1#configure terminal
R1(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
```

Table de routage

Avant

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 is directly connected, Serial0/1/0
S       10.240.30.0 is directly connected, Serial0/1/0
```

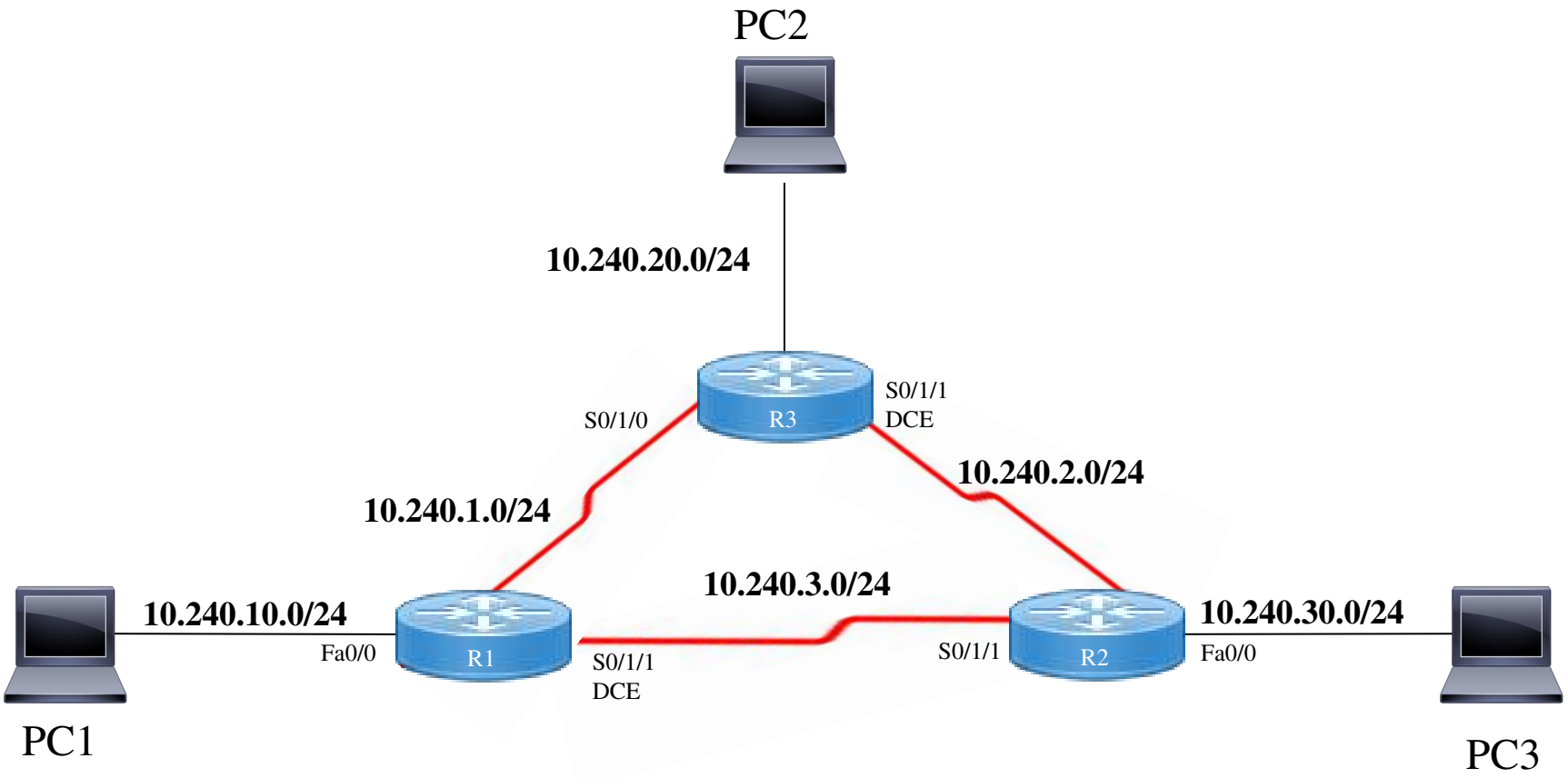
Après

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 2 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

Extension de la topologie



Ajout des nouvelles routes

Commande : (config)*#ip route 10.240.20.0 255.255.255.0 s0/1/0*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 0.0.0.0 0.0.0.0 s0/1/0
R1(config)#ip route 10.240.20.0 255.255.255.0 s0/1/0
R1(config)#ip route 10.240.2.0 255.255.255.0 s0/1/0
R1(config)#ip route 10.240.30.0 255.255.255.0 s0/1/0
R1(config)#ip route 10.240.20.0 255.255.255.0 s0/1/1
R1(config)#ip route 10.240.2.0 255.255.255.0 s0/1/1
R1(config)#ip route 10.240.30.0 255.255.255.0 s0/1/1
```

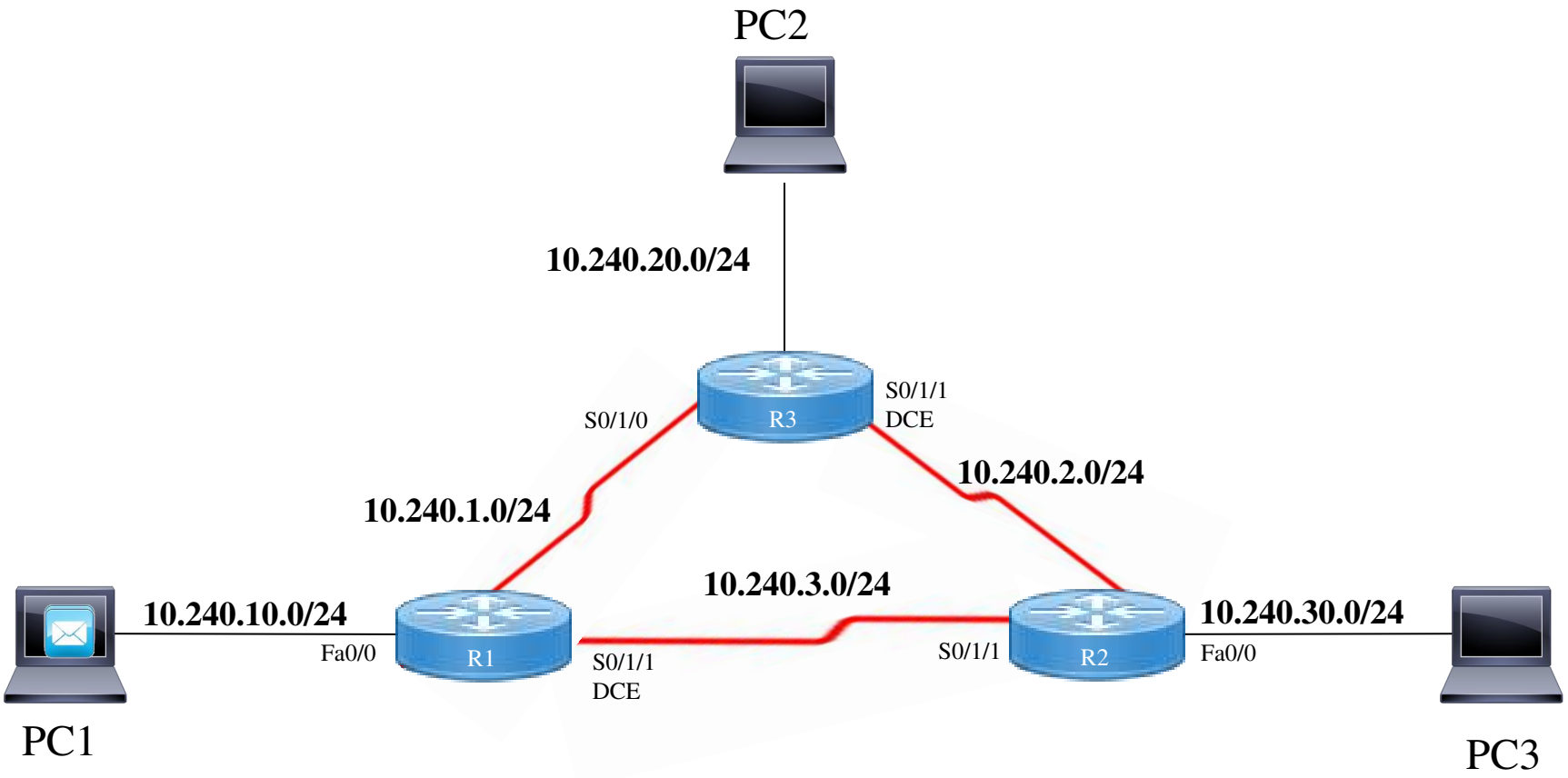
Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

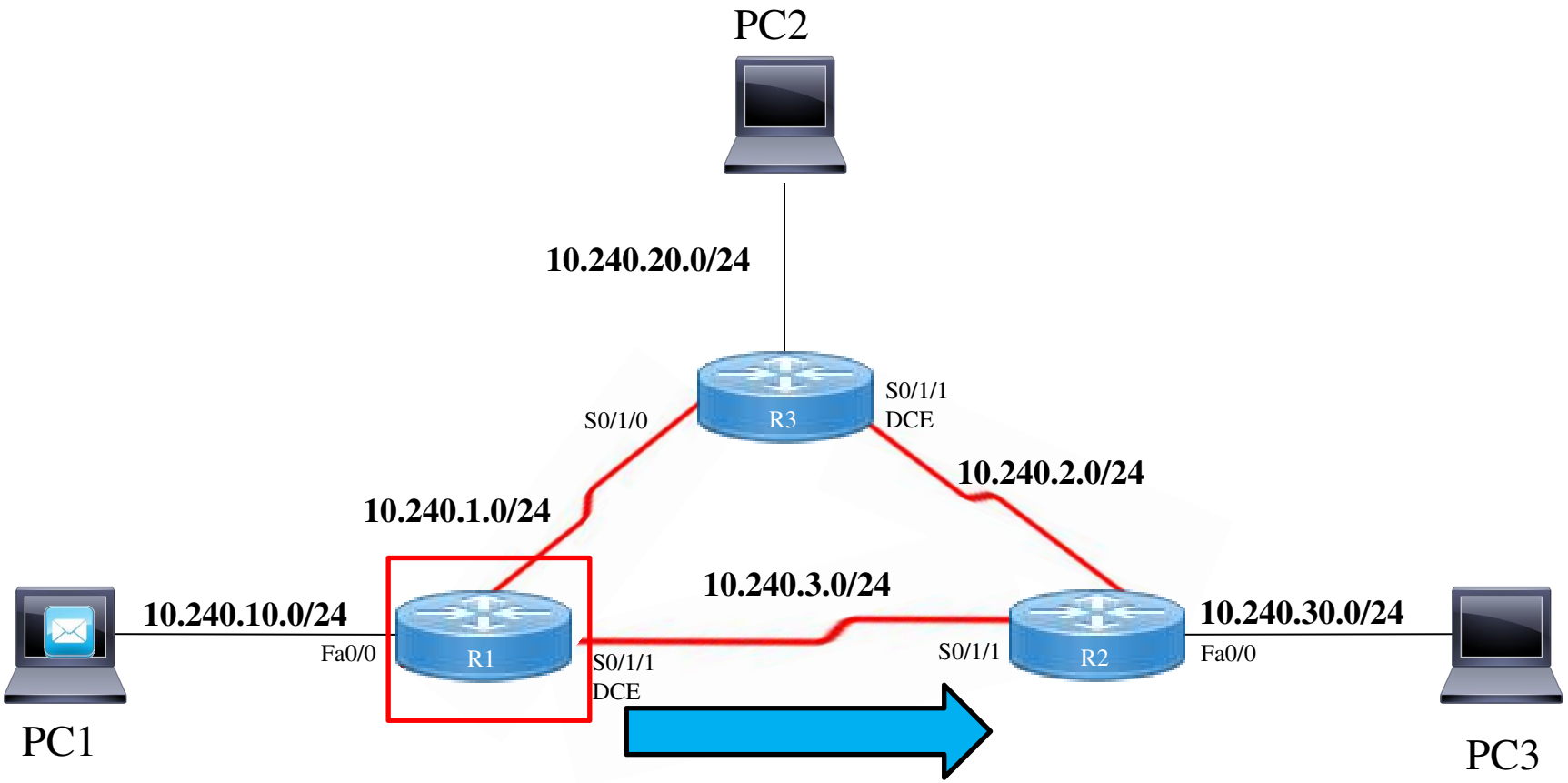
Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 6 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 is directly connected, Serial0/1/0
                   is directly connected, Serial0/1/1
C       10.240.3.0 is directly connected, Serial0/1/1
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 is directly connected, Serial0/1/0
                    is directly connected, Serial0/1/1
S       10.240.30.0 is directly connected, Serial0/1/0
                    is directly connected, Serial0/1/1
```

Répartition de charge



Gestion de la metric



Gestion des métriques

Commande : (config)*#ip route 10.240.20.0 255.255.255.0 s0/1/1* ***1***

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 10.240.30.0 255.255.255.0 s0/1/0 2
R1(config)#ip route 10.240.30.0 255.255.255.0 s0/1/1 1
```

Commande : *#show ip route*

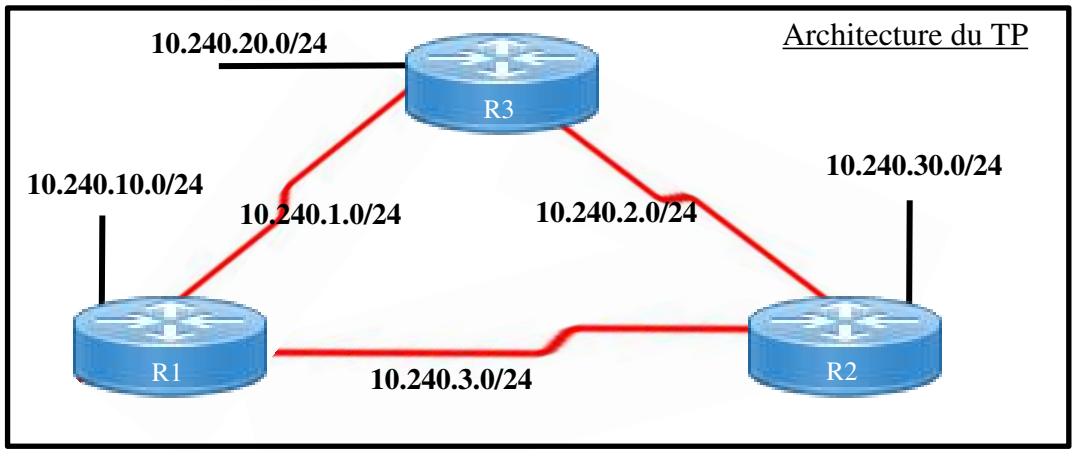
```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 6 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
S       10.240.2.0 is directly connected, Serial0/1/0
                   is directly connected, Serial0/1/1
C       10.240.3.0 is directly connected, Serial0/1/1
C       10.240.10.0 is directly connected, FastEthernet0/0
S       10.240.20.0 is directly connected, Serial0/1/0
                    is directly connected, Serial0/1/1
S       10.240.30.0 is directly connected, Serial0/1/1
```

TP : Routage statique





Configurez les interfaces FastEthernet comme indiqué dans le schéma

Visualisez la configuration des interfaces

Configurez les interfaces série comme indiqué dans le schéma

Visualisez la configuration des interfaces

Mettre en place le routage statique

Aficher la table de routage

Désactivez le CDP sur les interfaces FastEthrnet

Configurez une route par default sur R2 qui sort sur le réseau 10.240.3.0

Forcez les paquets a destination de 10.240.30.0/24 sortant de R1 à passer par le réseau 10.240.3.0/24

Afficher la table de routage



Routage dynamique



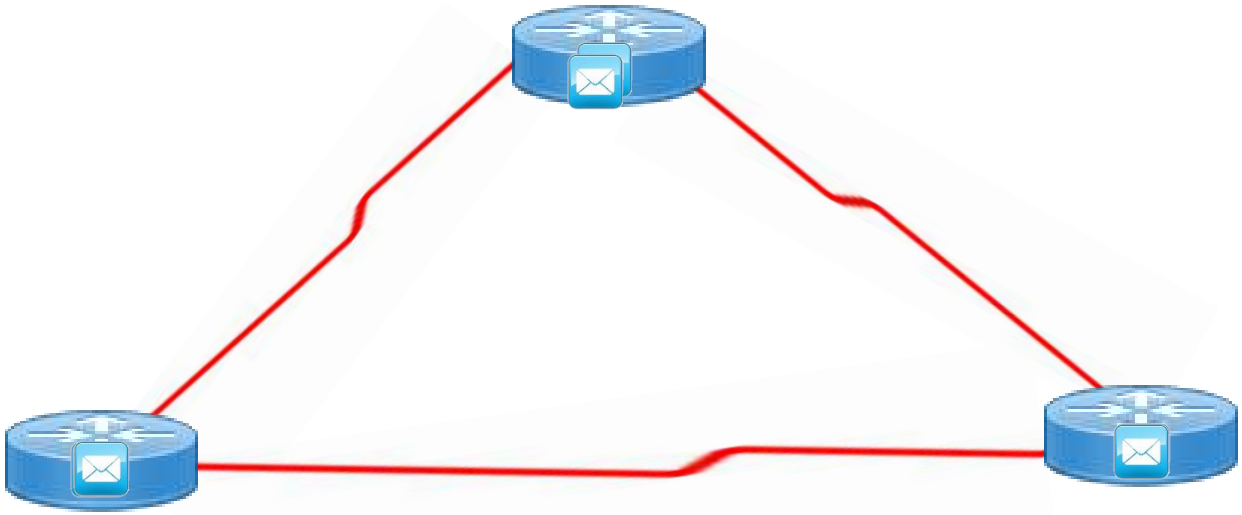


Les différents protocoles de routage

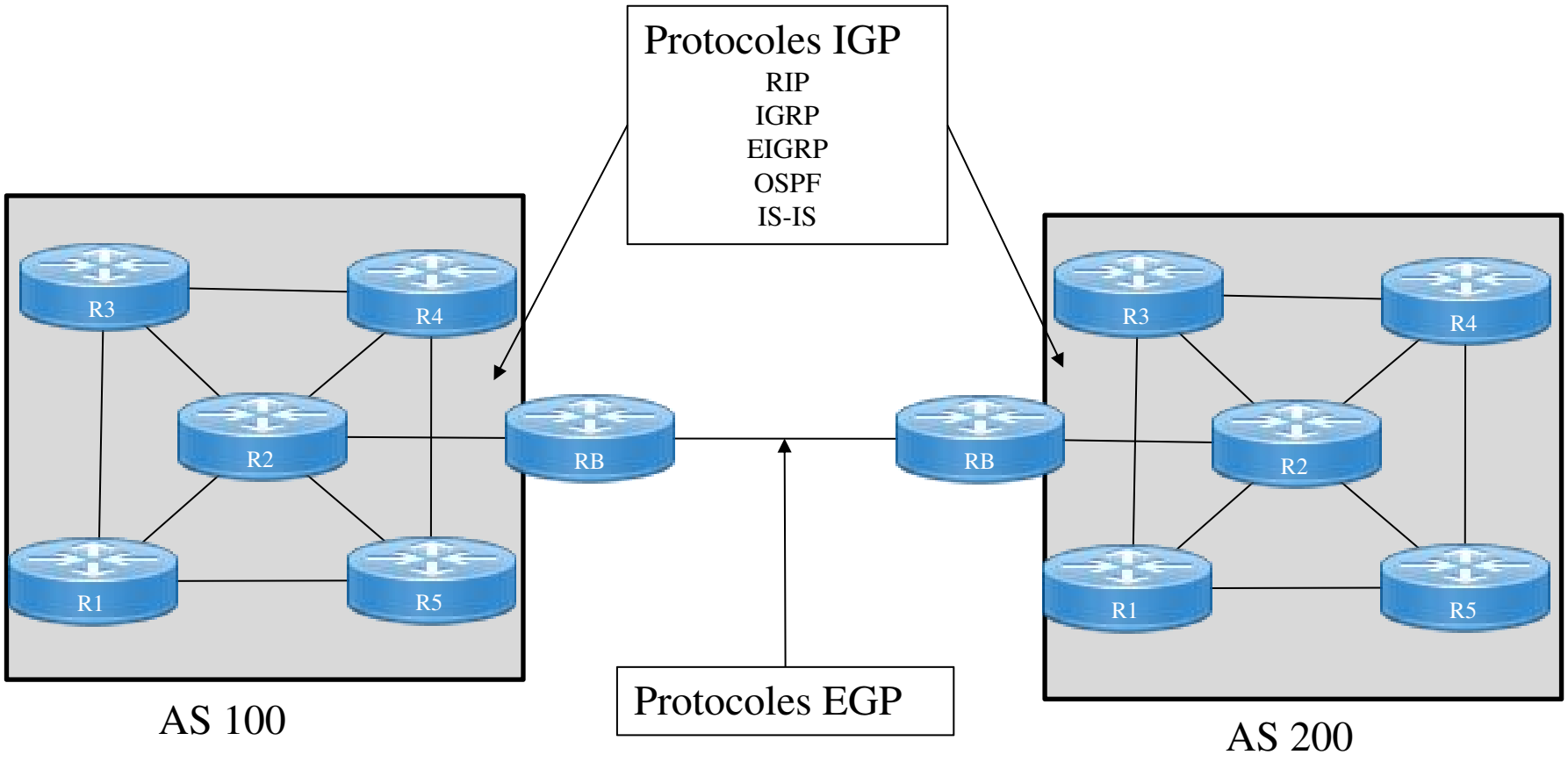
	Interior Gateway Protocols (IGP)				EGP
	vecteur de distance		Etat des liaisons		BGP
Class full	RIP	IGRP			EGP
Class less	RIP V2	EIGRP	OSPF v2	IS-IS	BGPv4
IPV6	RIPng	EIGRP IPV6	OSPFv3	IS-IS IPV6	BGPv6



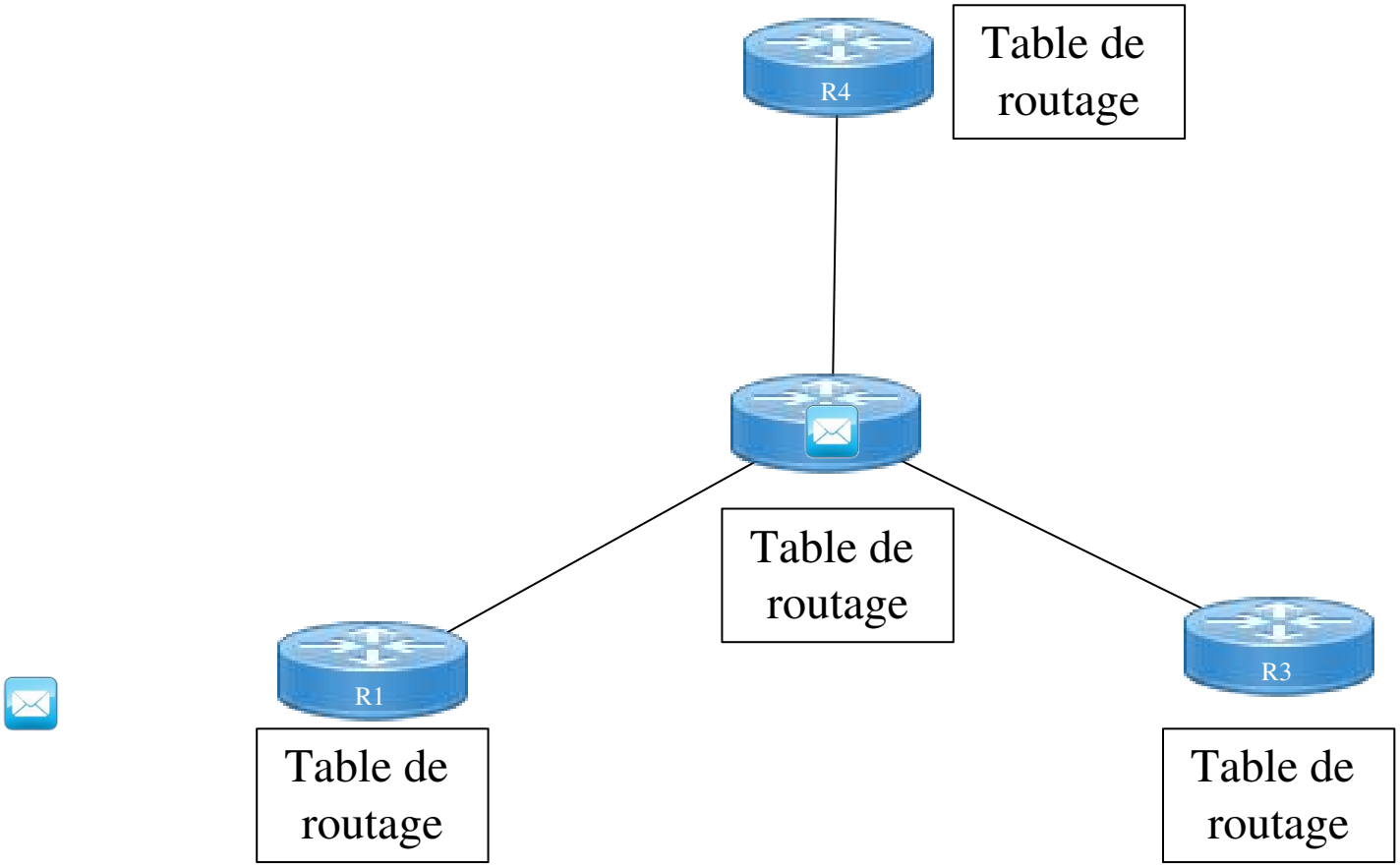
Mise à jours des tables de routage



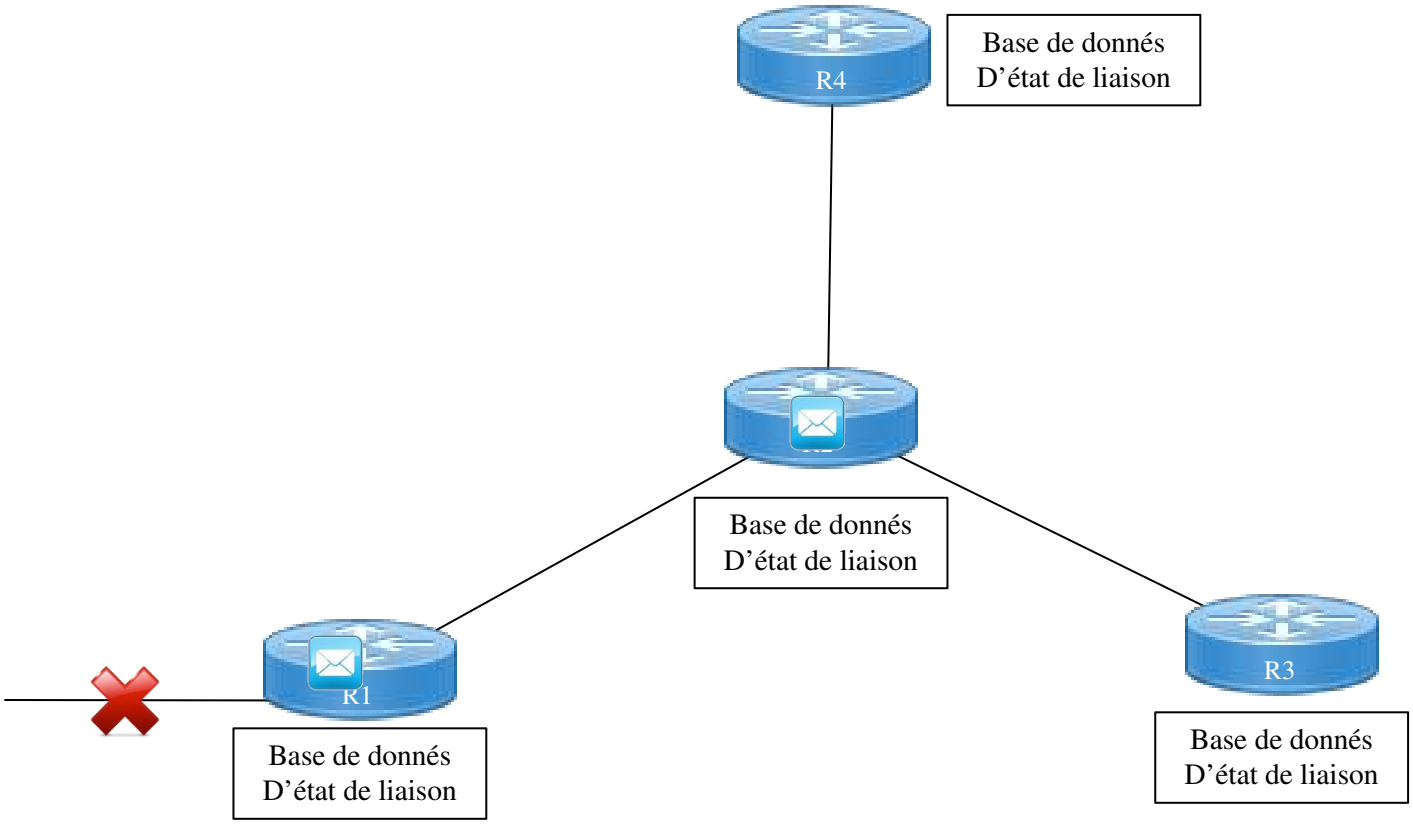
EGP et IGP


Protocoles IGP
RIP
IGRP
EIGRP
OSPF
IS-IS
R3
R4
R2
RB
R1
R5
RB
R3
R4
R2
R1
R5
Protocoles EGP
AS 100
AS 200

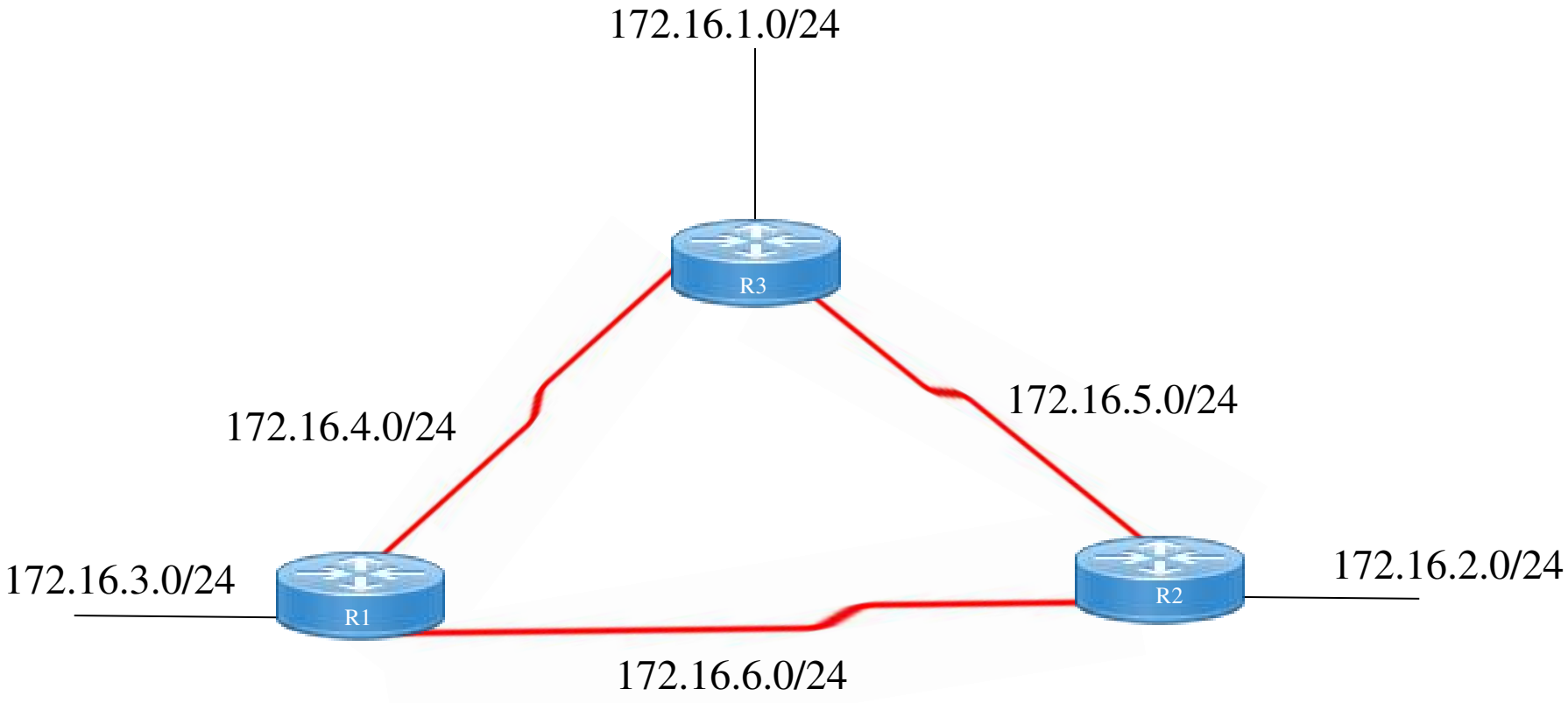
Fonctionnement du protocole à vecteur de distance



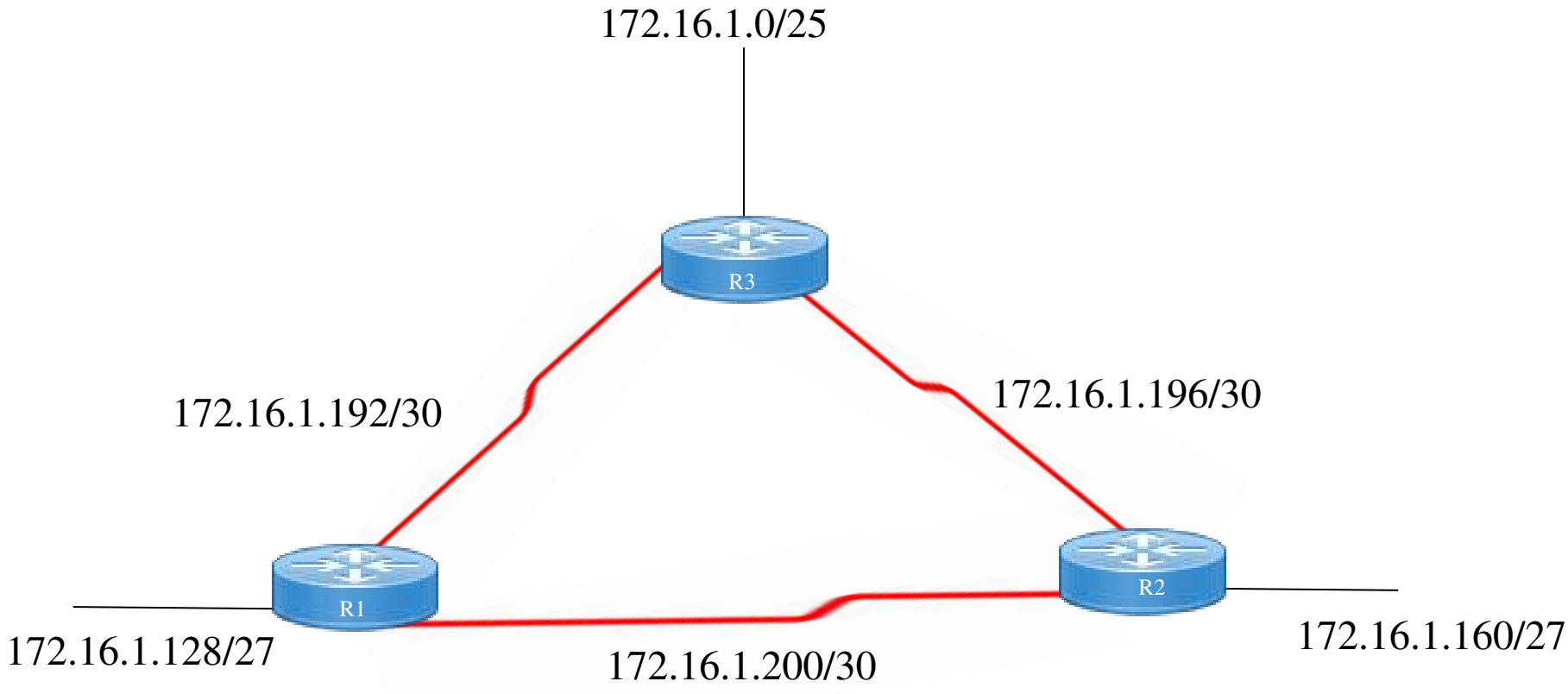
Fonctionnement du protocole à vecteur de distance



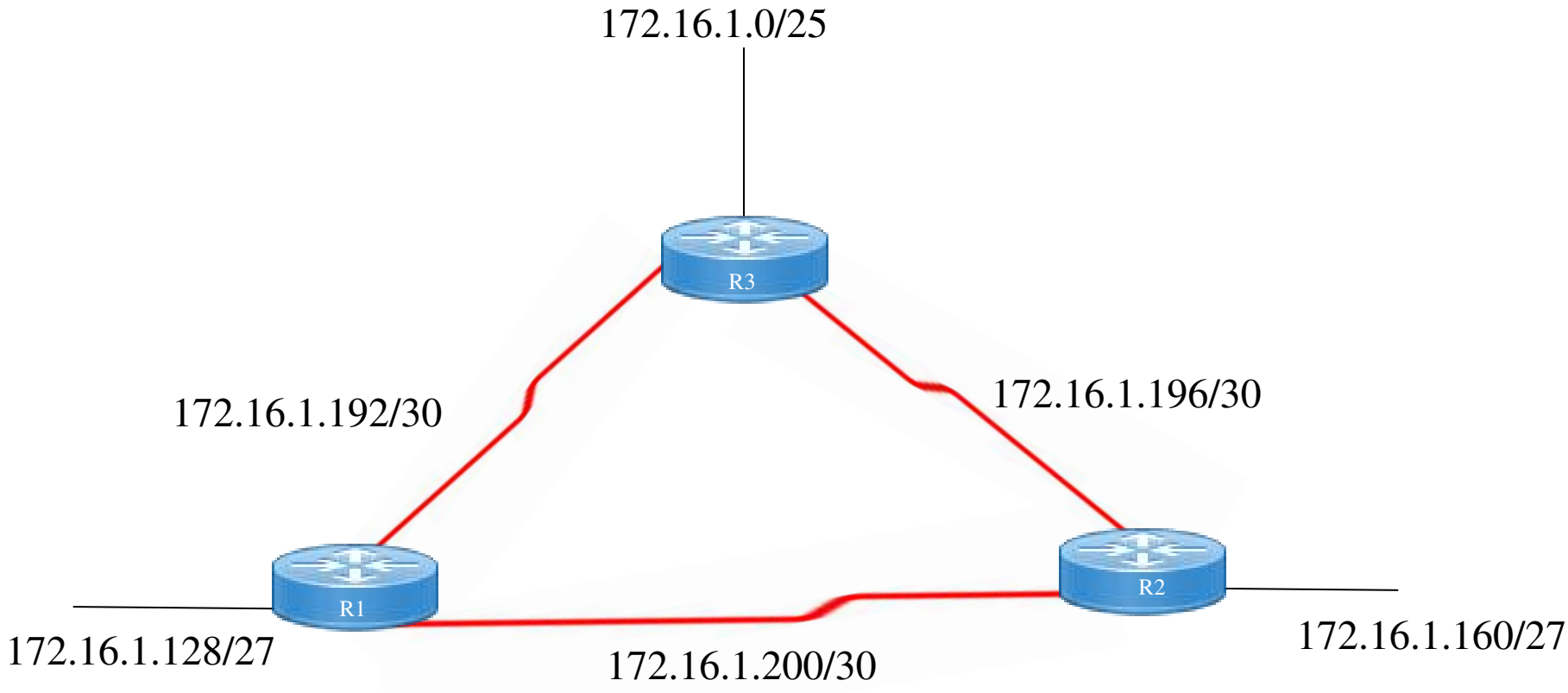
Routage class full



Routage class less



Convergence



Convergence plus lente : RIP IGRP
Convergence plus rapide : EIGRP et OSPF

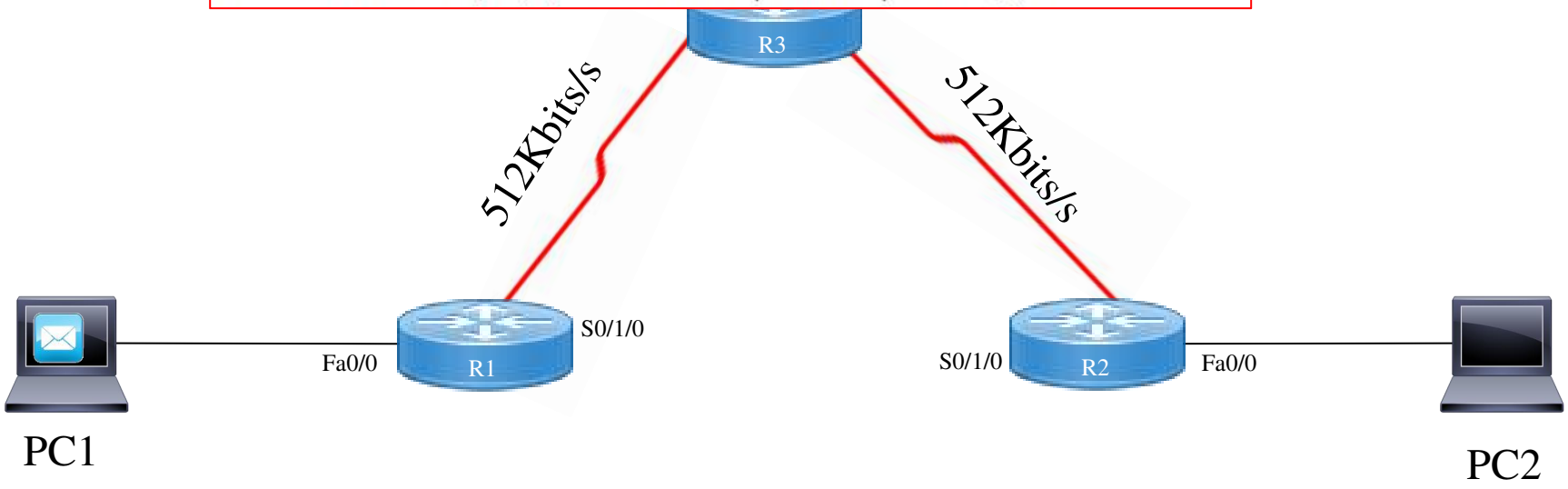
Metric

Nombre de saut

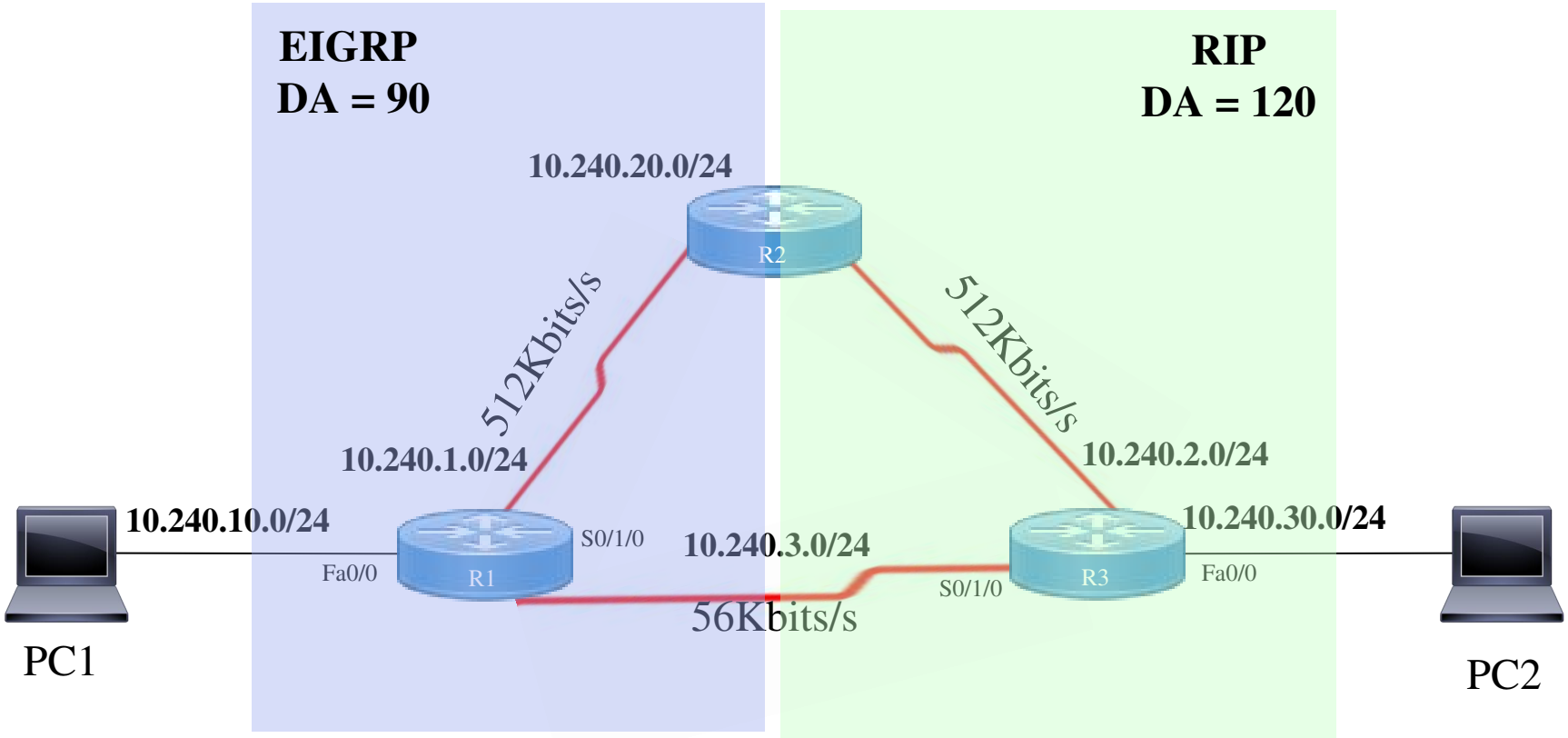
```
R1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
R       10.240.2.0 [120/1] via 10.240.1.2, 00:00:01, Serial0/1/0
C       10.240.10.0 is directly connected, FastEthernet0/0
R       10.240.20.0 [120/1] via 10.240.1.2, 00:00:01, Serial0/1/0
R       10.240.30.0 [120/2] via 10.240.1.2, 00:00:01, Serial0/1/0
```



Distance administrative



Distance administrative

Commande : *#show ip route*

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 6 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
C       10.240.2.0 is directly connected, Serial0/1/1
D       10.240.3.0 [90/2681856] via 10.240.1.1, 00:00:27, Serial0/1/0
D       10.240.10.0 [90/2172416] via 10.240.1.1, 00:00:27, Serial0/1/0
C       10.240.20.0 is directly connected, FastEthernet0/0
R       10.240.30.0 [120/1] via 10.240.2.2, 00:00:03, Serial0/1/1
```

Commande : *#show ip rip database*

```
R2#sh ip rip database
10.240.1.0/24    auto-summary
10.240.1.0/24    directly connected, Serial0/1/0
10.240.2.0/24    auto-summary
10.240.2.0/24    directly connected, Serial0/1/1
10.240.3.0/24    auto-summary
10.240.3.0/24
    [1] via 10.240.2.2, 00:00:17, Serial0/1/1
10.240.20.0/24    auto-summary
10.240.20.0/24    directly connected, FastEthernet0/0
10.240.30.0/24    auto-summary
10.240.30.0/24
    [1] via 10.240.2.2, 00:00:17, Serial0/1/1
```

Commande de visualisation des protocoles de routage

Commande : *#show ip route*

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 6 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
C       10.240.2.0 is directly connected, Serial0/1/1
D       10.240.3.0 [90/2681856] via 10.240.1.1, 00:00:27, Serial0/1/0
D       10.240.10.0 [90/2172416] via 10.240.1.1, 00:00:27, Serial0/1/0
C       10.240.20.0 is directly connected, FastEthernet0/0
R       10.240.30.0 [120/1] via 10.240.2.2, 00:00:03, Serial0/1/1
```

Commande : *#show ip protocols*

```
R2#show ip protocols

Routing Protocol is "eigrp  1 "
....
  Distance: internal 90 external 170

Routing Protocol is "rip"
.....
Distance: (default is 120)
R2#
```



Valeur des distances administratives

Origine de la route	Valeur de la DA
Connecté	0
Statique	1
Protocole BGP externe	20
EIGRP interne	90
Protocole IGRP	100
Protocole OSPF	110
Protocole de routage IS-IS	115
Protocole RIP	120
Protocole BGP interne	200



Valeur des routes directement connectés

Commande : *#show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.240.1.0 is directly connected, Serial0/1/0
D       10.240.2.0 [90/2681856] via 10.240.1.2, 00:13:00, Serial0/1/0
C       10.240.3.0 is directly connected, Serial0/1/1
C       10.240.10.0 is directly connected, FastEthernet0/0
D       10.240.20.0 [90/2172416] via 10.240.1.2, 00:13:00, Serial0/1/0
```

Commande : *#show ip route 10.240.1.0*

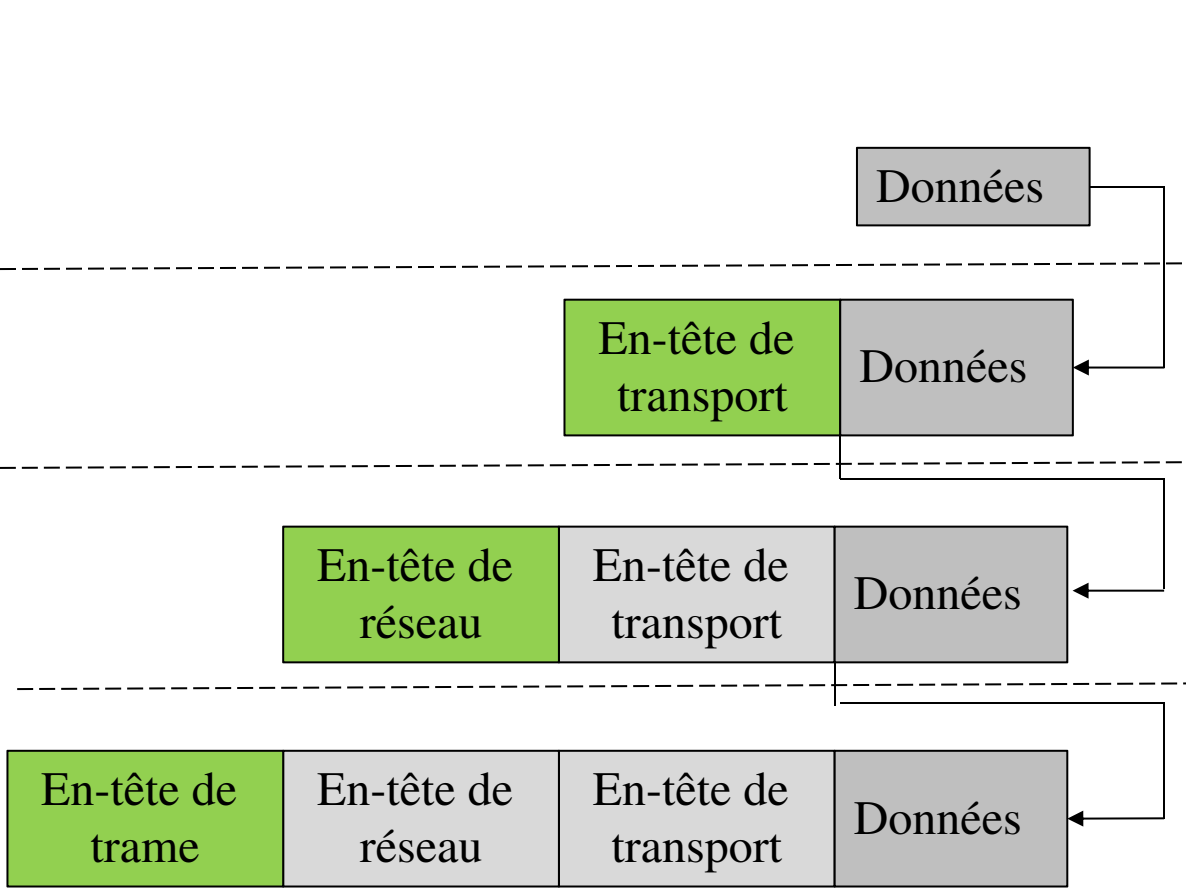
```
Routing entry for 10.240.1.0/24
Known via "connected", distance 0, metric 0 (connected, via interface)
  Redistributing via eigrp 1
  Routing Descriptor Blocks:
  * directly connected, via Serial0/1/0
      Route metric is 0, traffic share count is 1
```



RIP



Encapsulation de message RIP



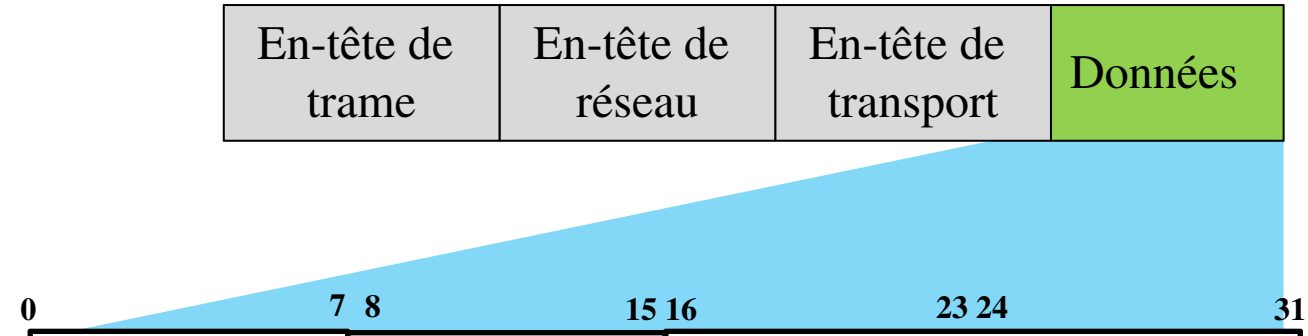
Message RIP
Commande : demande 1; Réponse 2
Version 1
Ide de famille d'adresse = 2 pour IP
Route adresse de réseau
Mesure nombre de saut

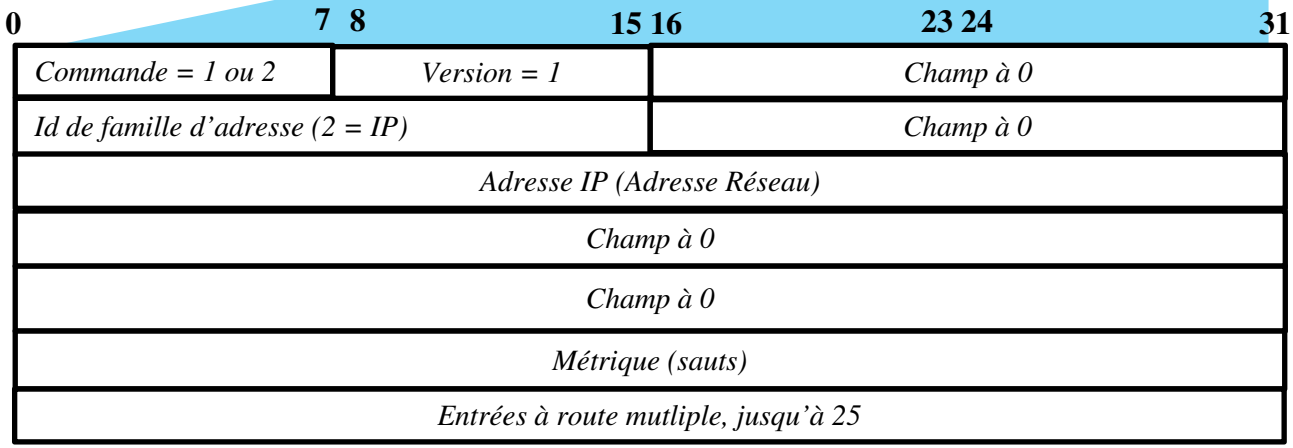
Segment UDP
Port source = 520
Port destination = 520

Paquet IP
IP source = Adresse ip de l'interface
IP destination = 255.255.255.255

Trame
MAC source = Adresse MAC de l'interface
MAC destination = FF:FF:FF:FF:FF:FF

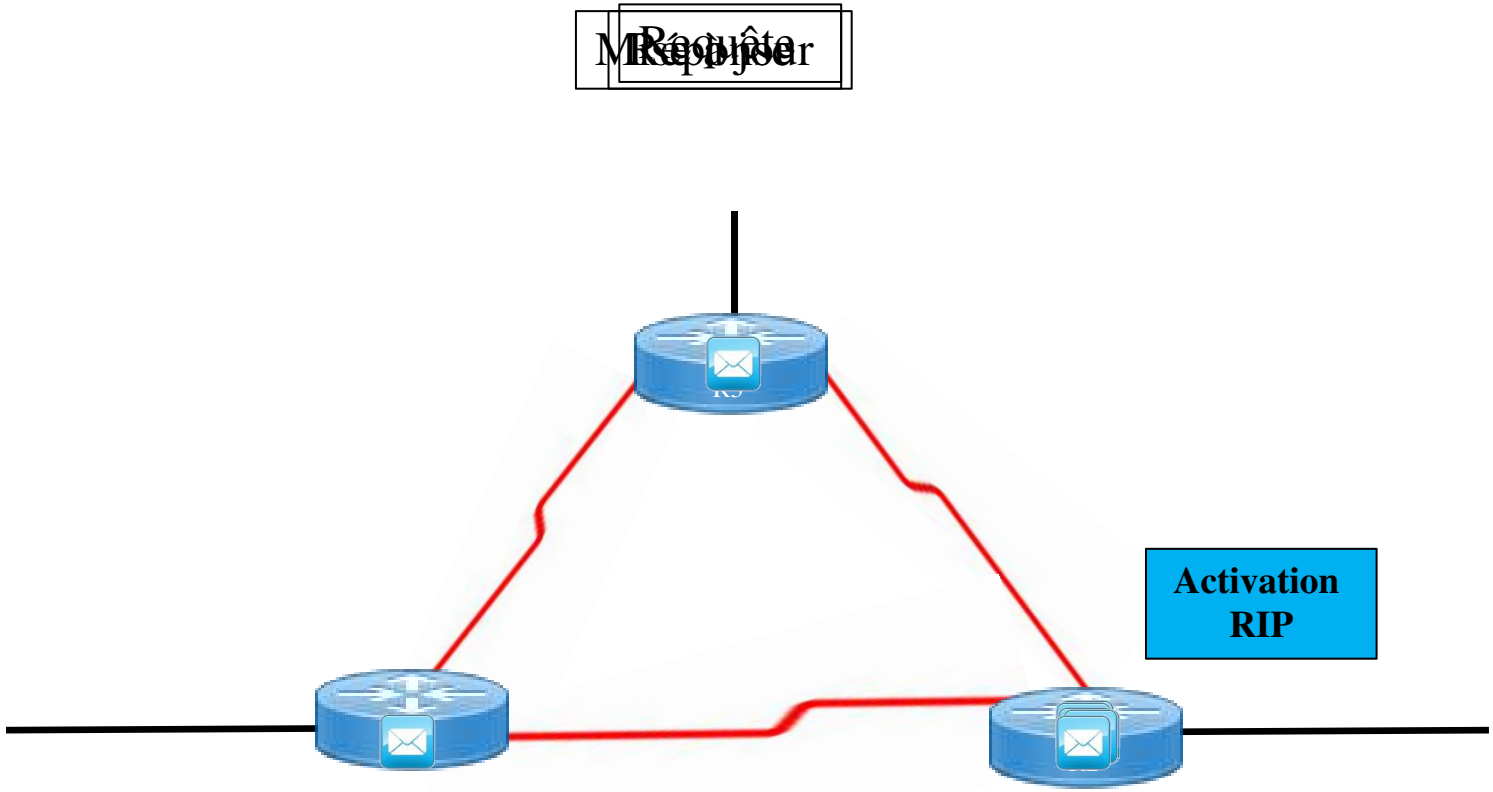
Détail du message RIP





Commande	1 pour une requête 2 pour une réponse
Version	A1pour rivp V1 2 pour rip V2
ID de famille d'adresse	2 pour une IP 0 pour un envoi de la table de routage
Adresse IP	Adresse de la route de destiantion (Réseau, S/s réseau, hôte)
Métrique	Nombre de saut de 1 à 16. Incrémentation de la métrique avant envoi u message

Processus de requête/ Réponse IP



Adressage par classe

Classe A

Réseau	Hôte	Hôte	Hôte
255 .	0 .	0 .	0

Classe B

Réseau	Réseau	Hôte	Hôte
255 .	255 .	0 .	0

Classe C

Réseau	Réseau	Réseau	Hôte
255 .	255 .	255 .	0

Classe	Début	Fin
A	0.0.0.0	127.255.255.255
B	128.0.0.0	191.255.255.255
C	224.0.0.0	223.0.0.0



Distance Administrative RIP

Commande : *#show ip route*

```
R2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:09, Serial0/1/0
C    192.168.2.0/24 is directly connected, Serial0/1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
C    192.168.4.0/24 is directly connected, Serial0/1/1
R    192.168.5.0/24 [120/1] via 192.168.4.2, 00:00:21, Serial0/1/1
R    192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:09, Serial0/1/0
                    [120/1] via 192.168.4.2, 00:00:21, Serial0/1/1
```

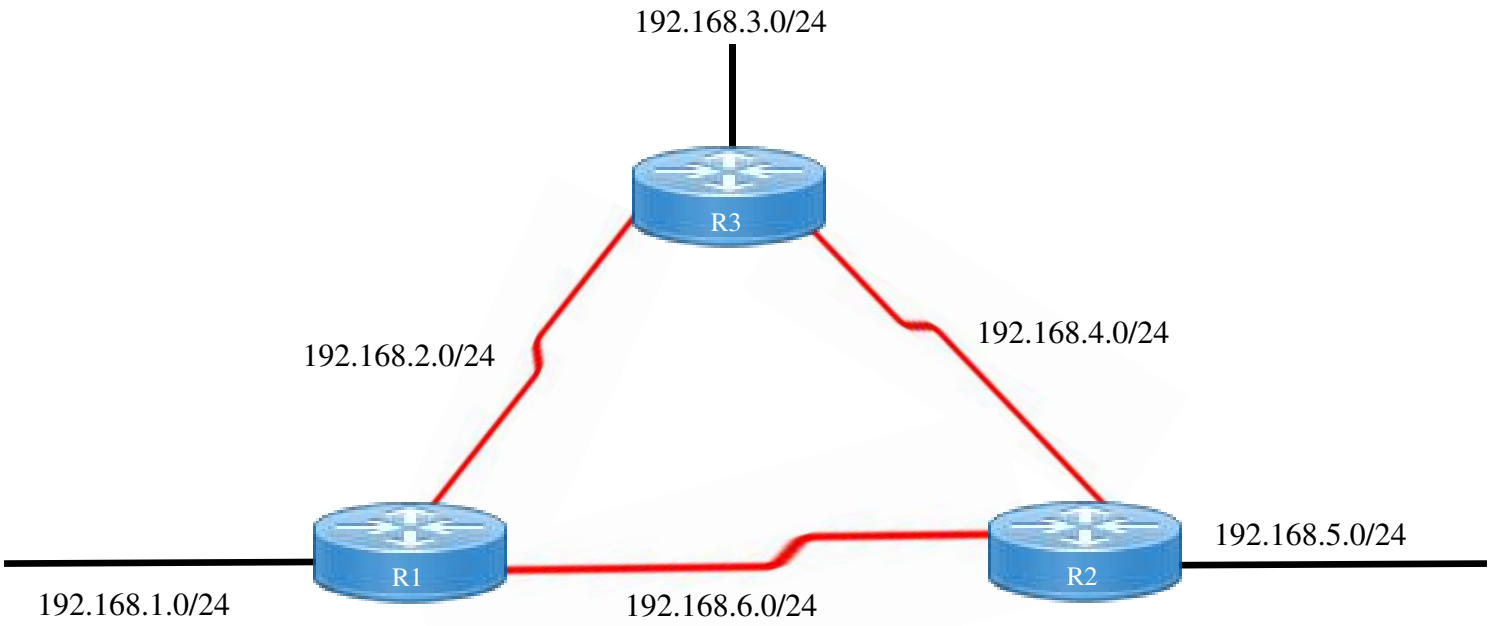
Commande : *#show ip Protocols*

```
R2#show ip protocols
....
Routing Information Sources:
        Gateway         Distance      Last Update
        192.168.2.1          120      00:00:10
        192.168.4.2          120      00:00:21
Distance: (default is 120)
```

Activation du protocole RIP

Commande : (config) *#router rip*

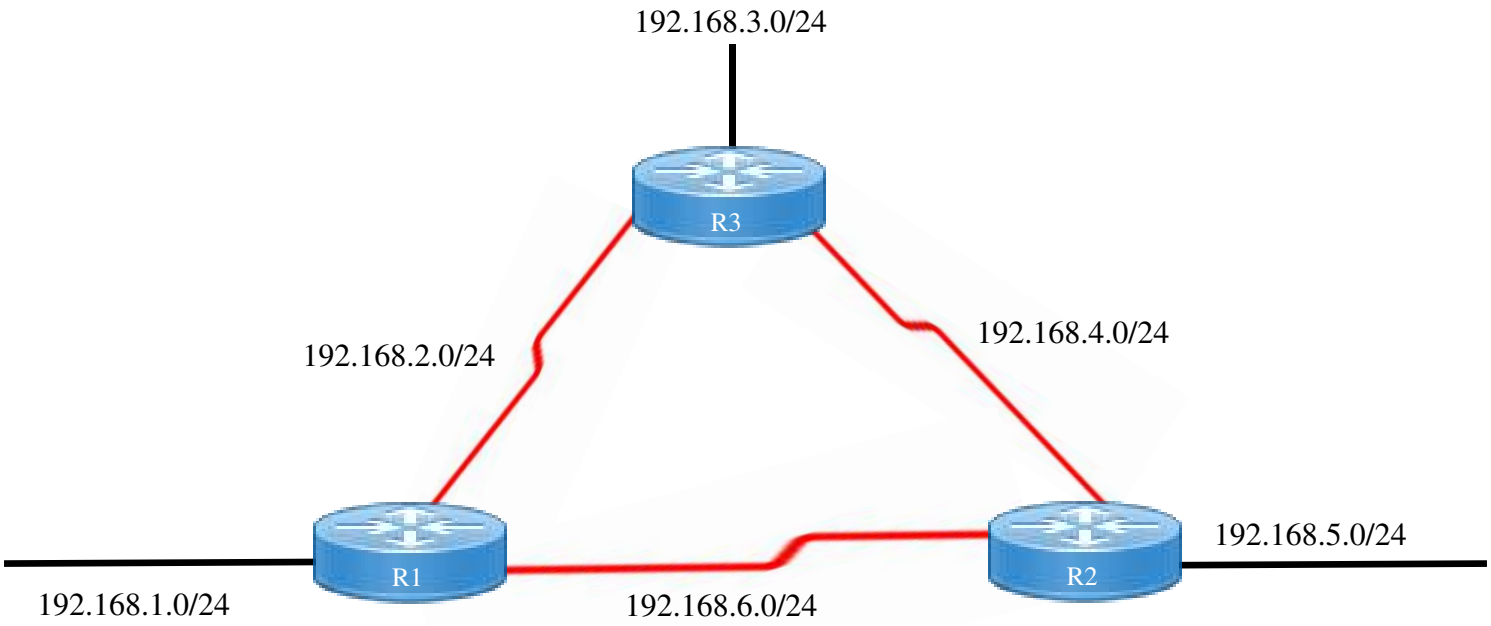
```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
R3(config)#router rip
```



Activation du protocole RIP

Commande : (config) *#router rip*

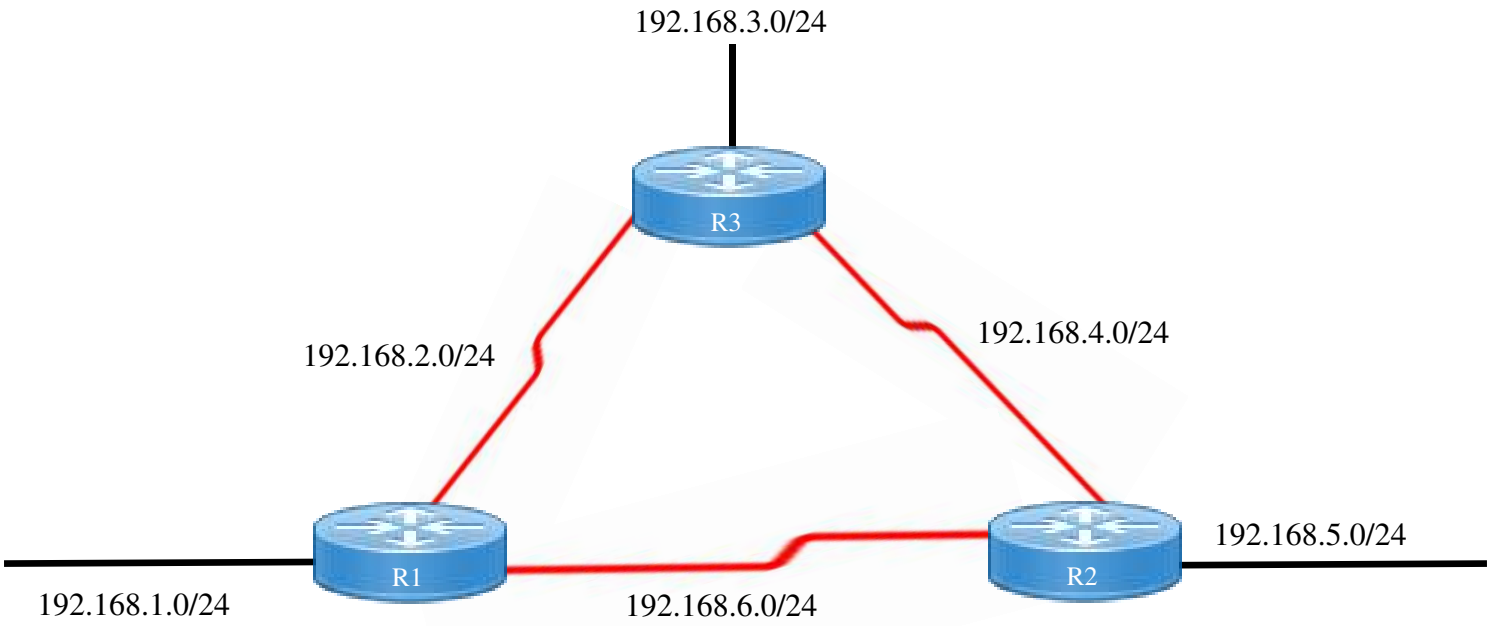
```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
R3(config)#router rip
```



Déclaration des routes

Commande : (config-router) *#network 192.168.1.0*

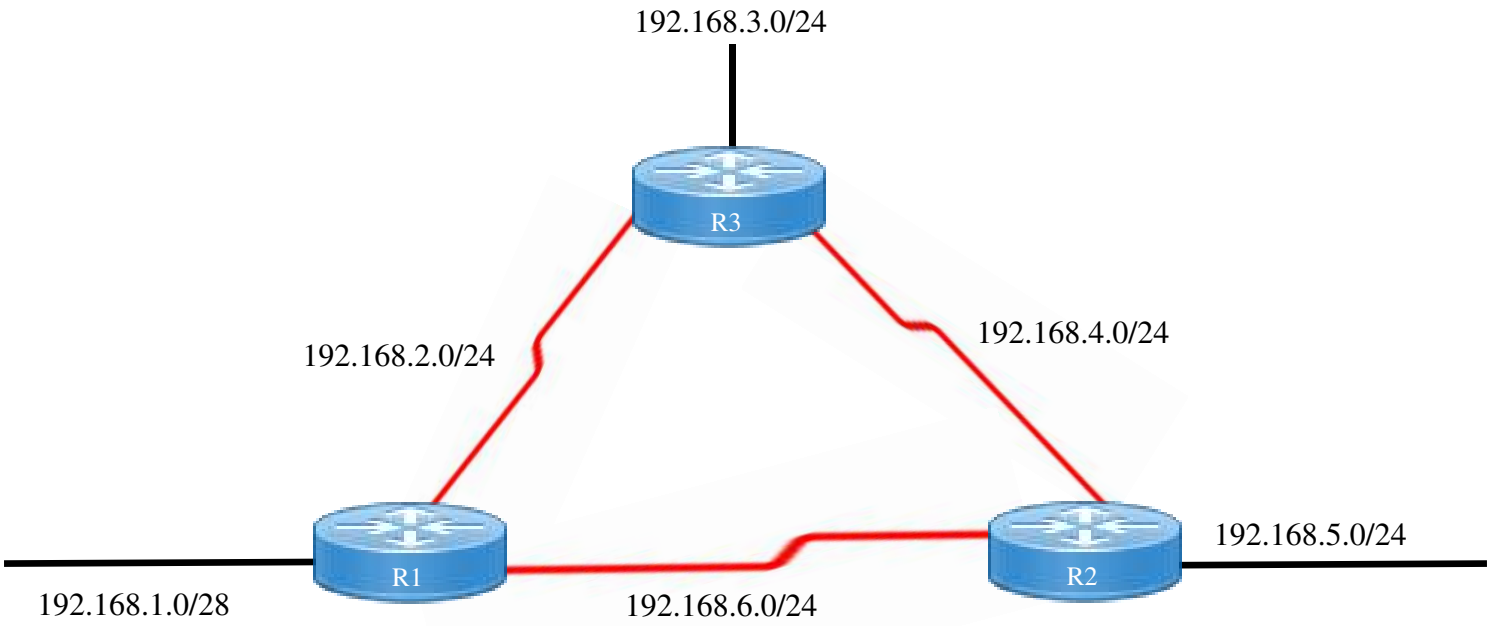
```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config-router)#network 192.168.1.0
```



Déclaration des routes

Commande : (config-router) *#network 192.168.1.0*

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config-router)#network 192.168.1.0
```





Vérification de la diffusion des routes

Commande : *#show ip route*

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:14, Serial0/1/0
C    192.168.2.0/24 is directly connected, Serial0/1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
C    192.168.4.0/24 is directly connected, Serial0/1/1
R    192.168.5.0/24 [120/1] via 192.168.4.2, 00:00:06, Serial0/1/1
R    192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:14, Serial0/1/0
                    [120/1] via 192.168.4.2, 00:00:06, Serial0/1/1
```

Analyse du protocole RIP

Commande : *#show ip protocols*

```
R3#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 3 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        1     2 1
  Serial0/1/0            1     2 1
  Serial0/1/1            1     2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
            192.168.4.0
            192.168.5.0
            192.168.6.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            192.168.6.1           120      00:00:10
            192.168.4.1           120      00:00:00
Distance: (default is 120)
```

Mode debug RIP

Commande : *#debug ip rip*

```
R3#debug ip rip
RIP protocol debugging is on
R3#RIP: received v1 update from 192.168.6.1 on Serial0/1/1
      192.168.1.0 in 1 hops
      192.168.2.0 in 1 hops
      192.168.3.0 in 2 hops

RIP: received v1 update from 192.168.4.1 on Serial0/1/0
      192.168.1.0 in 2 hops
      192.168.2.0 in 1 hops
      192.168.3.0 in 1 hops

RIP: sending  v1 update to 255.255.255.255 via FastEthernet0/0 (192.168.5.1)
RIP: build update entries
      network 192.168.1.0 metric 2
      network 192.168.2.0 metric 2
      network 192.168.3.0 metric 2
      network 192.168.4.0 metric 1
      network 192.168.6.0 metric 1

RIP: sending  v1 update to 255.255.255.255 via Serial0/1/0 (192.168.4.2)
RIP: build update entries
      network 192.168.1.0 metric 2
      network 192.168.5.0 metric 1
      network 192.168.6.0 metric 1
RIP: sending  v1 update to 255.255.255.255 via Serial0/1/1 (192.168.6.2)

RIP: received v1 update from 192.168.4.1 on Serial0/1/0
      192.168.1.0 in 16 hops
RIP: received v1 update from 192.168.6.1 on Serial0/1/1
      192.168.1.0 in 16 hops

RIP: received v1 update from 192.168.4.1 on Serial0/1/0
      192.168.1.0 in 2 hops
```



Retrait d'une interface au processus RIP

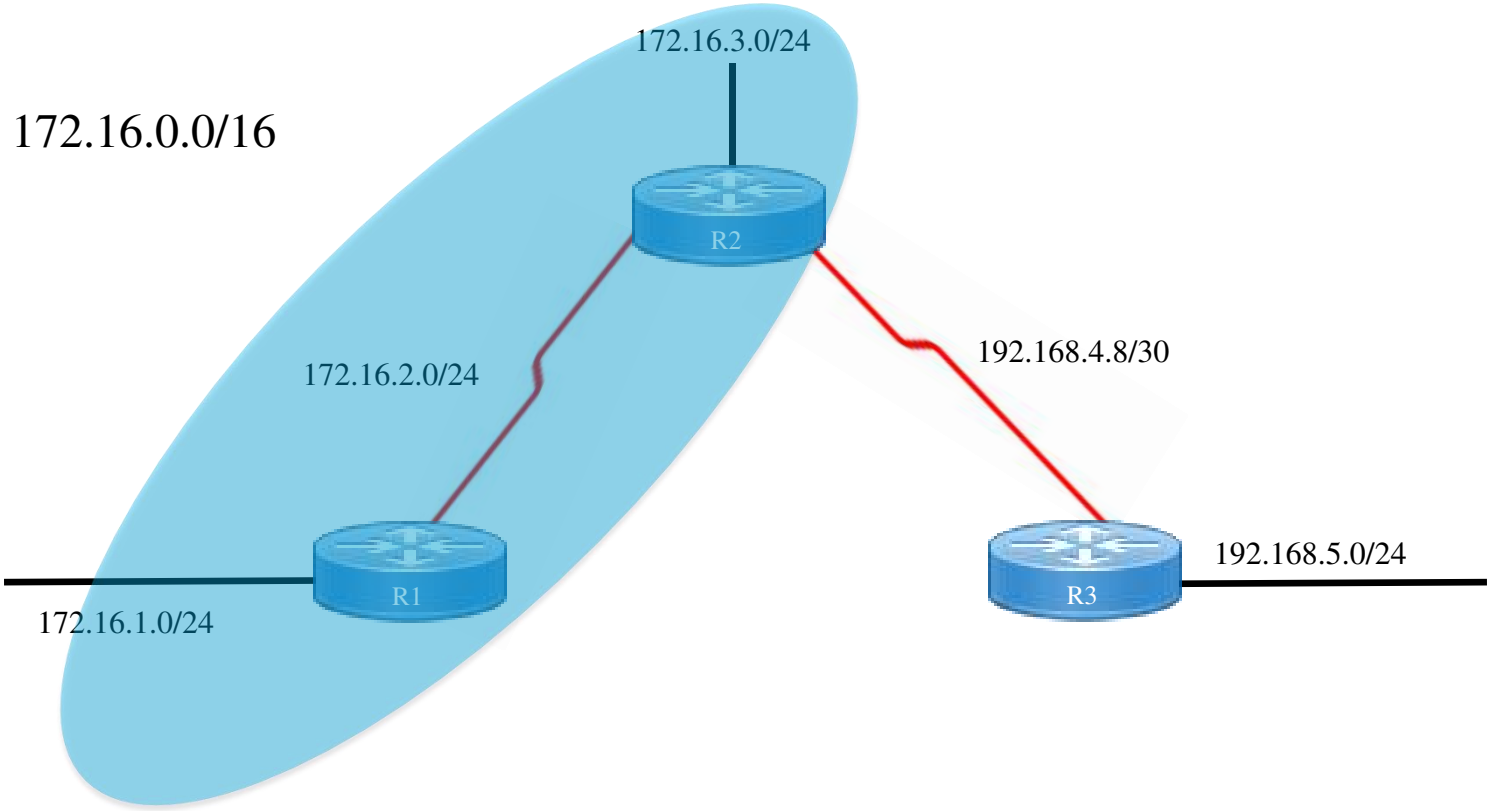
Commande : (config-router) #passive-interface fa 0/0

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#passive-interface fa 0/0
```

Commande : *#show ip protocols*

```
R1#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 12 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface              Send  Recv  Triggered RIP  Key-chain
  Serial0/1/1            1     2 1
  Serial0/1/0            1     2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
            192.168.1.0
            192.168.2.0
            192.168.6.0
```

Résumé de route





Résumé de route

Table de routage de R1

```
R1(config-router)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/1/0
R       172.16.3.0 [120/1] via 172.16.2.2, 00:00:07, Serial0/1/0
R    192.168.4.0/24 [120/1] via 172.16.2.2, 00:00:07, Serial0/1/0
R    192.168.5.0/24 [120/2] via 172.16.2.2, 00:00:07, Serial0/1/0
```

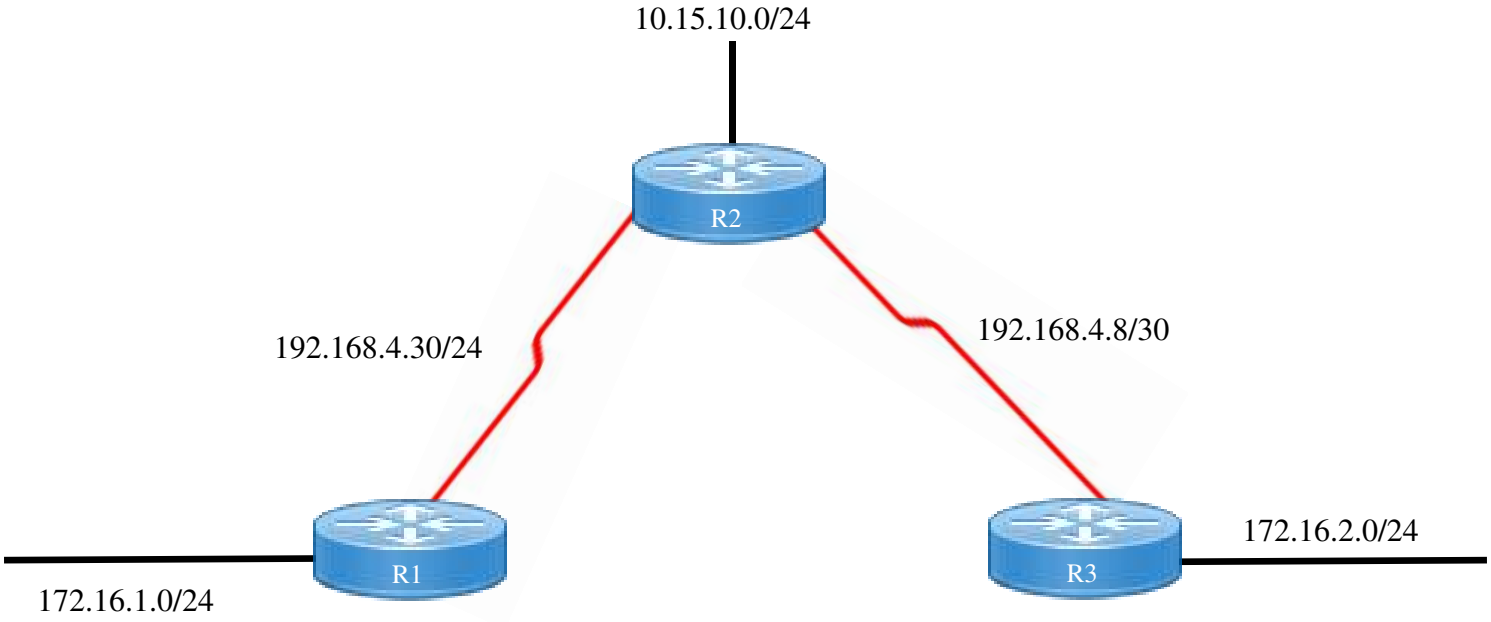
Table de routage de R3

```
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    172.16.0.0/16 [120/1] via 192.168.4.9, 00:00:04, Serial0/1/0
C    192.168.4.0/24 is directly connected, Serial0/1/0
C    192.168.5.0/24 is directly connected, FastEthernet0/0
```

Résumé de route





Problème de résumé de route

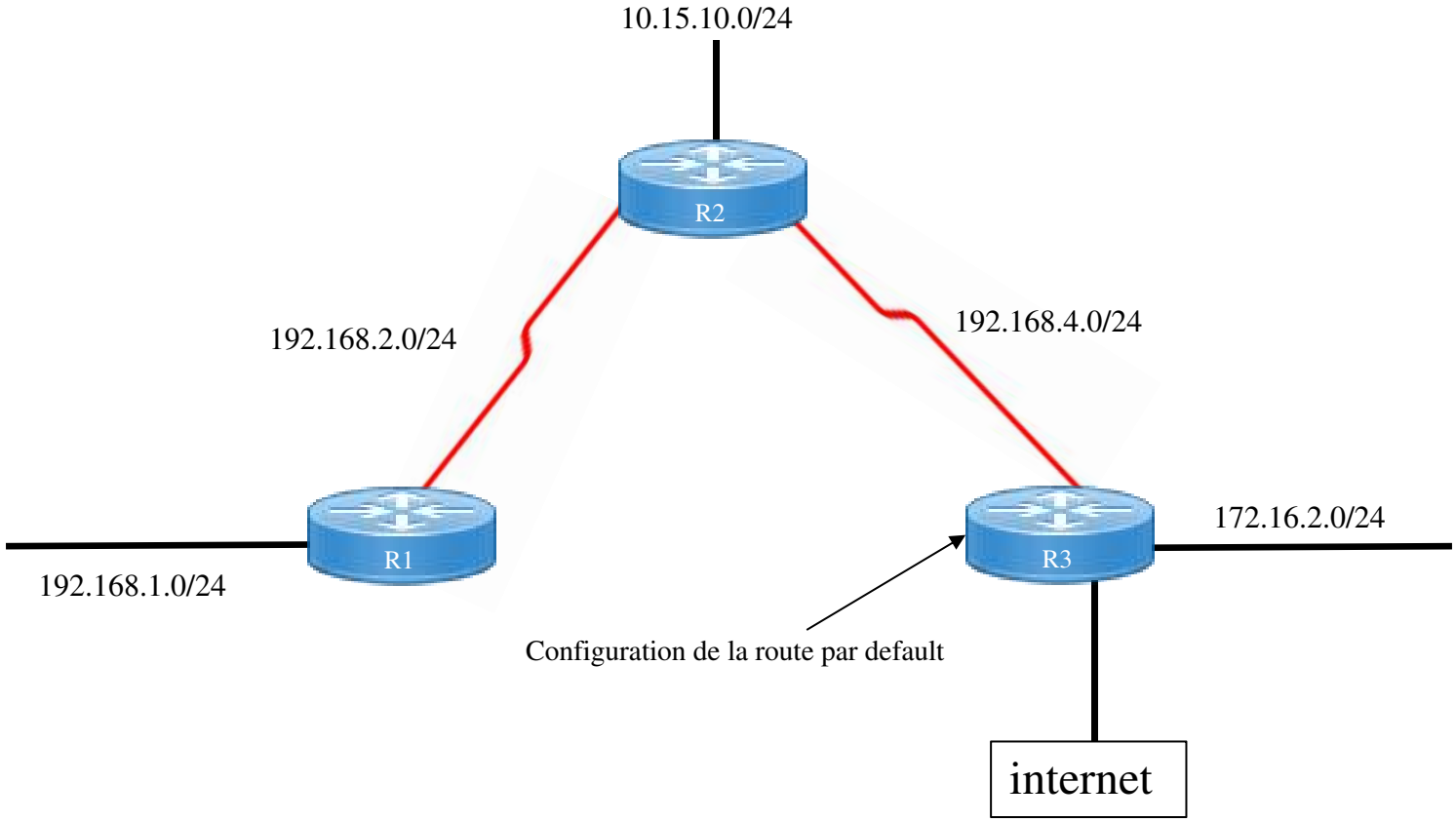
Table de routage de R2

```
R2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.15.10.0 is directly connected, FastEthernet0/0
R    172.16.0.0/16 [120/1] via 192.168.4.13, 00:00:03, Serial0/1/0
                   [120/1] via 192.168.4.10, 00:00:19, Serial0/1/1
     192.168.4.0/30 is subnetted, 2 subnets
C       192.168.4.8 is directly connected, Serial0/1/1
C       192.168.4.12 is directly connected, Serial0/1/0
```

Diffusion de la route par default





Diffusion de la route par default

Commande : *(config-router)#default-information originate*

```
R3(config)#router rip
R3(config-router)#default-information originate
```

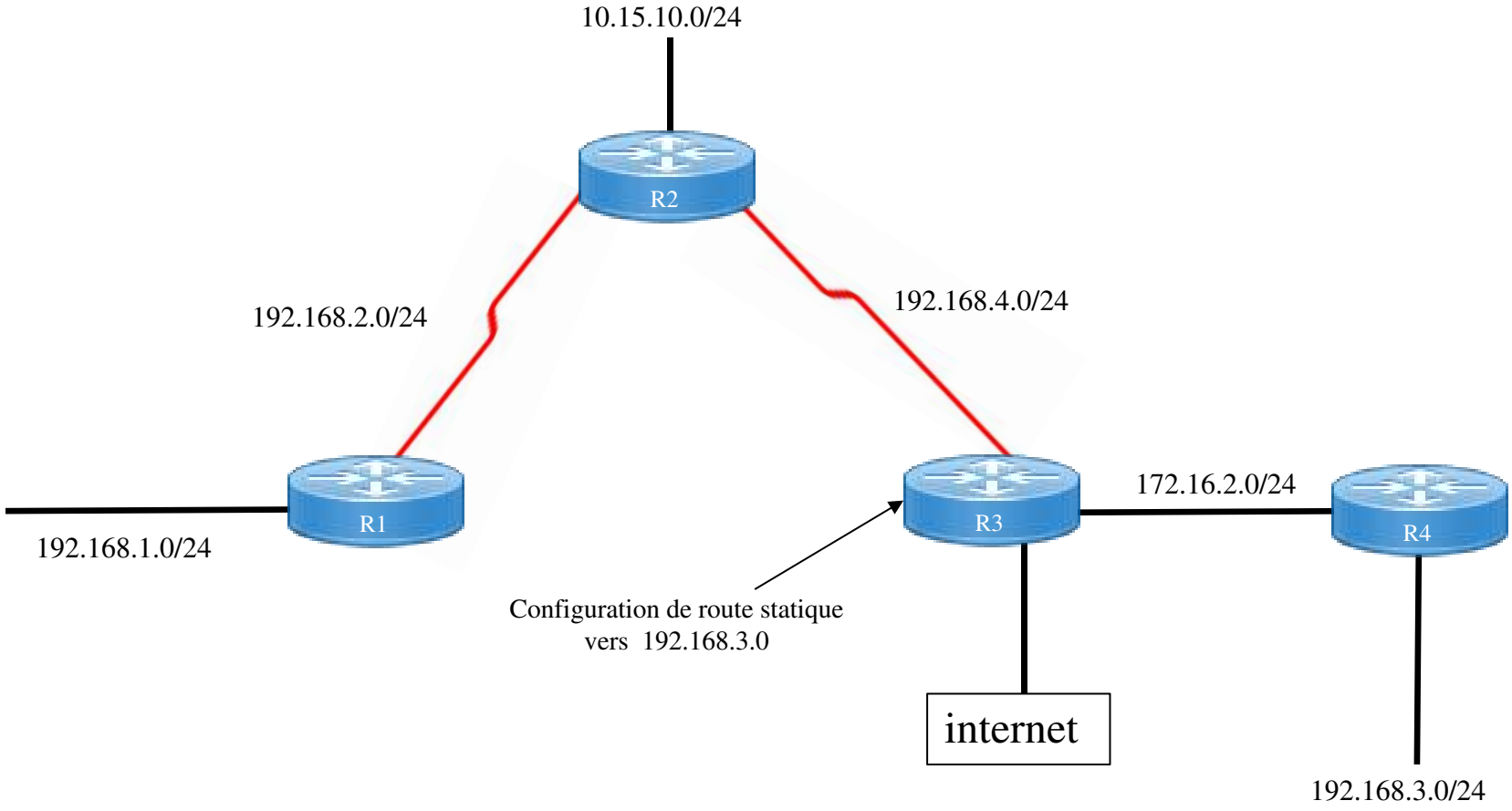
Table de routage de R2

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.4.1 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 1 subnets
C       10.15.10.0 is directly connected, FastEthernet0/0
R    172.16.0.0/16 [120/1] via 192.168.4.1, 00:00:16, Serial0/1/1
R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:25, Serial0/1/0
C    192.168.2.0/24 is directly connected, Serial0/1/0
C    192.168.4.0/24 is directly connected, Serial0/1/1
R*   0.0.0.0/0 [120/1] via 192.168.4.1, 00:00:16, Serial0/1/1
```

Redistribution de route statique



Redistribution de route statique

Commande : *(config-router)#redistribute static*

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router rip
R3(config-router)#redistribute static
```

Table de routage de R3

```
R3(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

R    10.0.0.0/8 [120/1] via 192.168.4.2, 00:00:19, Serial0/1/0
     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.2.0 is directly connected, FastEthernet0/0
R    192.168.1.0/24 [120/2] via 192.168.4.2, 00:00:19, Serial0/1/0
R    192.168.2.0/24 [120/1] via 192.168.4.2, 00:00:19, Serial0/1/0
S    192.168.3.0/24 is directly connected, FastEthernet0/0
C    192.168.4.0/24 is directly connected, Serial0/1/0
C    210.80.15.0/24 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 is directly connected, FastEthernet0/1
```

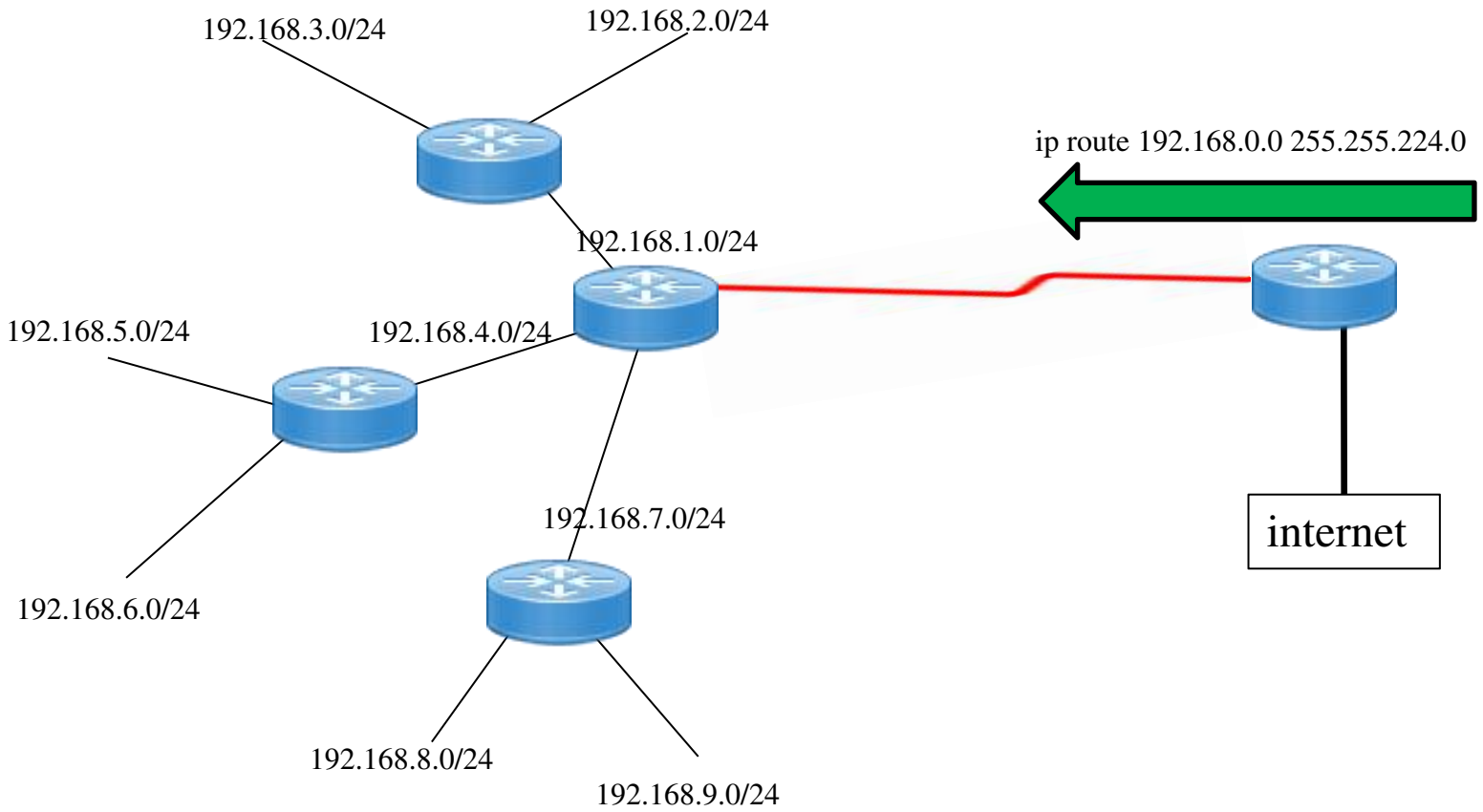
Table de routage de R2

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

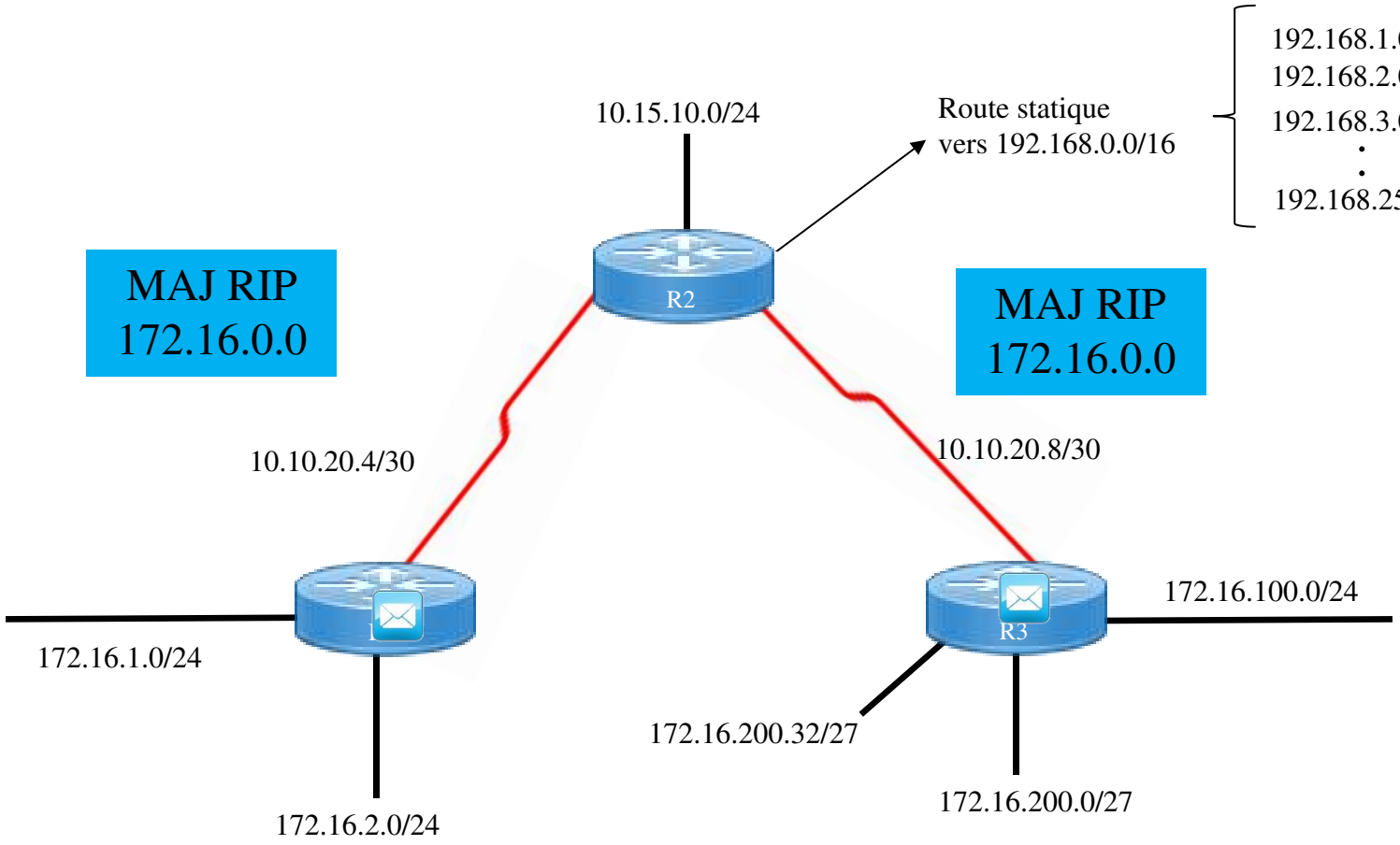
Gateway of last resort is 192.168.4.1 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 1 subnets
C       10.15.10.0 is directly connected, FastEthernet0/0
R    172.16.0.0/16 [120/1] via 192.168.4.1, 00:00:15, Serial0/1/1
R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:19, Serial0/1/0
C    192.168.2.0/24 is directly connected, Serial0/1/0
R    192.168.3.0/24 [120/1] via 192.168.4.1, 00:00:15, Serial0/1/1
C    192.168.4.0/24 is directly connected, Serial0/1/1
R*   0.0.0.0/0 [120/1] via 192.168.4.1, 00:00:15, Serial0/1/1
```

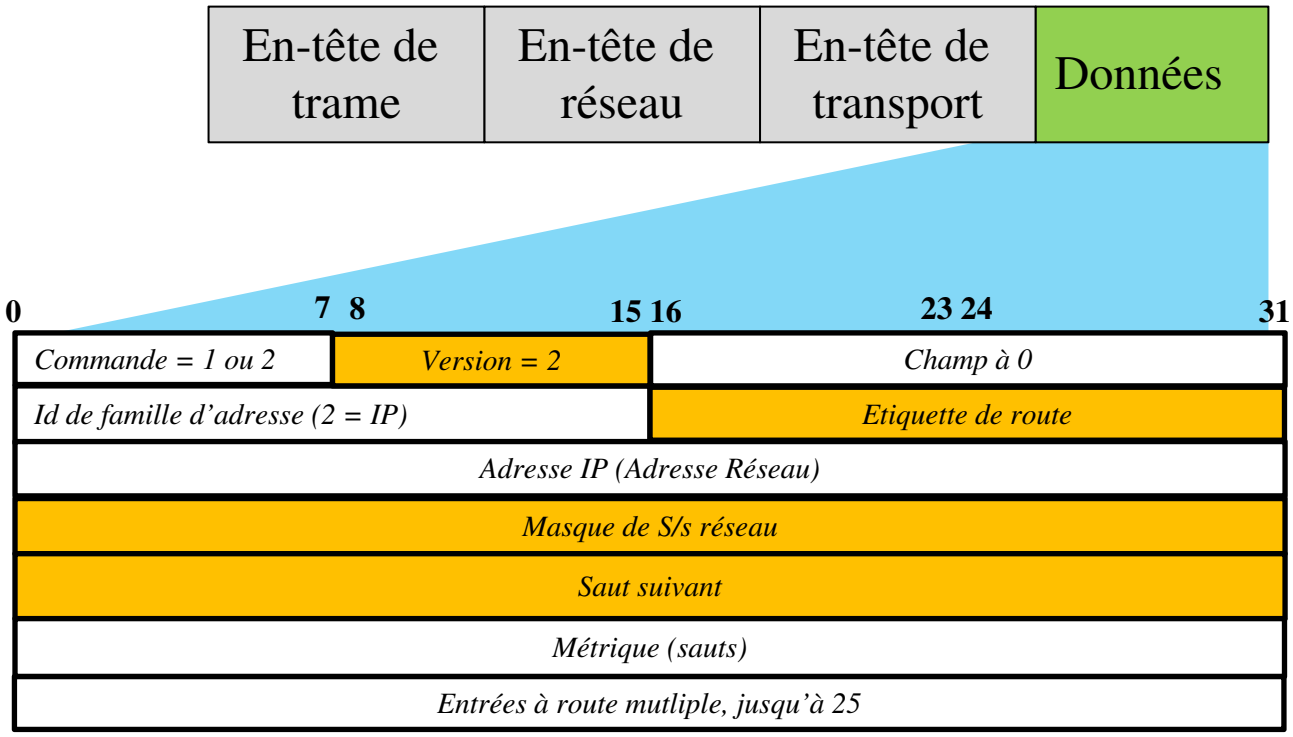
Résumé de route



Limite du protocole RIP V1



Détail du message RIP V2





Activation de RIP V2

Commande : (config-router)*#version 2*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
```

Commande : *#show ip protocols*

```
R2#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 9 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, static
Default version control: send version 2, receive 2
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        2     2
  Serial0/1/0            2     2
  Serial0/1/1            2     2
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
           10.0.0.0
Passive Interface(s):
Routing Information Sources:
           Gateway         Distance      Last Update
           10.10.20.10          120      00:00:21
           10.10.20.5           120      00:00:14
Distance: (default is 120)
```



Vérification des tables de routage

Table de routage de R1

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.10.20.4/30 is directly connected, Serial0/1/0
R       10.10.20.8/30 [120/1] via 10.10.20.6, 00:00:29, Serial0/1/0
R       10.15.10.0/24 [120/1] via 10.10.20.6, 00:00:29, Serial0/1/0
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, FastEthernet0/1
R    192.168.0.0/16 [120/1] via 10.10.20.6, 00:00:29, Serial0/1/0
```

Table de routage de R2

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.10.20.4/30 is directly connected, Serial0/1/0
C       10.10.20.8/30 is directly connected, Serial0/1/1
C       10.15.10.0/24 is directly connected, FastEthernet0/0
R    172.16.0.0/16 [120/1] via 10.10.20.10, 00:00:21, Serial0/1/1
                   [120/1] via 10.10.20.5, 00:00:13, Serial0/1/0
S    192.168.0.0/16 is directly connected, Null0
```



Vérification de la configuration du protocoles

Table de routage de R1

```
R3#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 17 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        2     2
  FastEthernet0/1        2     2
  Ethernet0/0/0          2     2
  Serial0/1/0            2     2
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
            172.16.0.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            10.10.20.9           120      00:00:09
Distance: (default is 120)
```



Suppression du résumé des réseau

Commande : (config-router)*#no auto-summary*

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router rip
R3(config-router)#no auto-summary
R3(config-router)#
```

Commande : *#show ip protocols*

```
R3#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 23 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        2     2
  FastEthernet0/1        2     2
  Ethernet0/0/0          2     2
  Serial0/1/0            2     2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
            172.16.0.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            10.10.20.9           120      00:00:06
Distance: (default is 120)
```



Vérification de la configuration

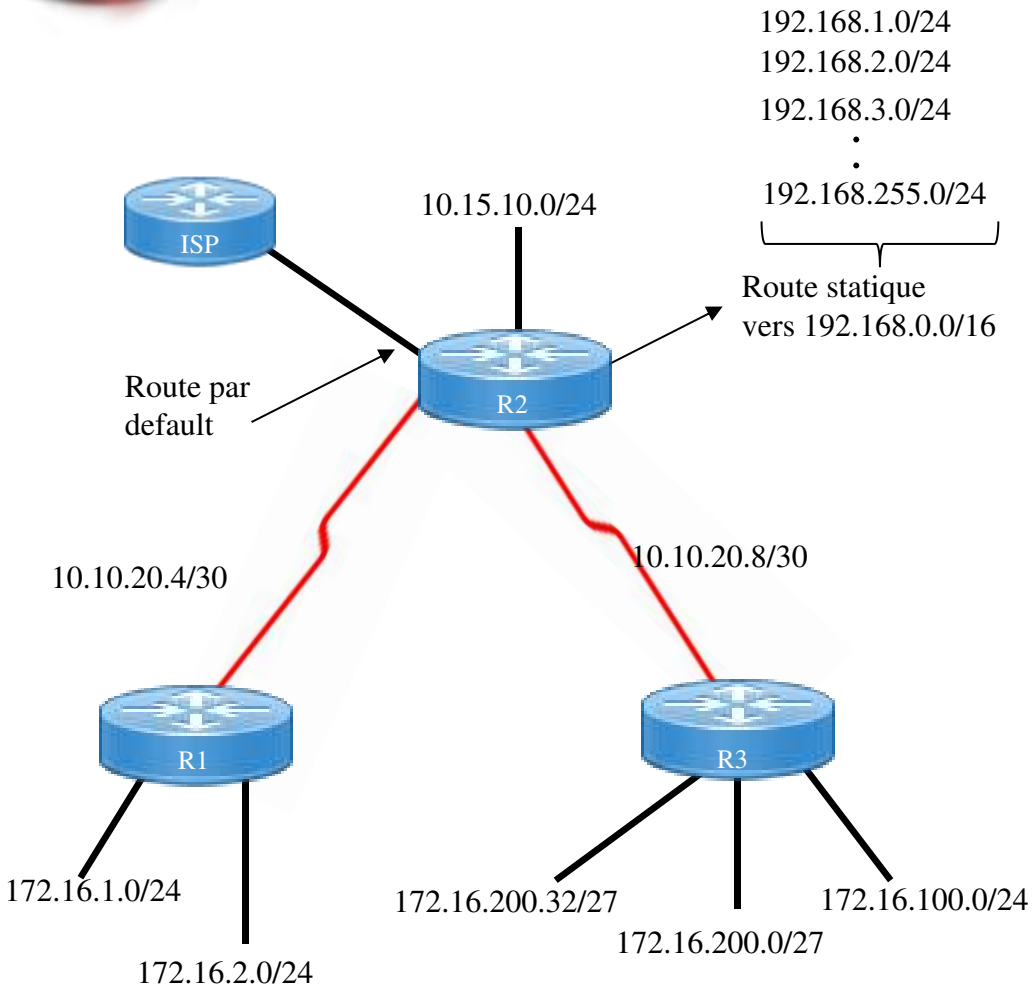
Table de routage de R1

```
R2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.10.20.4/30 is directly connected, Serial0/1/0
C       10.10.20.8/30 is directly connected, Serial0/1/1
C       10.15.10.0/24 is directly connected, FastEthernet0/0
     172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
R       172.16.0.0/16 [120/1] via 10.10.20.10, 00:02:58, Serial0/1/1
                      [120/1] via 10.10.20.5, 00:01:31, Serial0/1/0
R       172.16.1.0/24 [120/1] via 10.10.20.5, 00:00:07, Serial0/1/0
R       172.16.2.0/24 [120/1] via 10.10.20.5, 00:00:07, Serial0/1/0
R       172.16.100.0/24 [120/1] via 10.10.20.10, 00:00:11, Serial0/1/1
R       172.16.200.0/27 [120/1] via 10.10.20.10, 00:00:11, Serial0/1/1
R       172.16.200.32/27 [120/1] via 10.10.20.10, 00:00:11, Serial0/1/1
S    192.168.0.0/16 is directly connected, Null0
R2#
```

TP : Routage RIP



Configurez les interfaces comme indiqué dans le schéma

Configurer le protocole de routage RIP

Activer la version 2 du protocoles RIP

Ajouter la route statique comme indiqué dans la schéma

Redistribuer la route statique par le protocole RIP

Configurer la route par default qui sort vers le routeur de du FAI

Redistribuer la route par default avec le protocole RIP

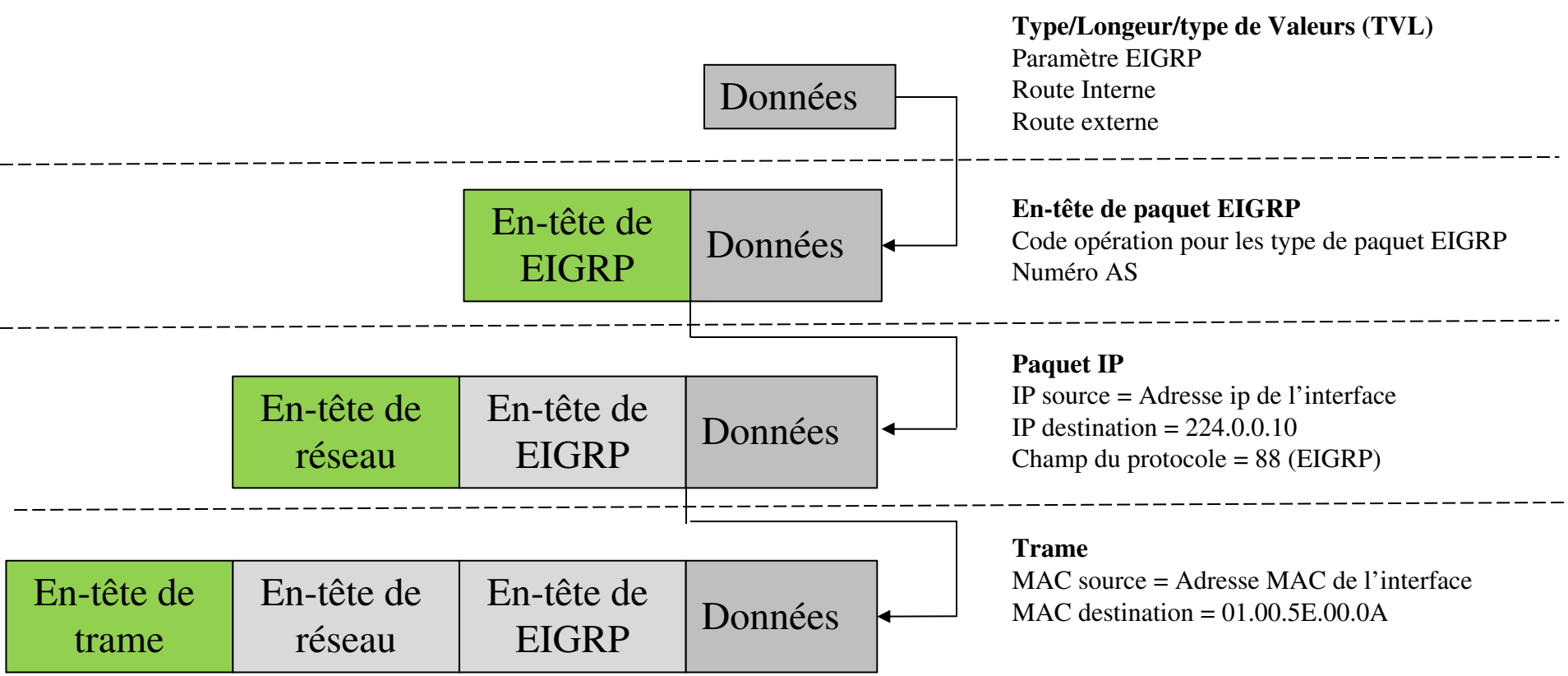
Vérifier l'ensemble de vos configuration et valider le bon fonctionnement de votre architecture



EIGRP



Encapsulation de message EIGRP



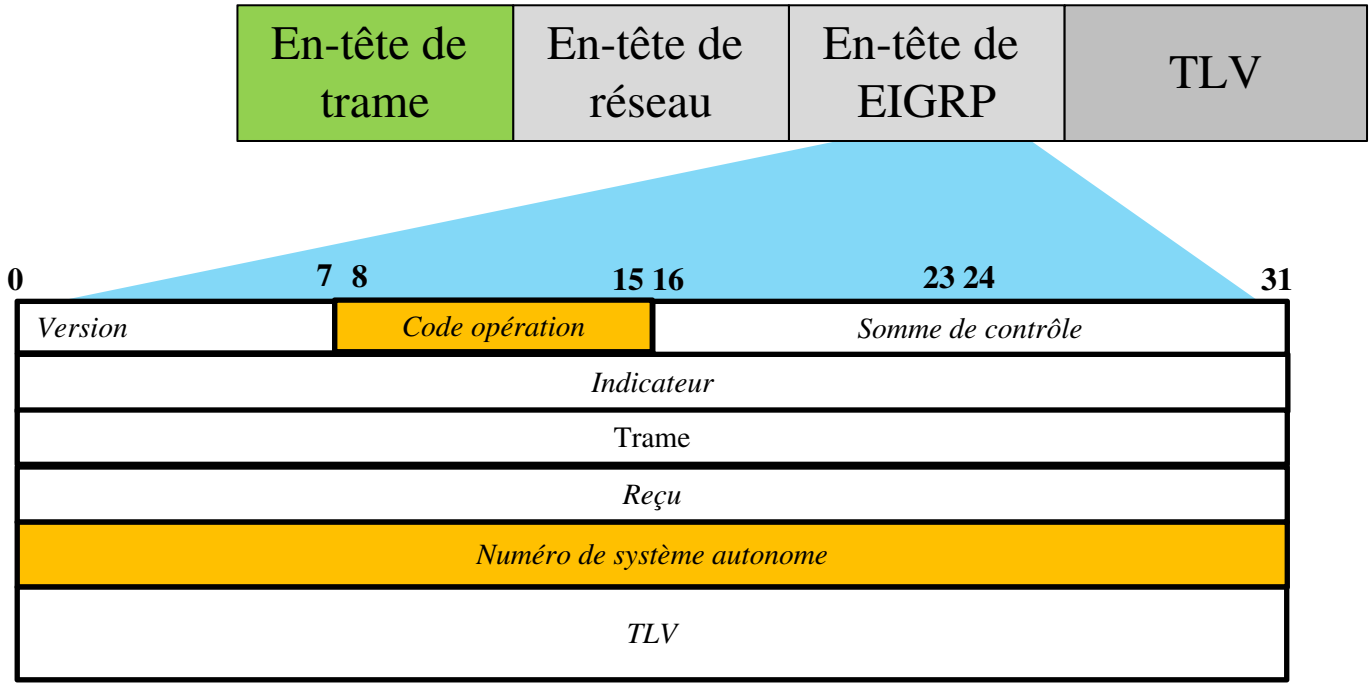
Type/Longeur/type de Valeurs (TVL)
Paramètre EIGRP
Route Interne
Route externe

En-tête de paquet EIGRP
Code opération pour les type de paquet EIGRP
Numéro AS

Paquet IP
IP source = Adresse ip de l'interface
IP destination = 224.0.0.10
Champ du protocole = 88 (EIGRP)

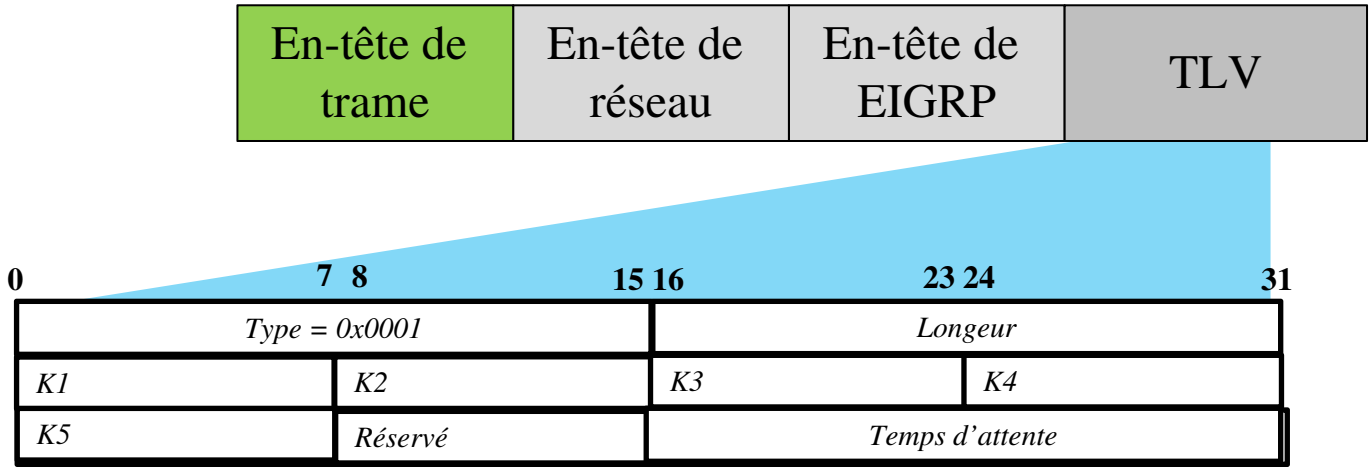
Trame
MAC source = Adresse MAC de l'interface
MAC destination = 01.00.5E.00.0A

Détail du de l'en-tête EIGRP



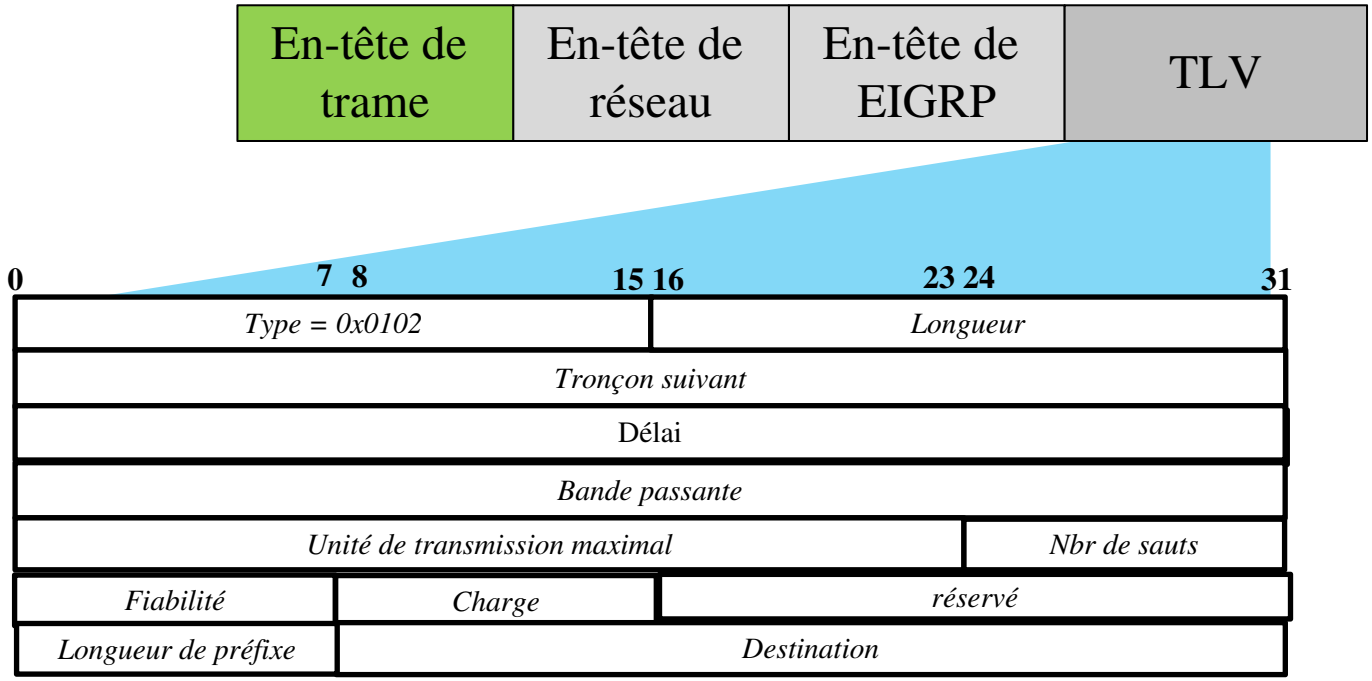
Code opération	Type de paquet EIGRP : Mise à jour (1), Demande (3), Réponse (4), Hello (5)
Numéro de système autonome	ID pour ce processus de routage EIGRP

Détail du du champ TLV paramêtre



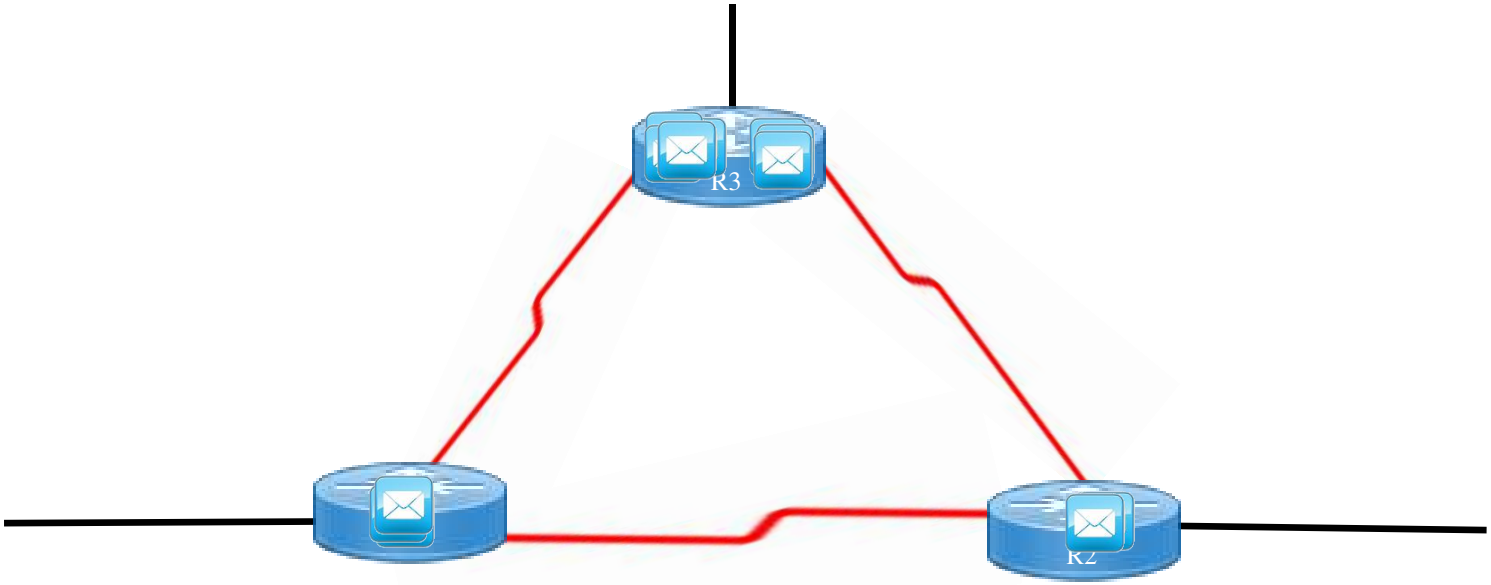
K1 et K3	Valeurs de bande passante et de délai ; fixées à 1 par default
Temps d'attente	Temps d'attente maximum pendant lequel le routeur doit attendre le Hello suivant

Détail du du message TLV IP interne



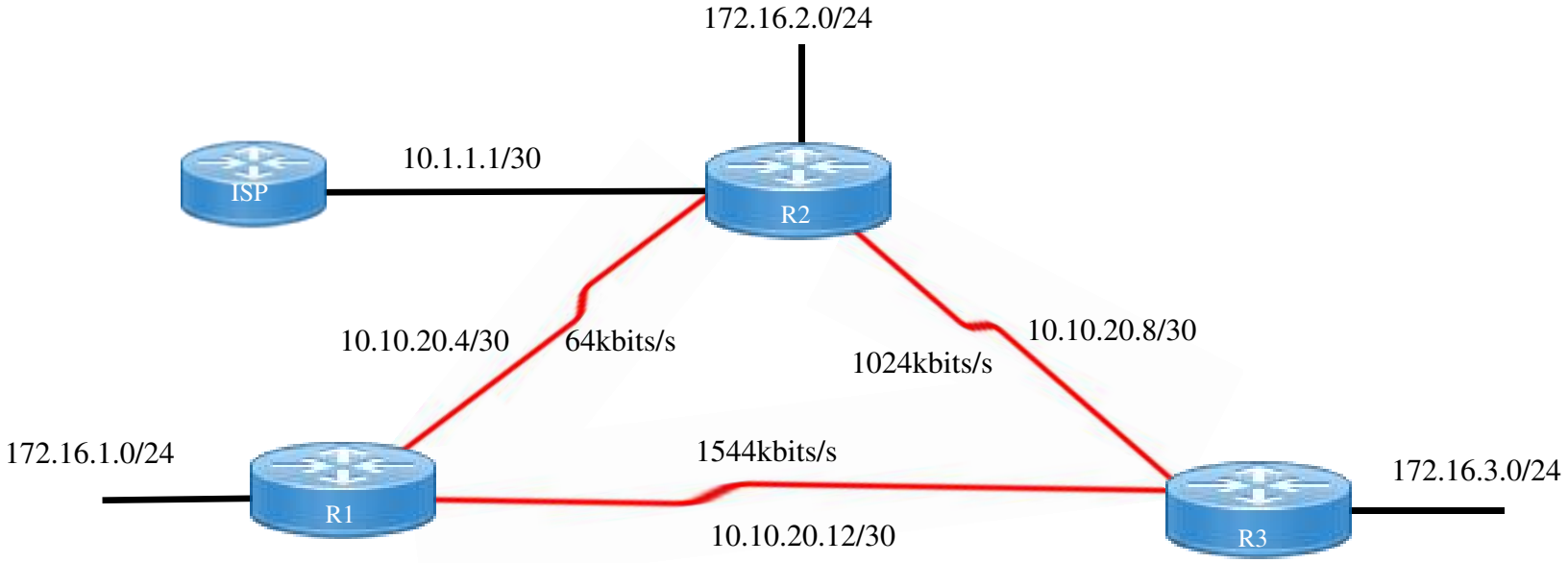
Délai	somme des délais en unités de 10 microsecondes entre la source et la destination ; 0xFFFFFFFF indique une route inaccessible
Bande passante	configuration de bande passante la plus basse de toute interface située le long de la route
Longueur de préfixe	spécifie le nombre de bits réseau du masque de sous-résea
Destination	adresse de destination de la route

Les échanges EIGRP

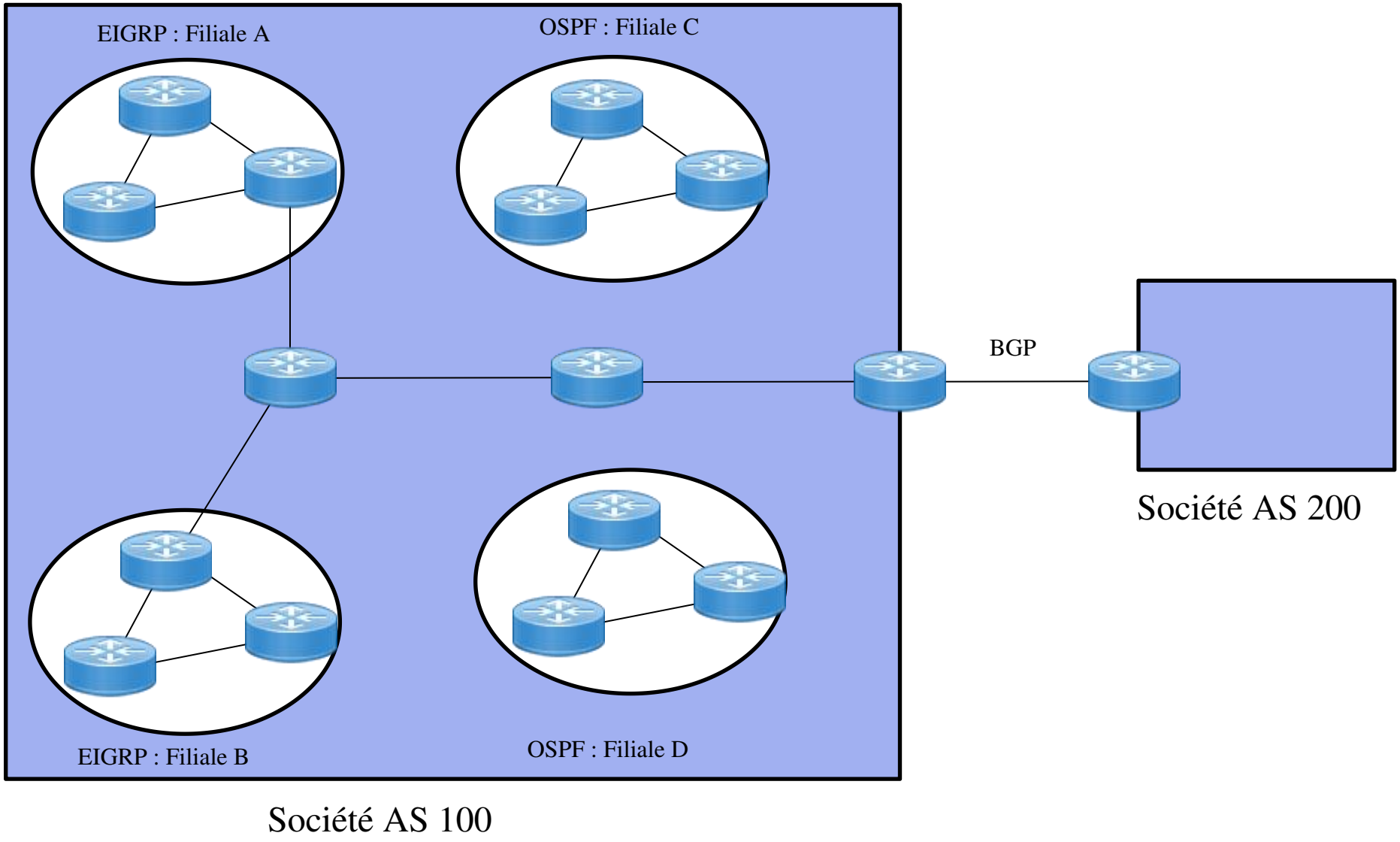


Bande passante	Exemple de liaison	Intervalle HELLO	Temps d'attente
1,544 Mbits/s	Frame relay Multipoint	60s	180s
Supérieure à 1,544Mbit/s	T1 Ethernet	5s	15s

Topologie EIGRP

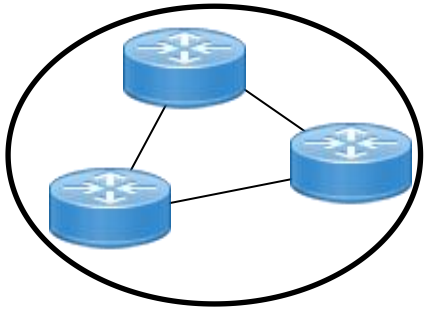


Système autonome



ID de processus

EIGRP 1



Commande : (config) *#router eigrp 1*

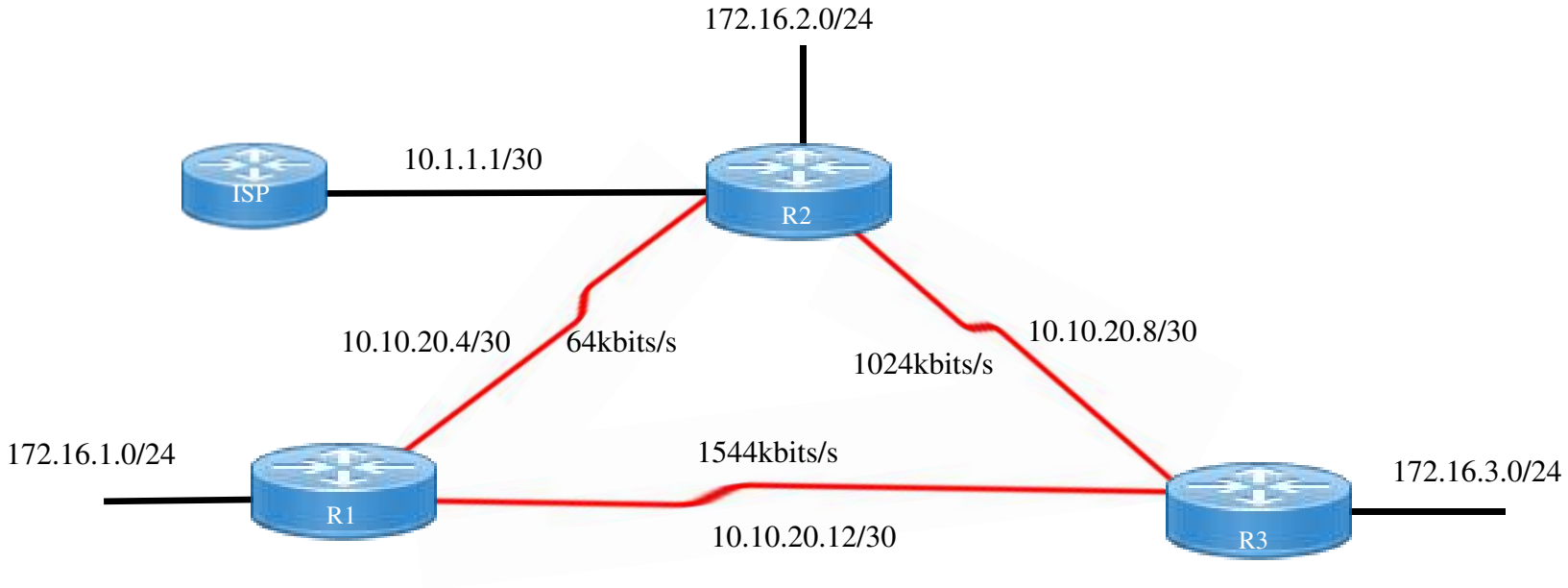
```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router eigrp 1
```

Activation du protocole EIGRP

Commande : (config-router) *#network 10.10.240.4 0.0.0.3*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router eigrp 1
R1(config-router)#network 10.10.240.4 0.0.0.3
R1(config-router)#network 10.10.240.20 0.0.0.3
R1(config-router)#network 172.16.1.0 0.0.0.255
```

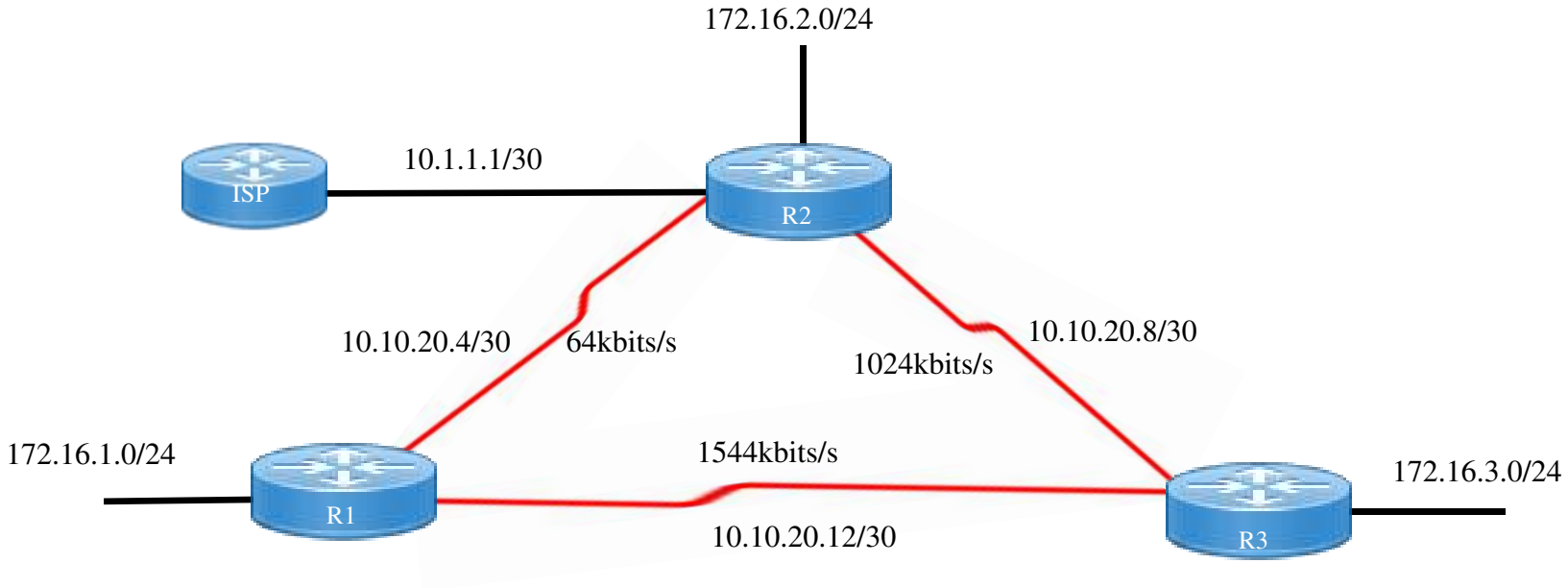
```
R3(config-router)#network 172.16.3.0 0.0.0.255
R3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.10.20.10 (Serial0/1/0) is up: new
adjacency
```



Visualisation des voisin EIGRP

Commande : (config-router) *#show ip eigrp neighbors*

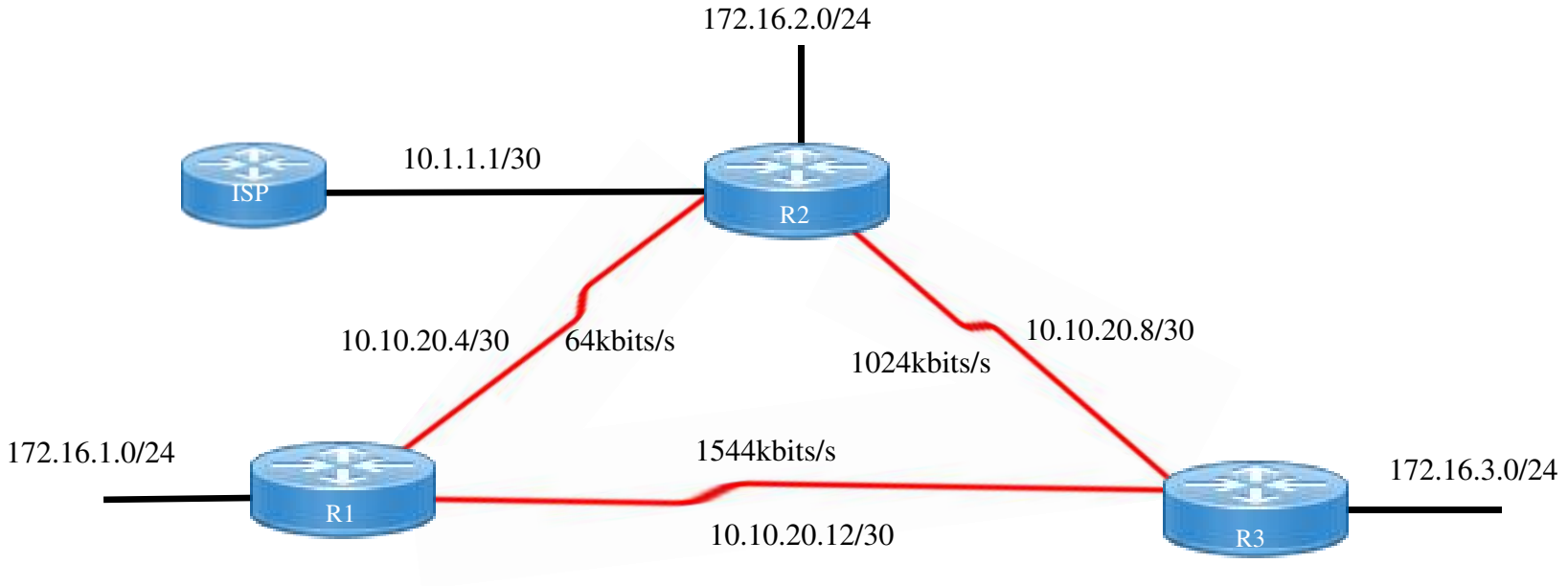
```
R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address         Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                    (sec)         (ms)       Cnt Num
0   10.10.20.13     Se0/1/1         10   00:03:25 40     1000 0   4
1   10.10.20.6      Se0/1/0         13   00:00:05 40     1000 0   14
```



Visualisation de la DA EIGRP

Commande : (config-router) *#Show ip protocols*

```
R1#sh ip protocols
Routing Protocol is "eigrp  1 "
....
Distance: internal 90 external 170
```





Visualisation de la table de routage

Commande : *#Show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
D       10.0.0.0/8 is a summary, 00:08:16, Null0
C       10.10.20.4/30 is directly connected, Serial0/1/0
D       10.10.20.8/30 [90/2681856] via 10.10.20.13, 00:08:16, Serial0/1/1
                      [90/2681856] via 10.10.20.6, 00:04:56, Serial0/1/0
C       10.10.20.12/30 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
D       172.16.0.0/16 is a summary, 00:08:16, Null0
C       172.16.1.0/24 is directly connected, FastEthernet0/0
```

Suppression du résumé des route

Commande : (config-router)*#no auto-summary*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router eigrp 1
R1(config-router)#no auto summary
```

Commande : *#Show ip route*

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter are
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.10.20.4 is directly connected, Serial0/1/0
D       10.10.20.8 [90/41024000] via 10.10.20.6, 00:02:12, Serial0/1/0
C       10.10.20.12 is directly connected, Serial0/1/1
     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
D       172.16.2.0 [90/2684416] via 10.10.20.13, 00:02:11, Serial0/1/1
D       172.16.3.0 [90/2172416] via 10.10.20.13, 00:02:11, Serial0/1/1
```



Calcul du métrique

Formule par default :
Métrique = [K1 x Bande passante + K3 x délai]*253

Formule par complété :
Métrique = [K1 x Bande passante (k2*bande passante) / (256 –charge)+ K3 x délai] x [k5/(fiabilité + k4)]

K1 = Bande Passant
K2 = Charge
K 3= Délai
K4 = Fiabilité
K5 = Fiabilité

Modification de la formule de calcul du métrique

Commande : (config-router) *#metric weights tos k1 k2 k3 k4 k5*

```
R1#configure terminal
R1(config)#router eigrp 1
R1(config-router)#metric weights 0 1 1 1 1 1
```



Affichage des indicateur de metrique

Commande : (config-router) *#Show ip protocols*

```
R1#show ip protocols

Routing Protocol is "eigrp  1 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 1
  Automatic network summarization is in effect
  Automatic address summarization:
    10.0.0.0/8 for FastEthernet0/0
      Summarizing with metric 2169856
    172.16.0.0/16 for Serial0/1/1, Serial0/1/0
      Summarizing with metric 28160
  Maximum path: 4
  Routing for Networks:
     172.16.1.0/24
     10.10.20.12/30
     10.10.20.4/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.20.13     90            2099347
    10.10.20.6      90            2099542
  Distance: internal 90 external 170
```

Affichage des indicateur de métrique d'une interface

Commande : (config-router) *#show interface S0/1/0*

```
R1#show interfaces s0/1/0
Serial0/1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.5/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Valeur du delai en microseconde

Commande : (config-router) *#show nterface S0/1/0*

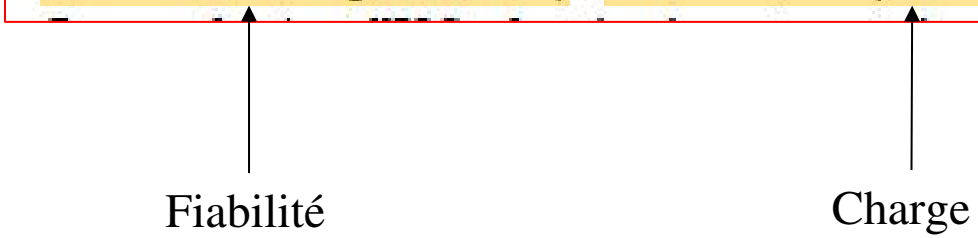
```
R1#show interfaces s0/1/0
Serial0/1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.5/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Origine de la route	Valeur de la DA
Fastethernet	100 us
Ethernet	1000us
T1 (série par default)	20000us
512k	20000us
56k	20000us

Valeur de fiabilité et de charge

Commande : (config-router) *#show nterface S0/1/0*

```
R1#show interfaces s0/1/0
Serial0/1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.5/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```



Fiabilité

Charge

Modification de la bande passante

Commande : (config-if) *#Bandwidth 64*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#inter s 0/1/0
R1(config-if)#bandwdth 64
```

Commande : *#show interface S0/1/0*

```
R1#show interfaces s0/1/0
Serial0/1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.5/30
  MTU 1500 bytes, BW 64 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

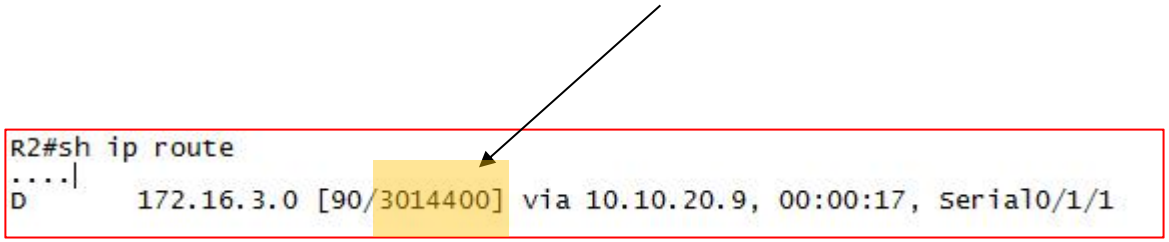
Calcul du métrique

Formule par default :
Métrique = [K1 x Bande passante + K3 x délai]*256

Bande passante (10 000 000/bande passante en kbits/s) * 256

Delai (somme des délai/10) * 256

= mesure EIGRP

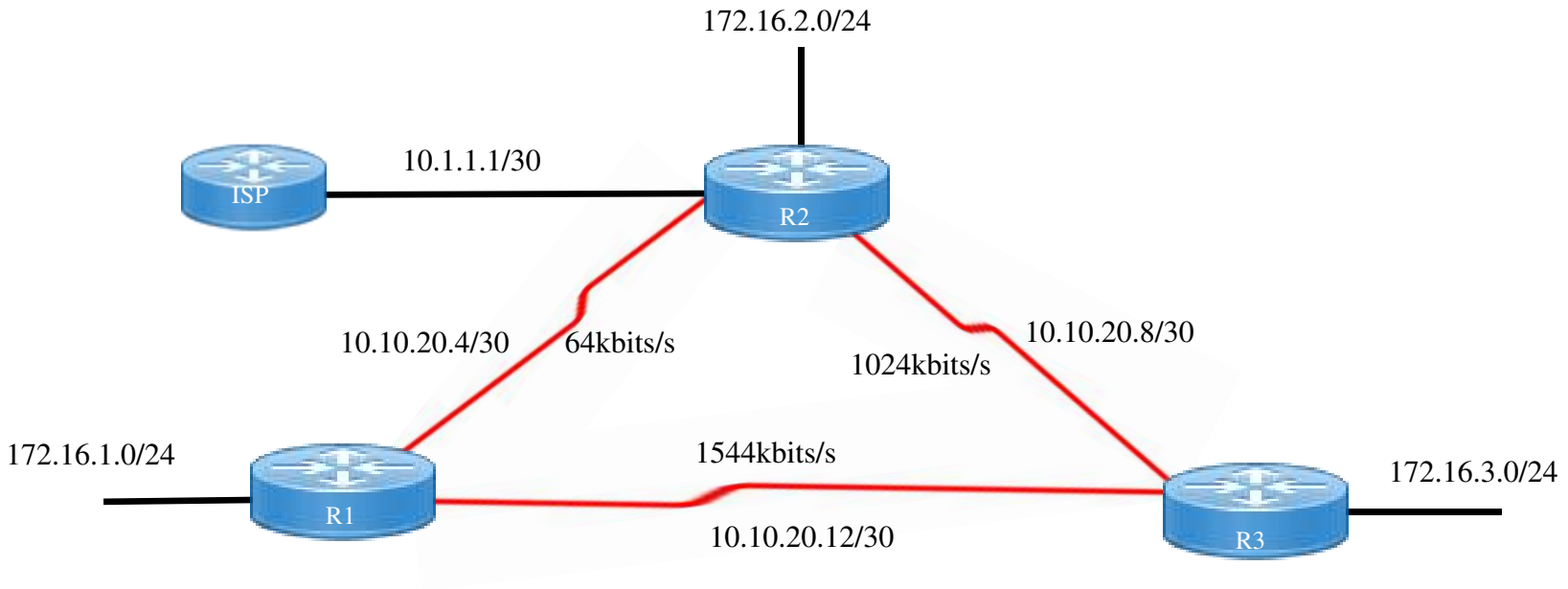


Calcul de la bande passante

```
R3#show interfaces fastEthernet 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.f701.c501 (bia 00e0.f701.c501)
  Internet address is 172.16.3.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

```
R2#show interfaces s0/1/1
Serial0/1/1 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.10/30
  MTU 1500 bytes, BW 1024 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Bande passante = (10000000/1024) = 9765 x 256 = **2499840**

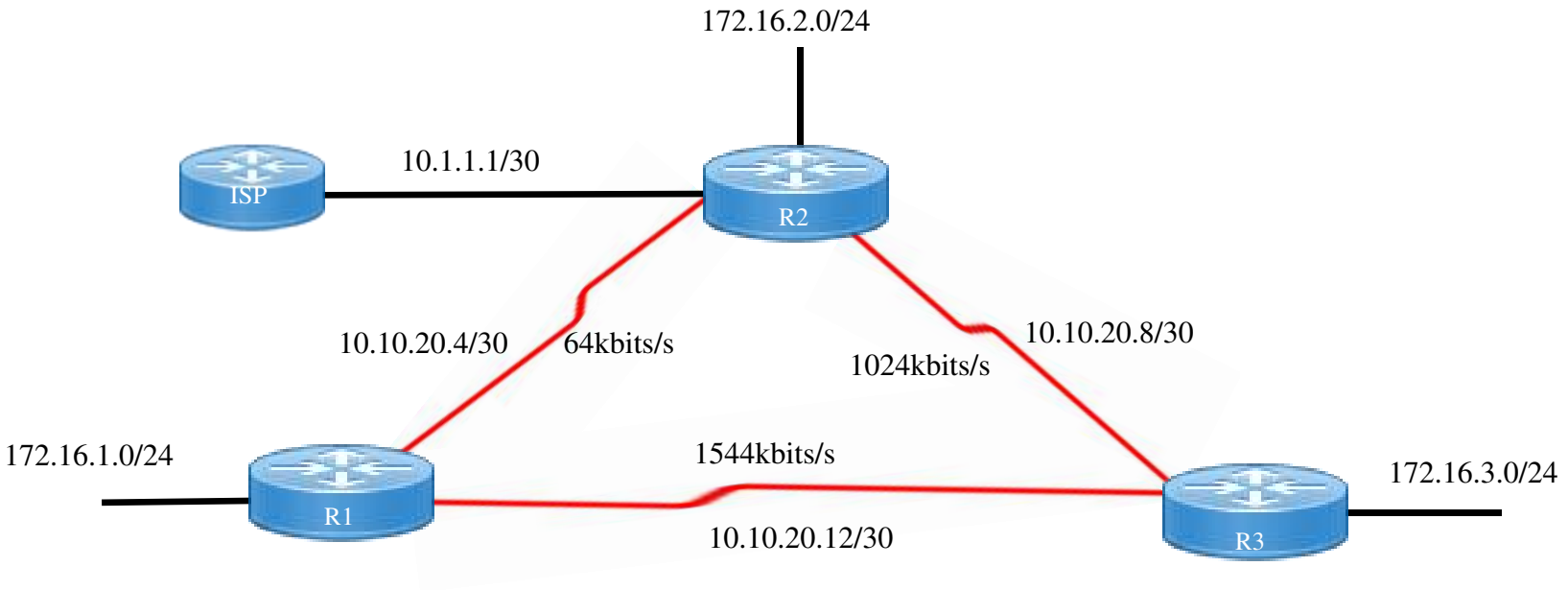


Calcul du délai

```
R3#show interfaces fastEthernet 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.f701.c501 (bia 00e0.f701.c501)
  Internet address is 172.16.3.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

```
R2#show interfaces s0/1/1
Serial0/1/1 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.10/30
  MTU 1500 bytes, BW 1024 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Délai = [(20000/10) + (100/10)] x 256 = **514560**



Métrique EIGRP

Métrique = 2499840 + 514560 = **3014400**

```
R2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

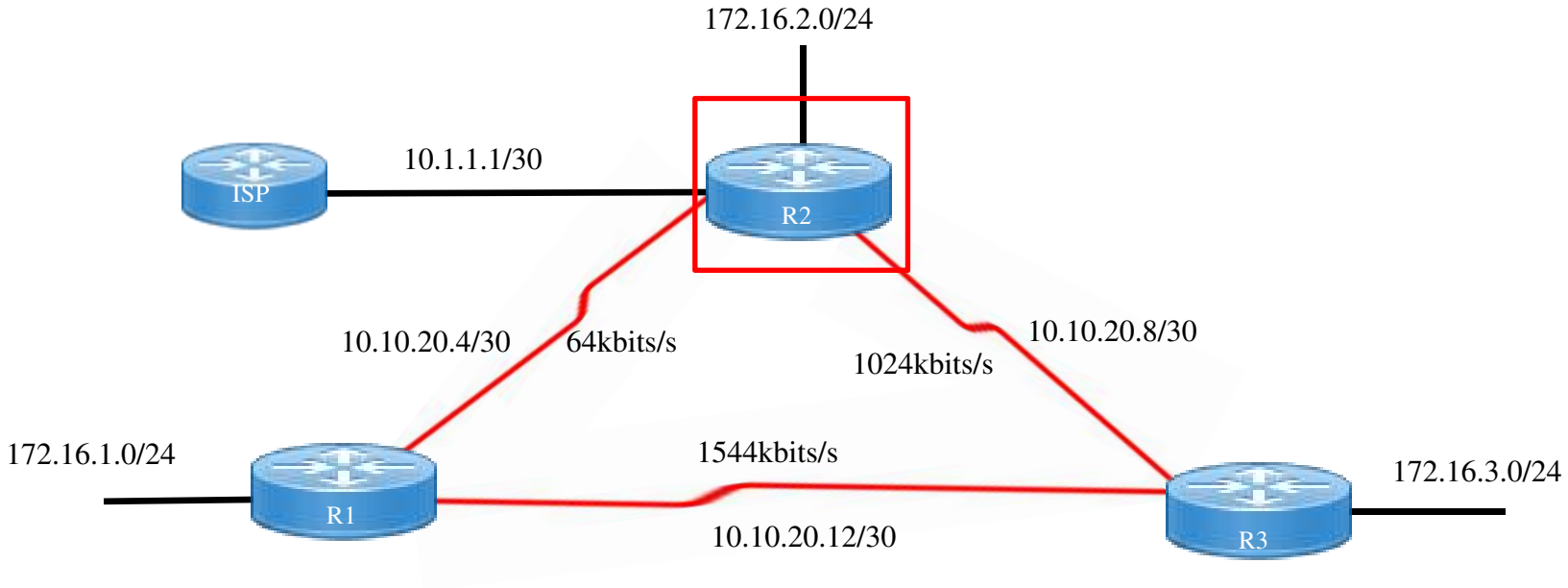
     10.0.0.0/30 is subnetted, 4 subnets
C       10.1.1.0 is directly connected, Loopback0
C       10.10.20.4 is directly connected, Serial0/1/0
C       10.10.20.8 is directly connected, Serial0/1/1
D       10.10.20.12 [90/3523840] via 10.10.20.9, 00:14:30, Serial0/1/1
     172.16.0.0/24 is subnetted, 3 subnets
D       172.16.1.0 [90/3526400] via 10.10.20.9, 00:14:18, Serial0/1/1
C       172.16.2.0 is directly connected, FastEthernet0/0
D       172.16.3.0 [90/3014400] via 10.10.20.9, 00:14:30, Serial0/1/1
```

Métrique EIGRP

```
D        172.16.1.0 [90/3526400] via 10.10.20.9, 00:14:18, Serial0/1/1
C        172.16.2.0 is directly connected, FastEthernet0/0
D        172.16.3.0 [90/3014400] via 10.10.20.9, 00:14:30, Serial0/1/1
```

Distance
De faisabilité

Successeur





Successeur potentiel

Commande : *#show ip eigrp topology*

```
R2#show ip eigrp topology
IP-EIGRP Topology Table for AS 1/ID(10.1.1.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.10.20.4/30, 1 successors, FD is 40512000
         via Connected, Serial0/1/0
P 10.10.20.8/30, 1 successors, FD is 3011840
         via Connected, Serial0/1/1
P 10.10.20.12/30, 1 successors, FD is 3523840
         via 10.10.20.9 (3523840/3011840), Serial0/1/1
         via 10.10.20.5 (41024000/2169856), Serial0/1/0
P 172.16.1.0/24, 1 successors, FD is 3526400
         via 10.10.20.9 (3526400/3014400), Serial0/1/1
         via 10.10.20.5 (40514560/28160), Serial0/1/0
P 172.16.2.0/24, 1 successors, FD is 28160
         via Connected, FastEthernet0/0
P 172.16.3.0/24, 1 successors, FD is 3014400
         via 10.10.20.9 (3014400/28160), Serial0/1/1
         via 10.10.20.5 (41026560/2172416), Serial0/1/0
R2#
```



Successeur potentiel

Commande : *#show ip eigrp topology 172.16.2.0 255.255.255.0*

```
R2#show ip eigrp topology 172.16.3.0 255.255.255.0
IP-EIGRP (AS 1): Topology entry for 172.16.3.0/24
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is 3014400
  Routing Descriptor Blocks:
  10.10.20.9 (Serial0/1/1), from 10.10.20.9, Send flag is 0x0
      Composite metric is (3014400/28160), Route is Internal
      Vector metric:
        Minimum bandwidth is 1024 Kbit
        Total delay is 20100 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
  10.10.20.5 (Serial0/1/0), from 10.10.20.5, Send flag is 0x0
      Composite metric is (41026560/2172416), Route is Internal
      Vector metric:
        Minimum bandwidth is 64 Kbit
        Total delay is 40100 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 2
```



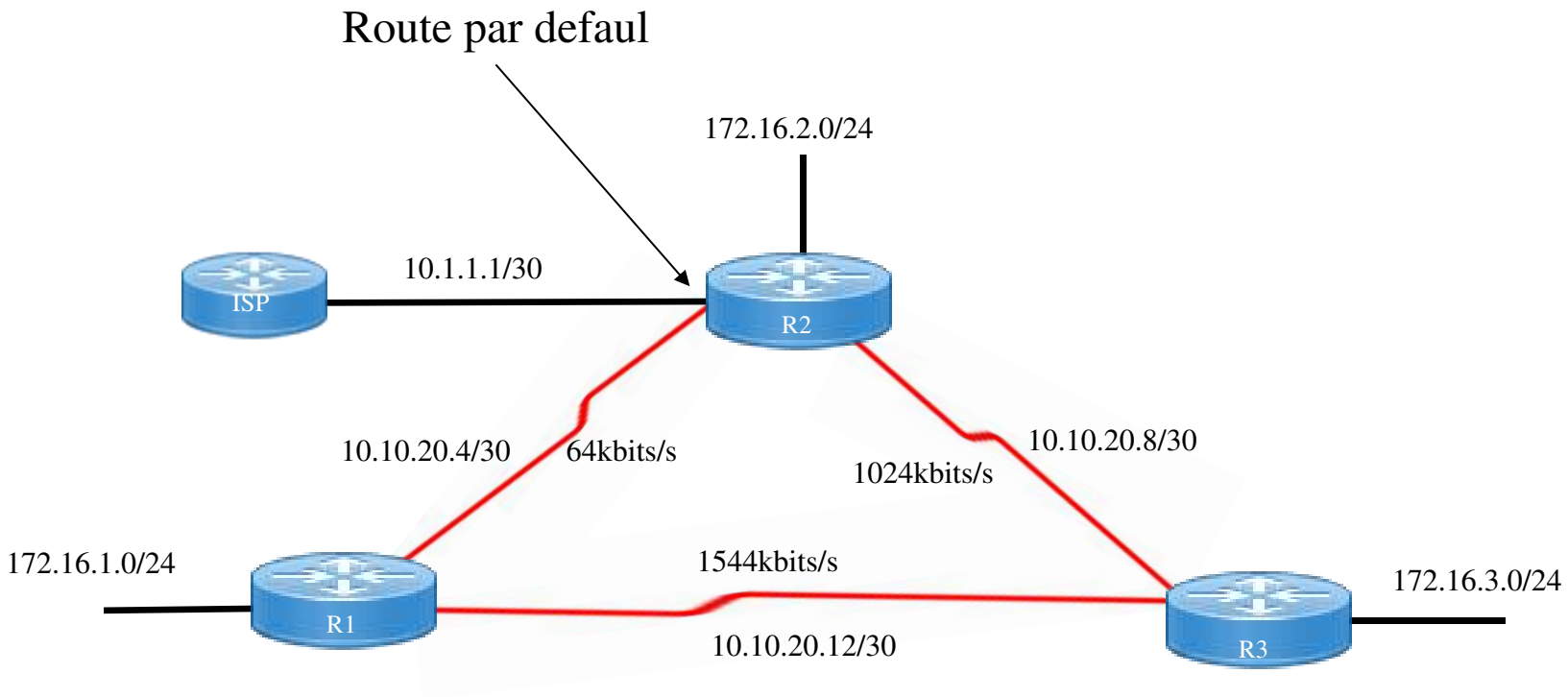
Successeur potentiel

Commande : *#show ip eigrp topology 172.16.2.0 255.255.255.0*

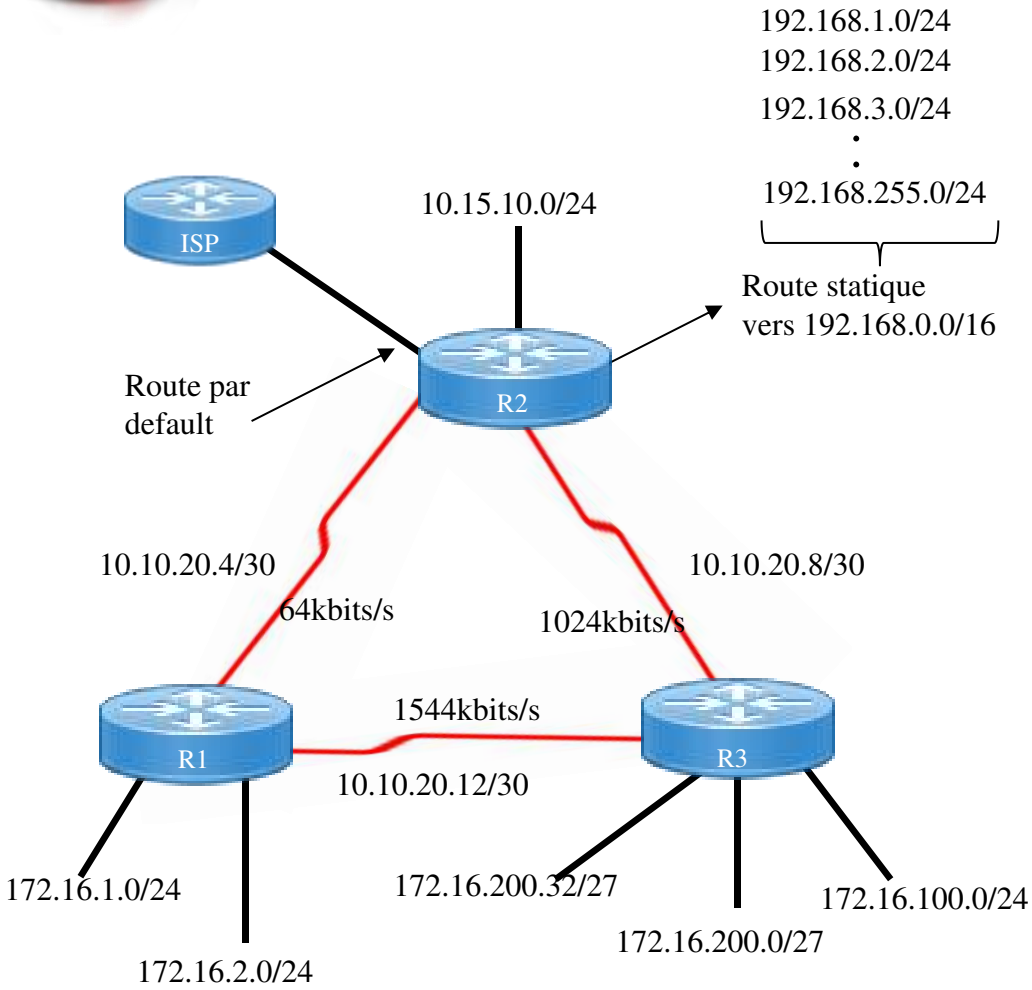
```
R2#show ip eigrp topology 172.16.3.0 255.255.255.0
IP-EIGRP (AS 1): Topology entry for 172.16.3.0/24
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is 3014400
  Routing Descriptor Blocks:
  10.10.20.9 (Serial0/1/1), from 10.10.20.9, Send flag is 0x0
      Composite metric is (3014400/28160), Route is Internal
      Vector metric:
        Minimum bandwidth is 1024 Kbit
        Total delay is 20100 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
  10.10.20.5 (Serial0/1/0), from 10.10.20.5, Send flag is 0x0
      Composite metric is (41026560/2172416), Route is Internal
      Vector metric:
        Minimum bandwidth is 64 Kbit
        Total delay is 40100 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 2
```

Redistribution de la route par default

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 10
R2(config)#router eigrp 1
R2(config-router)#redistribute static
```



TP : Routage EIGRP



Configurez les interfaces comme indiqué dans le schéma

Configurer le protocole de routage EIGRP

Ajouter la route statique comme indiqué dans la schéma

Configurer la route par default qui sort vers le routeur de du FAI

Redistribuer la route par default avec le protocole EIGRP

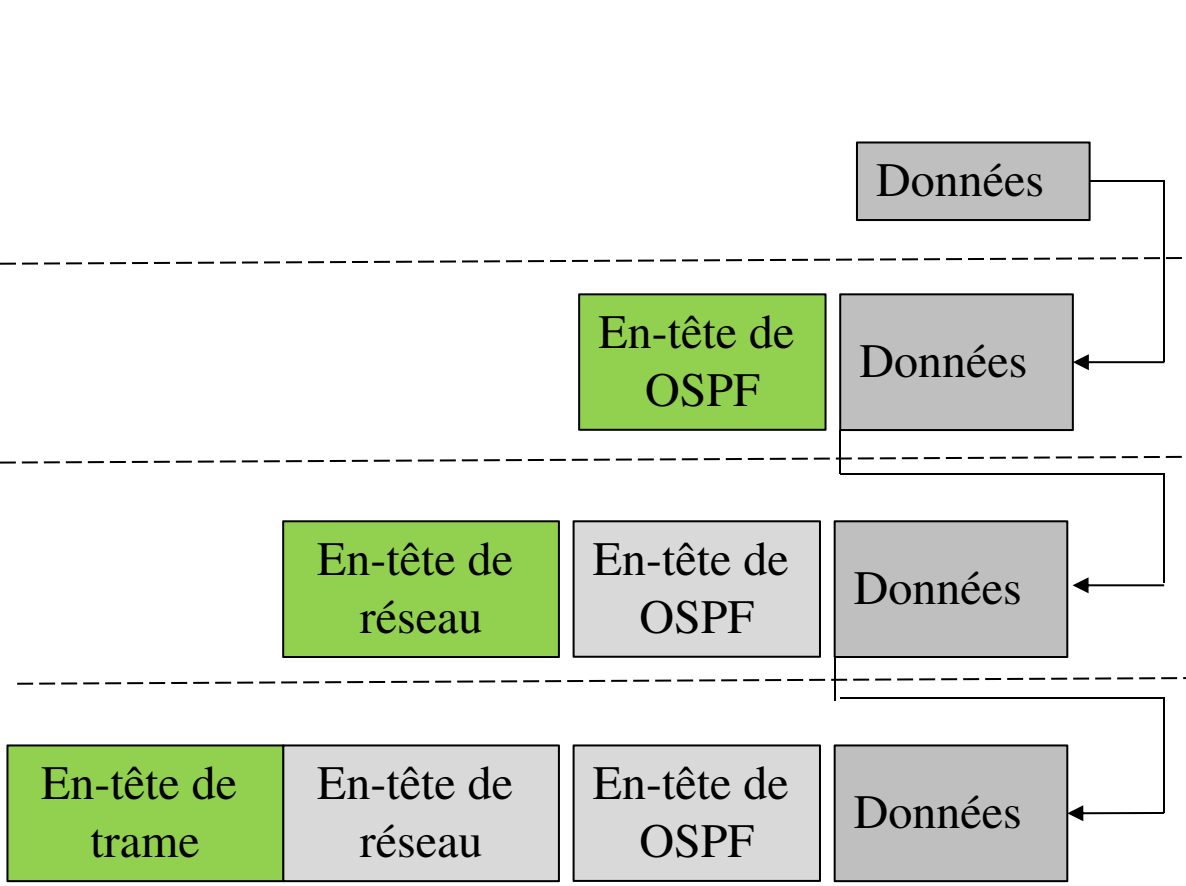
Vérifier l'ensemble de vos configuration et valider le bon fonctionnement de votre architecture



OSPF



Encapsulation de message OSPF



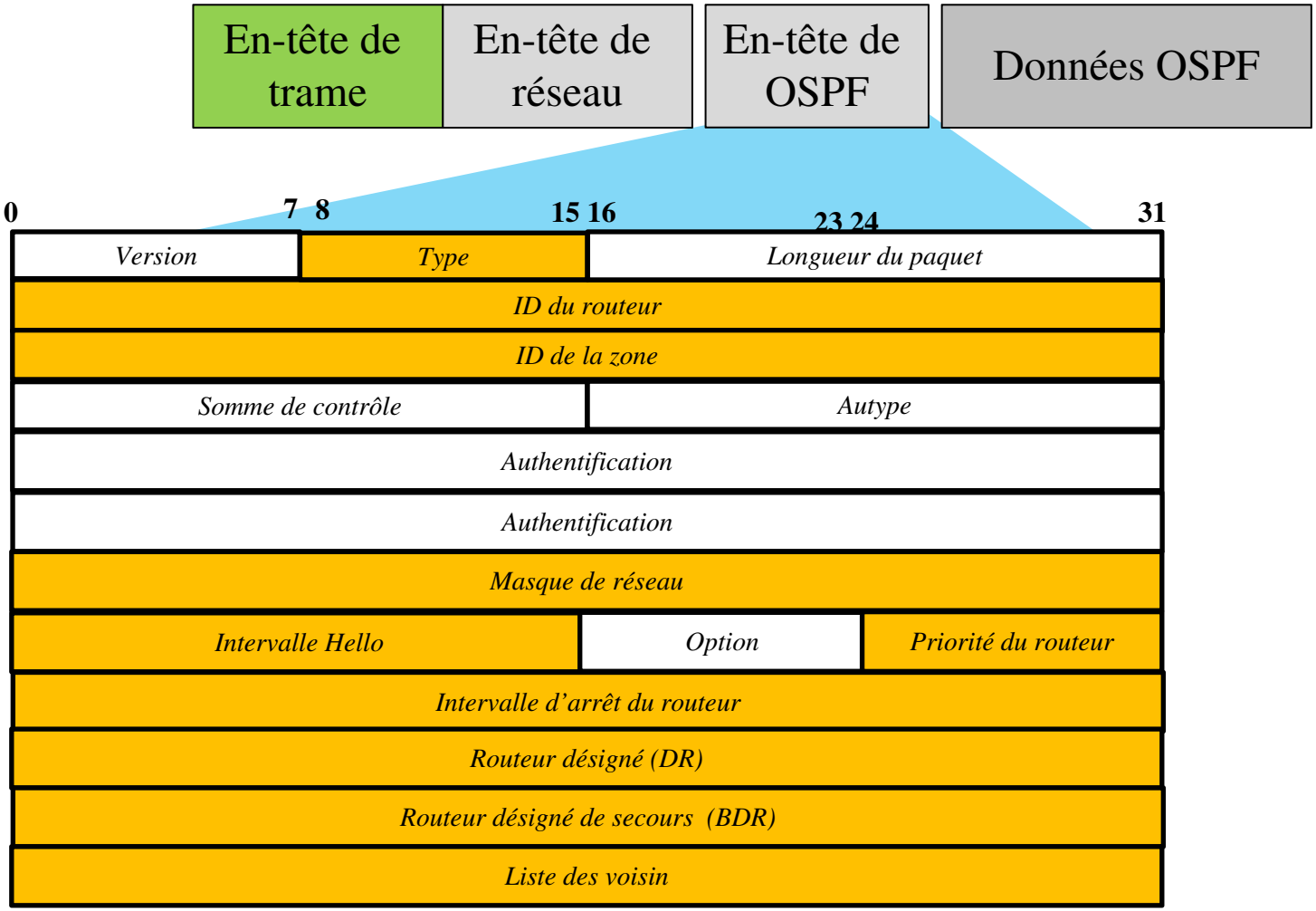
Type de paquet OSPF
0x01 Hello
0x02 Description de base de données (DD)
0x03 Requête d'état de lien s
0x04 Mise à jour d'état de liens
0x05 Accusé de réception d'état de liens

En-tête de paquet OSPF
Code du type de paquet OSPF
ID du routeur et ID de la zone

Paquet IP
IP source = Adresse ip de l'interface
IP destination Multidiffusion = 224.0.0.5
224.0.0.6
Champ du protocole = 89 (OSPF)

Trame
MAC source = Adresse MAC de l'interface
MAC destination Multidiffusion =
01:00:5E:00:00:05
01:00:5E:00:00:06

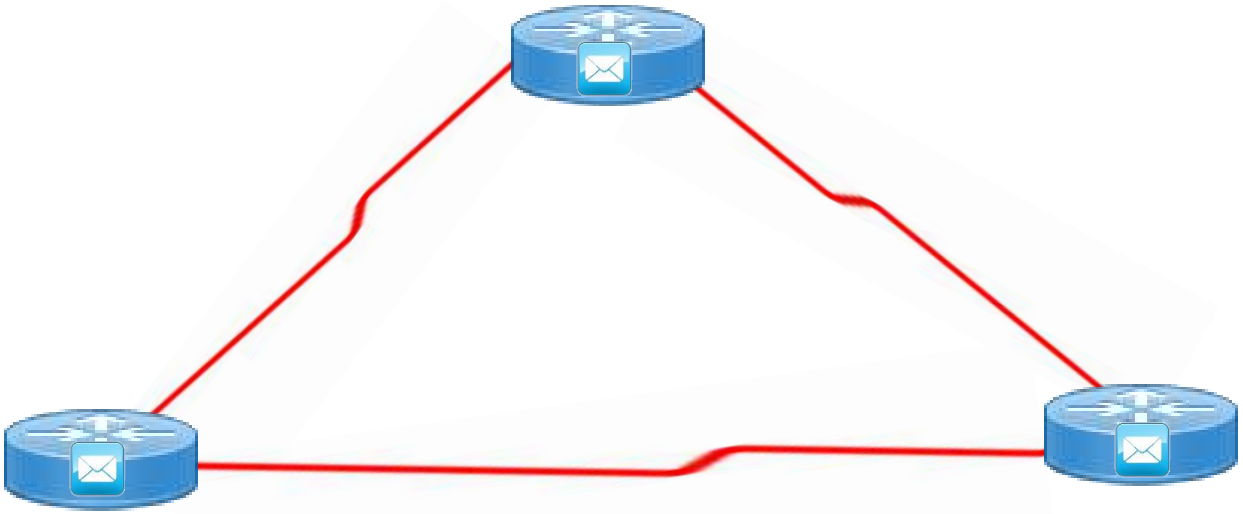
Détail du de l'en-tête OSPF



Hello Protocole

Intervalle Hello = 10 secondes

Intervalle Dead = 40 secondes

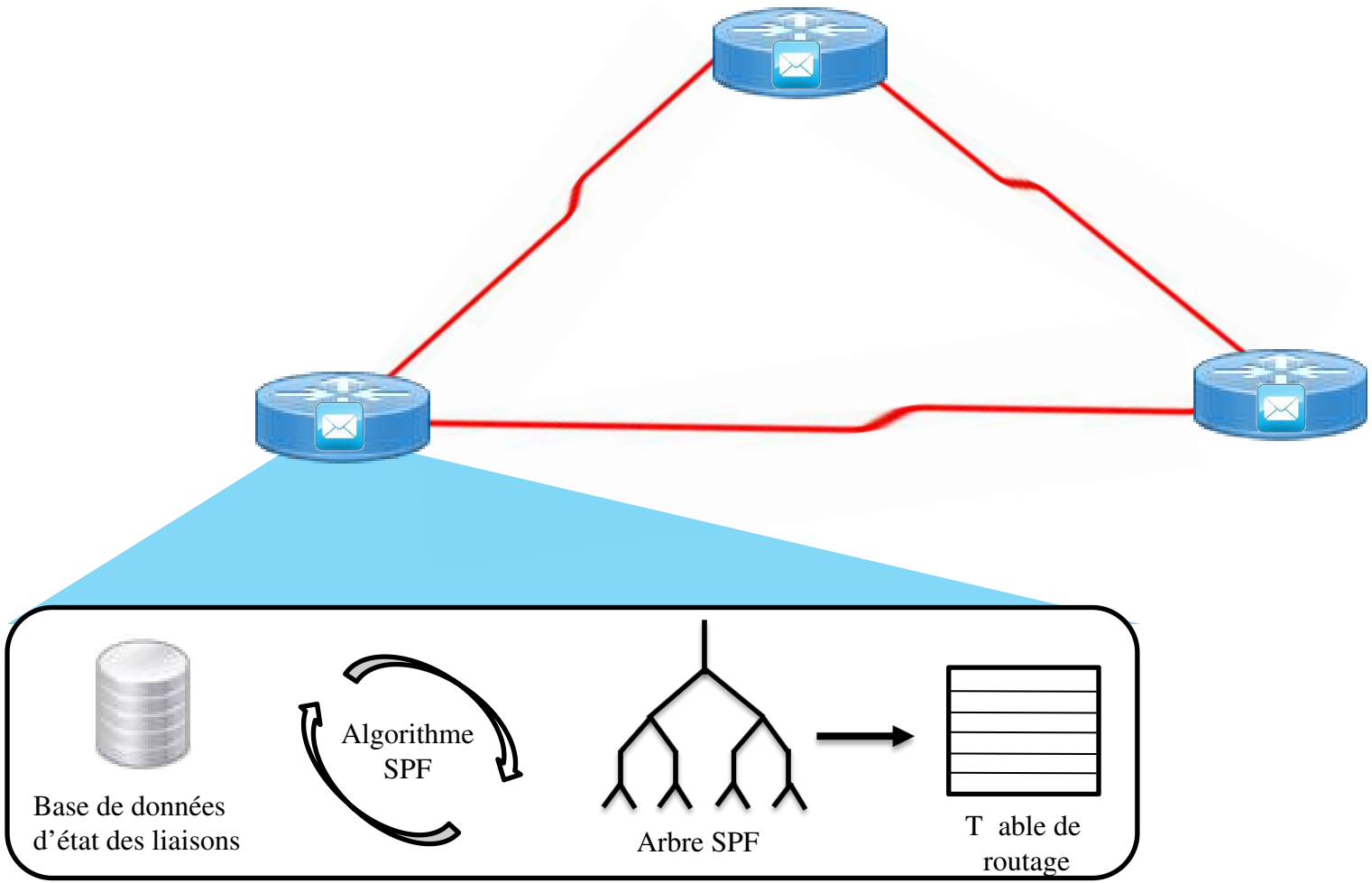


- Intervalle Hello
- Intervalle Dead
- Type de réseau

=

- Intervalle Hello
- Intervalle Dead
- Type de réseau

Construction de la table de routage



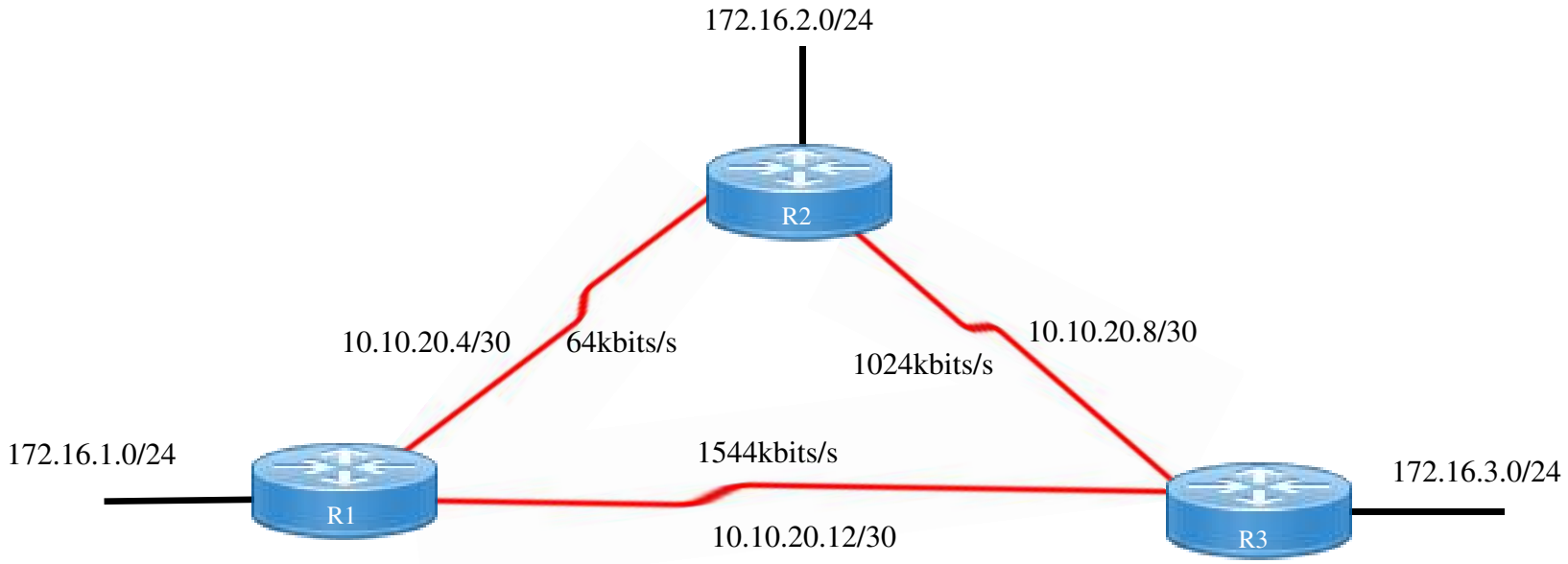


Les types de paquets OSPF

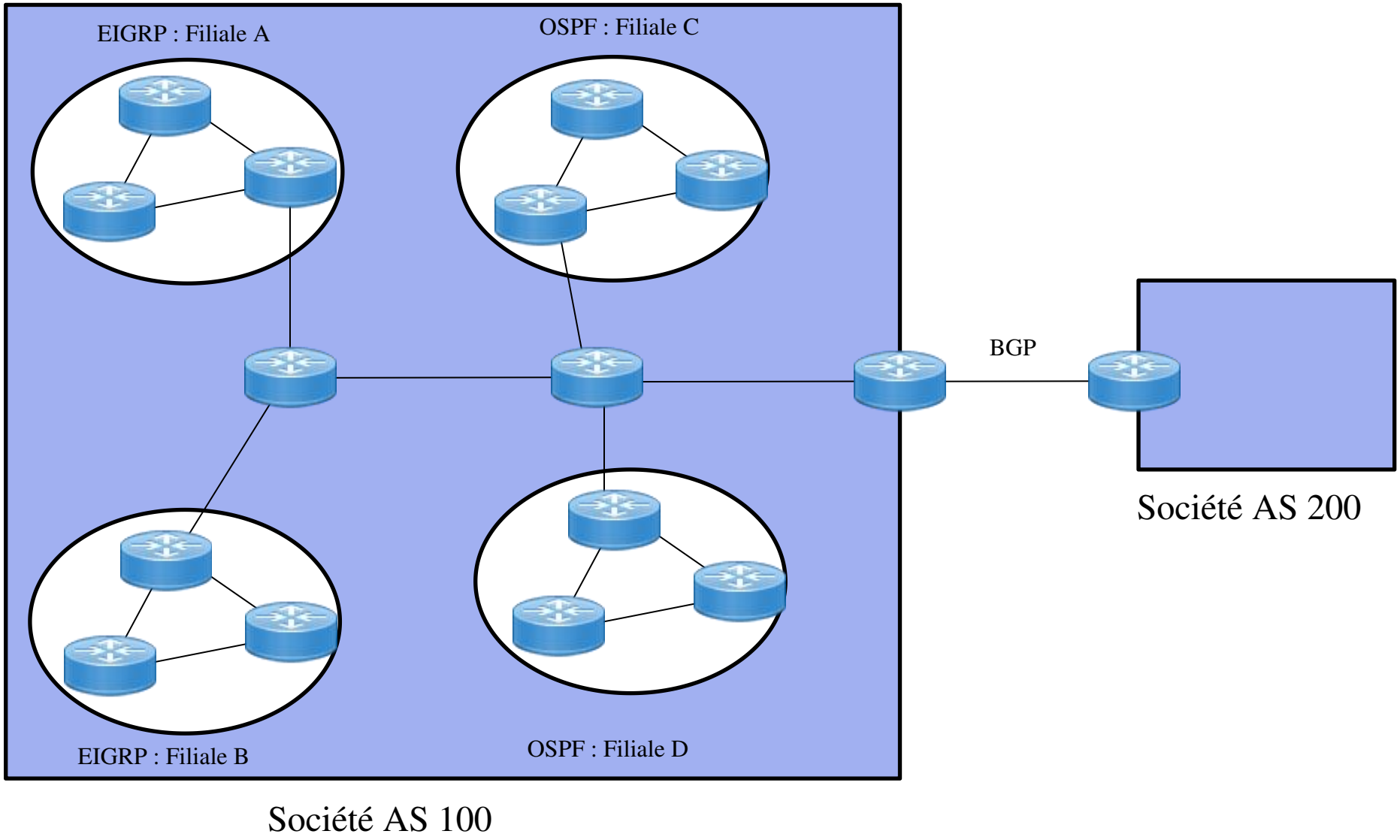
Type	Nom de paquet	Description
1	Hello	Découvre les voisin
2	DBD	Vérifie la synchronisation de la base de données avec le routeur
3	LSR	Demande des enregistrements d'état de liaisons spécifique d'un routeur à un autre
4	LSU	Envoi les enregistrements d'état de liaisons spécifiquement demandés
5	LSAck	Autres types de paquet

Type de LSA	Nom de paquet
1	LSA du routeur
2	LSA du réseau
3 ou 4	LSA résumées
5	LSA externes du système autonome
6	LSA OSPF de multidifusion
7	Défini pour les zonez not-so-stubby
8	LSA d'attrubuts externe pour BGP
9,10,11	LSA opaque

Topologie OSPF

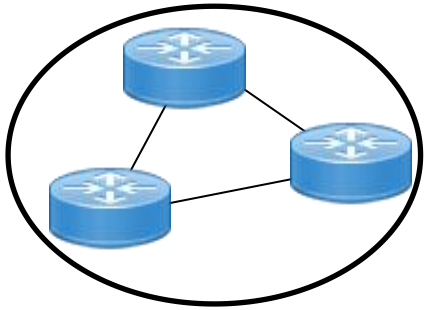


Zone OSPF


EIGRP : Filiale A
OSPF : Filiale C
BGP
Société AS 200
EIGRP : Filiale B
OSPF : Filiale D
Société AS 100

ID de processus

OSPF 1



Commande : (config) *#router ospf 1*

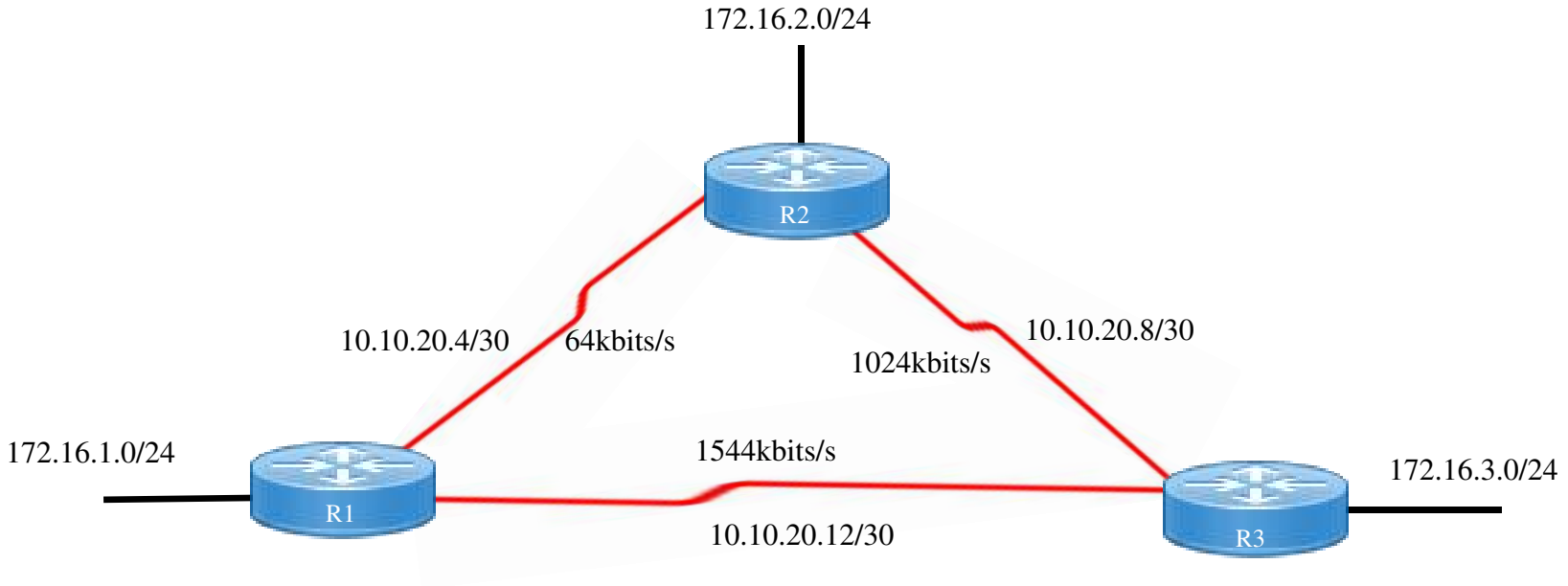
```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#
```

Déclaration des réseau ospf

Commande : (config-router) *#network 10.10.240.4 0.0.0.3 area 0*

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 10.10.24.4 0.0.0.3 area 0
R1(config-router)#network 10.10.24.12 0.0.0.3 area 0
R1(config-router)#network 172.16.1.0 0.0.0.255 area 0
```

```
R2(config-router)#network 10.10.20.8 0.0.0.3 area 0
00:12:44: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.3.1 on Serial0/1/1 from LOADING to
FULL, Loading Done
```

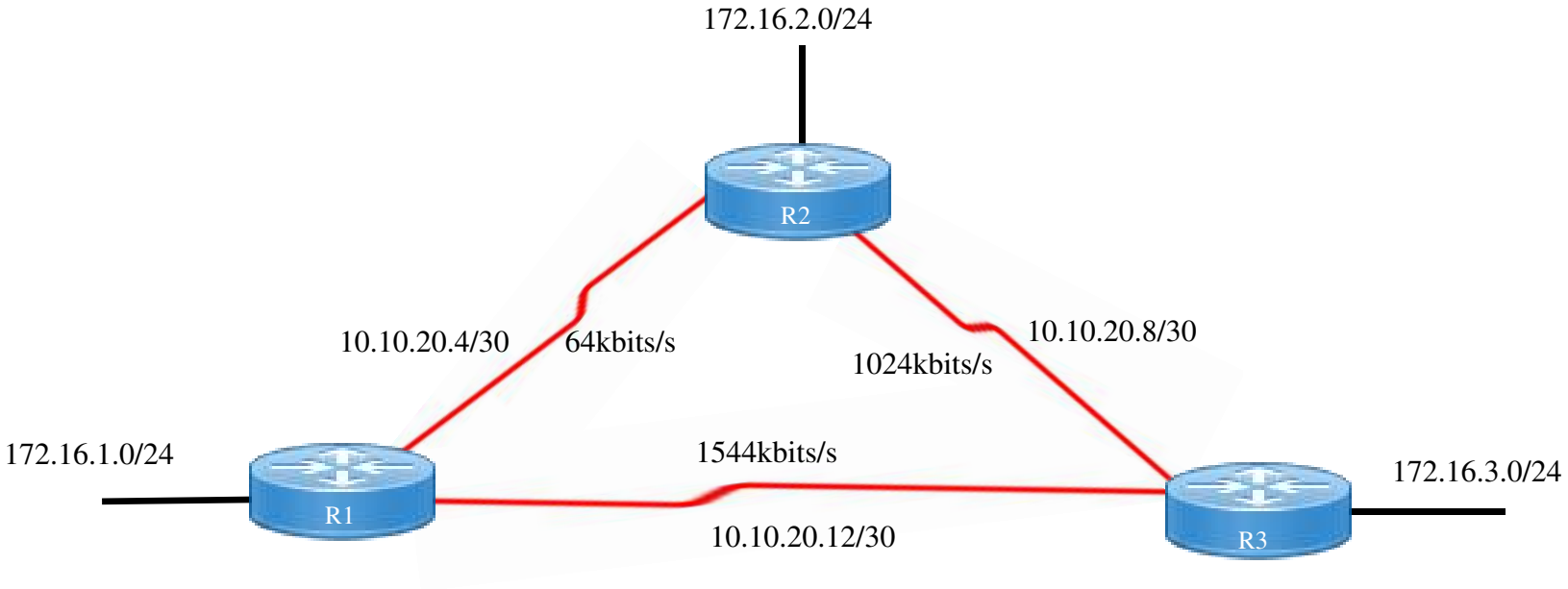


Router ID

Commande : *#show ip protocols*

```
R1#show ip protocols

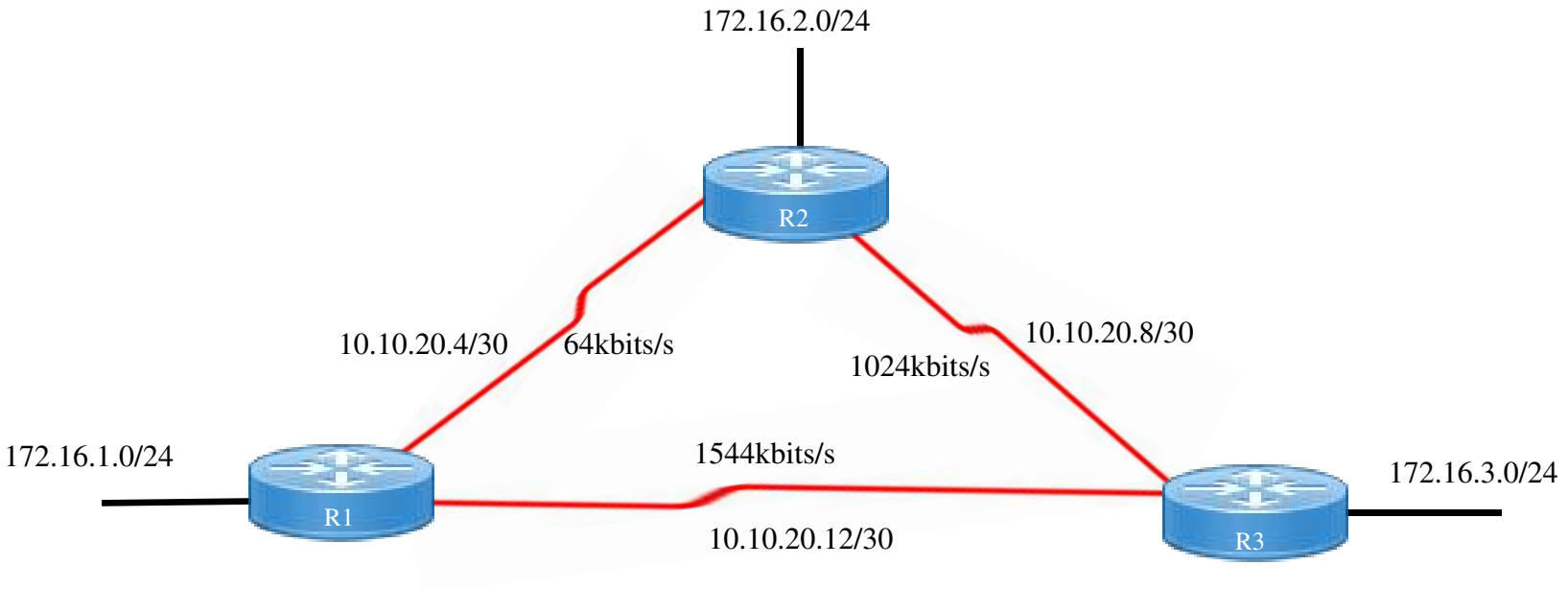
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 172.16.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
```



Définition du router ID

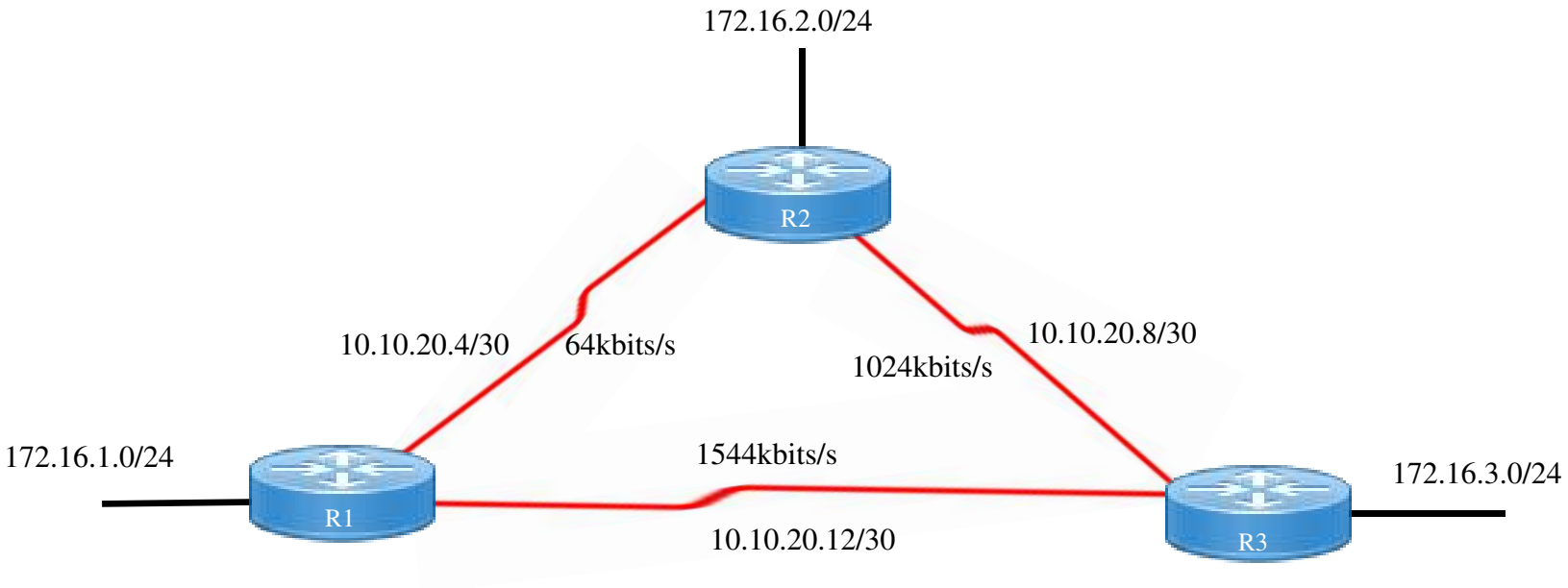
```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface loopback 0
R2(config-if)#ip address 10.1.1.1 255.255.255.255
R2(config-if)#exit
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```



Définition du router ID

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 10.2.2.2
```

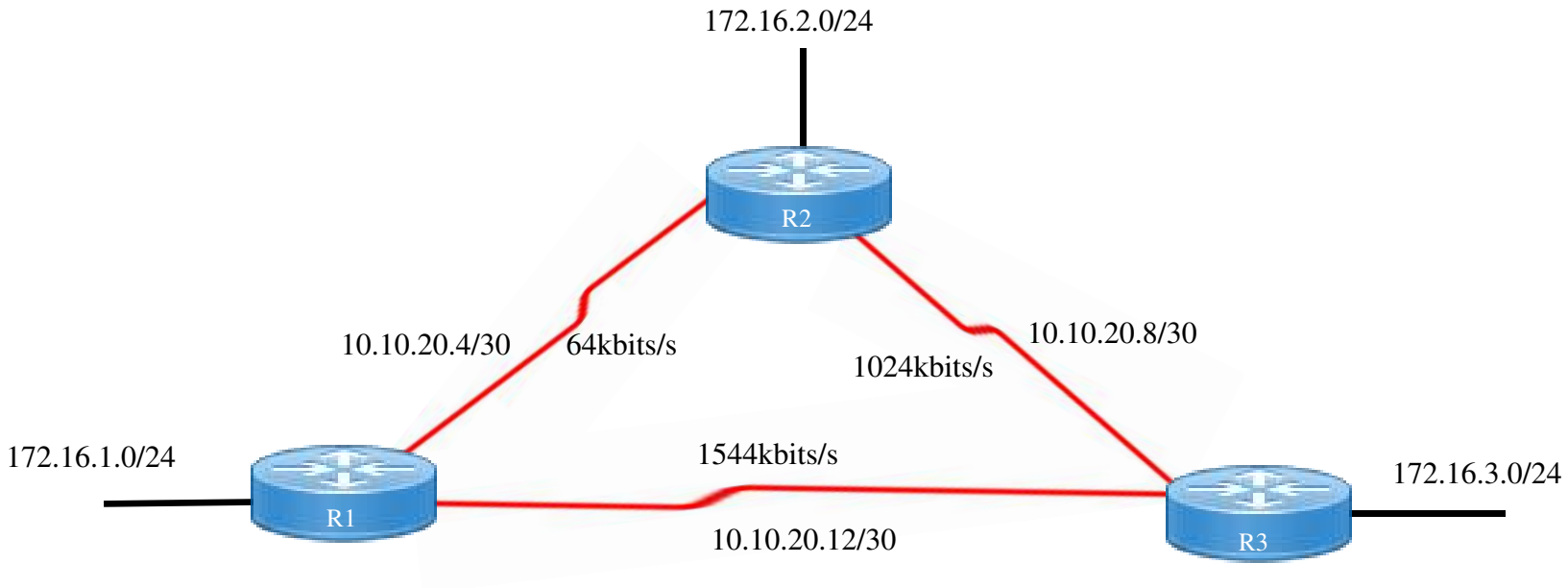


Visualisation des voisins ospf

Commande : #show ip ospf neighbor

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:30    10.10.20.6      Serial0/1/0
10.3.3.3          0   FULL/  -        00:00:37    10.10.20.13     Serial0/1/1
```





Visualisation de la configuration protocole

Commande : #show ip protocols

```
R1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.1.0 0.0.0.255 area 0
    10.10.20.12 0.0.0.3 area 0
    10.10.20.4 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.1.1.1              110      00:02:51
    10.2.2.2              110      00:02:51
    10.3.3.3              110      00:05:17
    172.16.1.1            110      00:21:12
    172.16.3.1            110      00:06:14
  Distance: (default is 110)
```



Visualisation de la configuration ospf

Commande : #show ip ospf

```
R2#show ip ospf
 Routing Process "ospf 1" with ID 10.2.2.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm executed 12 times
        Area ranges are
```



Visualisation de la configuration ospf d'interface

Commande : #show ip ospf interface

```
R2#show ip ospf interface

Serial0/1/1 is up, line protocol is up
  Internet address is 10.10.20.10/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 97
  Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 10.3.3.3
  Suppress hello for 0 neighbor(s)
```



Analyse de la table de routage

Commande : #show ip route

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.10.20.4 is directly connected, Serial0/1/0
O       10.10.20.8 [110/128] via 10.10.20.13, 00:22:52, Serial0/1/1
C       10.10.20.12 is directly connected, Serial0/1/1
     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
O       172.16.2.0 [110/129] via 10.10.20.13, 00:22:52, Serial0/1/1
O       172.16.3.0 [110/65] via 10.10.20.13, 00:22:52, Serial0/1/1
```



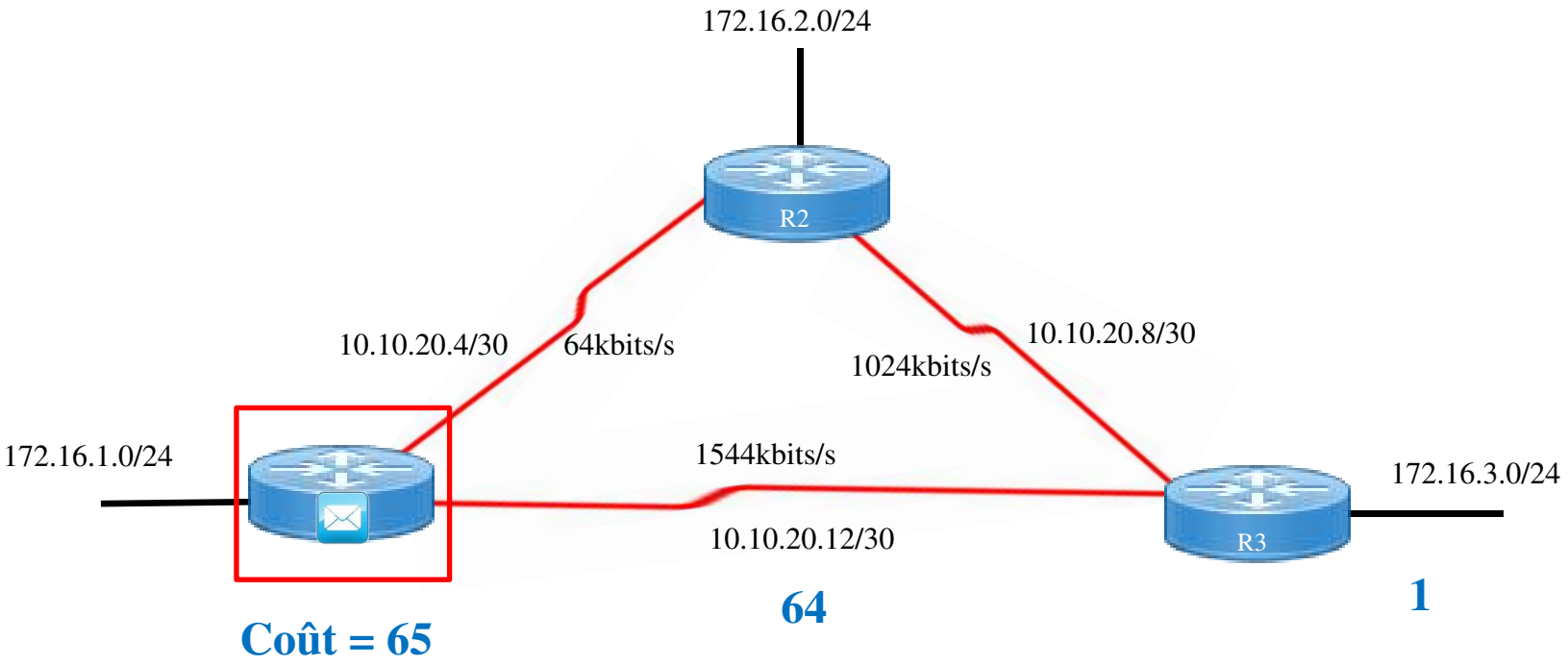
Calcul de la métrique

Type d'interface	10^8/bits/s = coût
Fast Ethernet	10^8/100 000 000 bits/s =1
Ethernet	10^8/10 000 000 bits/s =10
E1	10^8/2 048 000 bits/s =48
T1	10^8/1 544 000 bits/s =64
128 Kbits/s	10^8/128 000 bits/s =64
64 Kbits/s	10^8/64 000 bits/s =64
56 Kbits/s	10^8/56 000 bits/s =64

Calcul de la métrique

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

**********
D       172.16.3.0 [110/65] via 10.10.20.13, 00:22:52, Serial0/1/1
```



Visualisation du coût des routes

Commande : #show interfaces s0/1/1

```
R1#show interfaces s0/1/1
Serial0/1/1 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.20.14/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
```

Commande : #show ip route

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

**********

O       172.16.3.0 [110/65] via 10.10.20.13, 00:22:52, Serial0/1/1
```

Commande : #show ip ospf interface serial 0/1/1

```
R1#show ip ospf interface serial 0/1/1

Serial0/1/1 is up, line protocol is up
  Internet address is 10.10.20.14/30, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
```

Modification du coût des routes

Commande : (config-if)#bandwidth 64

```
R1(config)#interface s0/1/1
R1(config-if)#bandwidth 64
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ospf interface s0/1/0

Serial0/1/0 is up, line protocol is up
  Internet address is 10.10.20.5/30, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 1562
  Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
```

10^8/64 000 = 1562

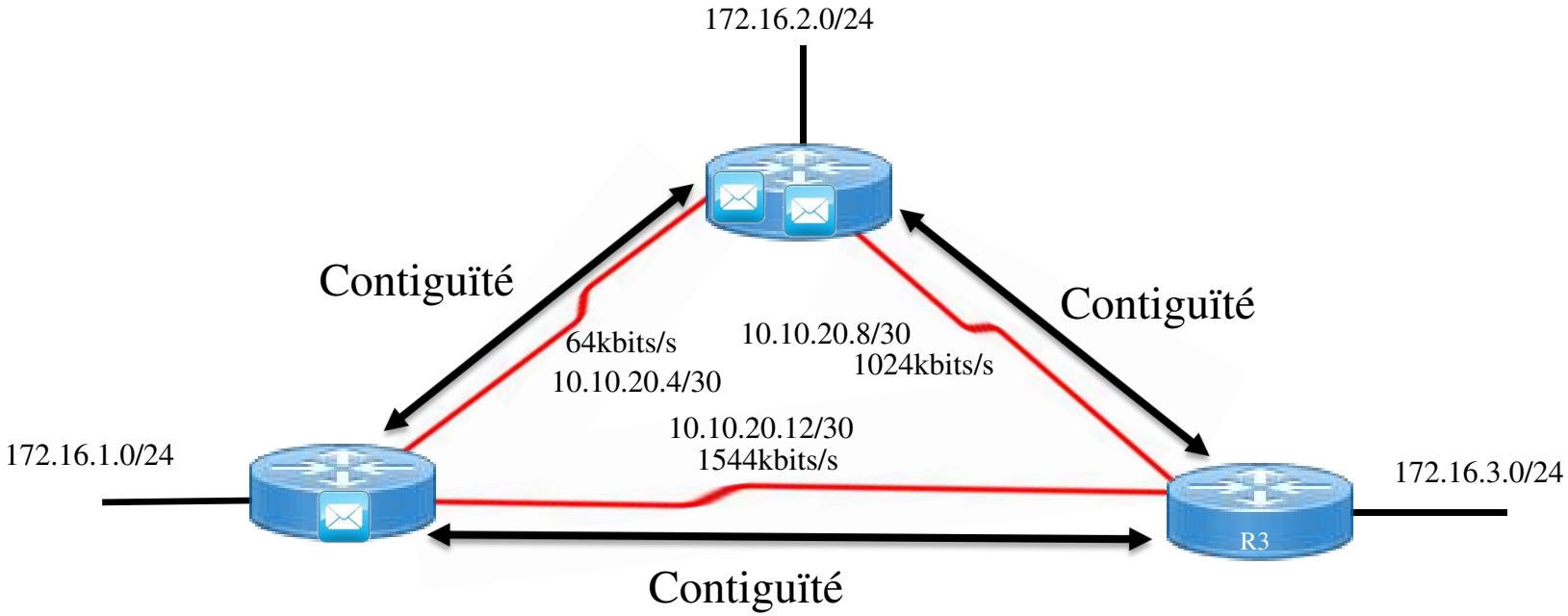
Commande : (config-if)#ip ospf cost 1562

```
R1(config)#inter s0/1/0
R1(config-if)#ip ospf cost 1562
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

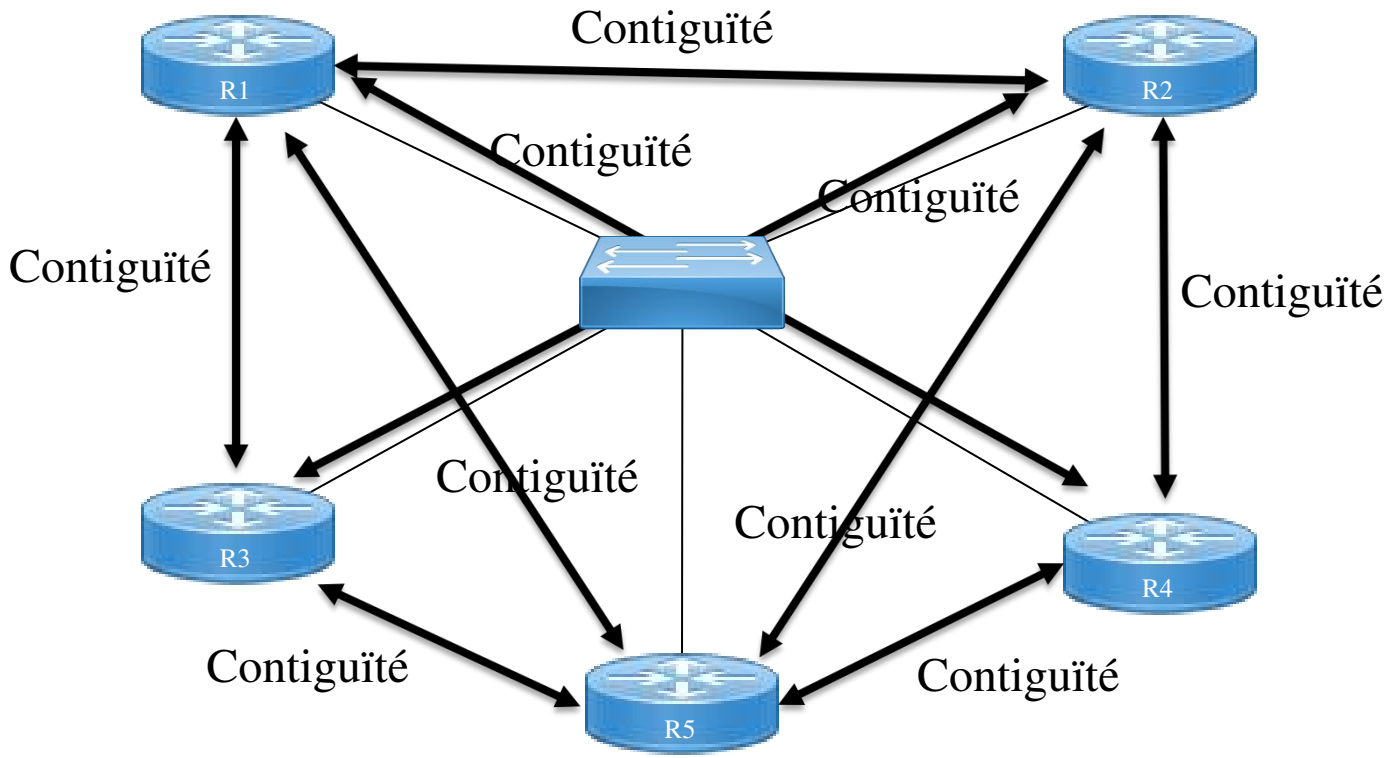
R1#show ip ospf interface s 0/1/0

Serial0/1/0 is up, line protocol is up
  Internet address is 10.10.20.5/30, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 1562
  Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
  No designated router on this network
```

Contiguïté sur les réseaux point à point



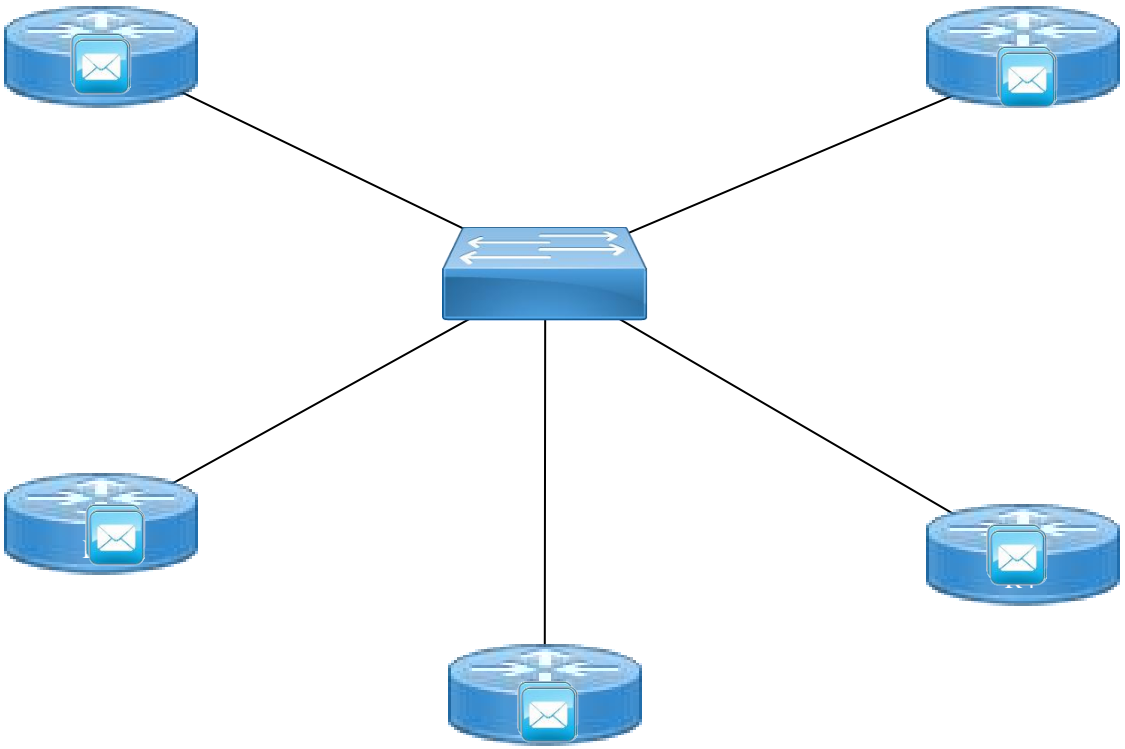
Contiguïté sur les réseaux à accès multiple



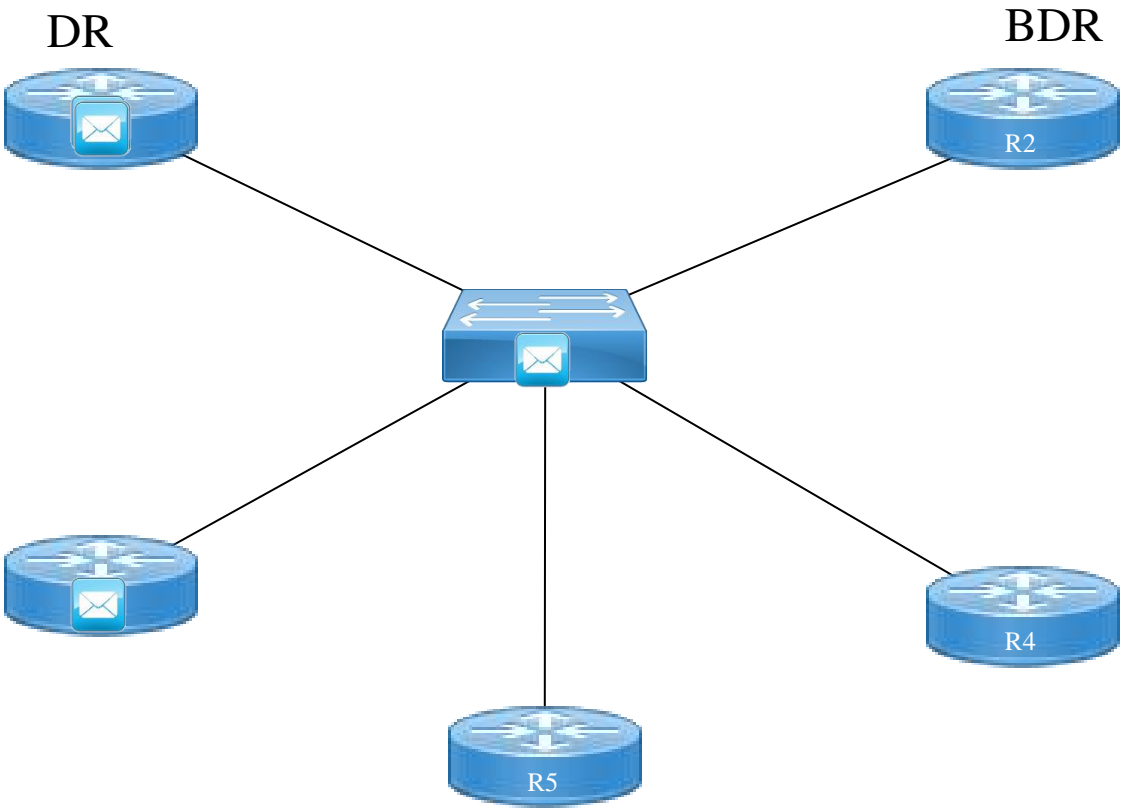
Routeurs	Contiguïtés
N	N(n-1)/2
5	10
10	45
20	190
100	4950

Nombre de contiguïtés = n (n-1)/2
Exemple = 5 x (5-1) = 10 contiguïtés

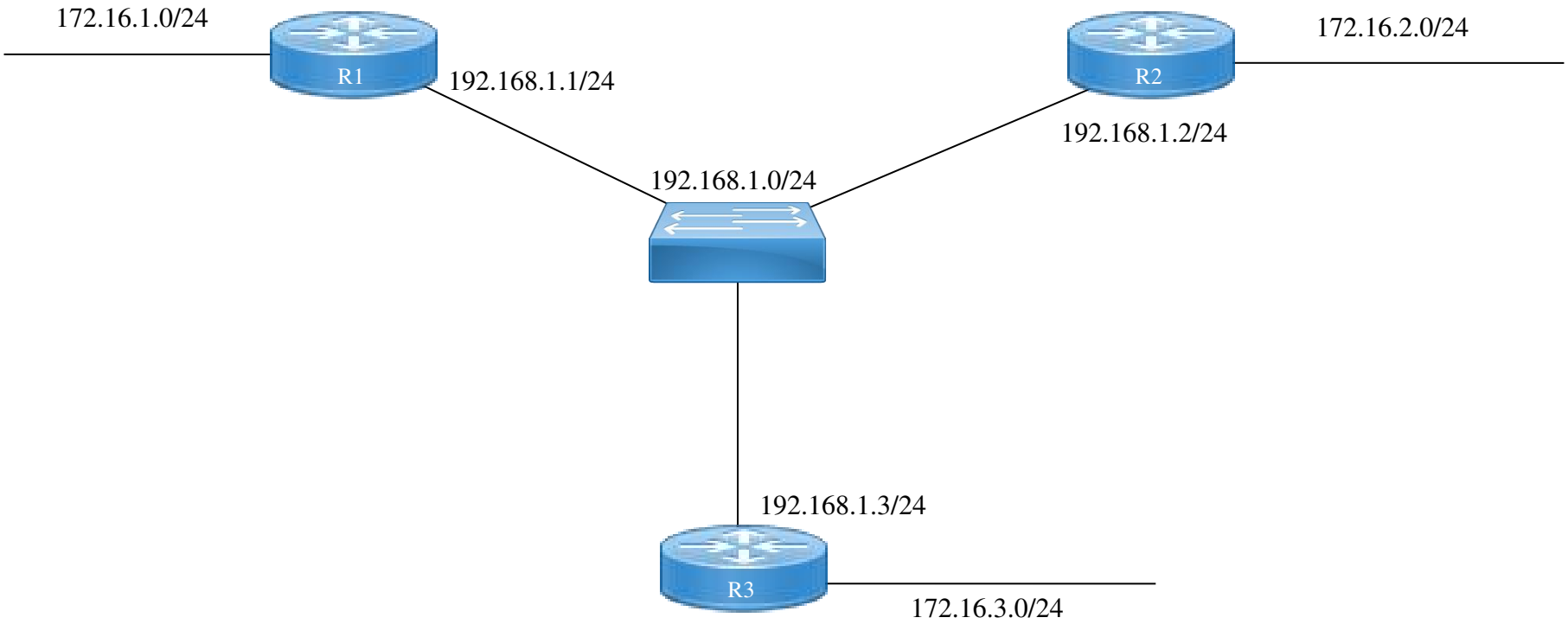
Diffusion des LSA



Designed Router et Backup Designed Router



Topologie à accès multiple





Visualisation des voisin OSPF

Voisin OSPF de R1

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.3       1   FULL/DR         00:00:39    192.168.1.3     FastEthernet0/0
192.168.1.2       1   FULL/BDR        00:00:30    192.168.1.2     FastEthernet0/0
```

Voisin OSPF de R2

```
r2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.1       1   FULL/DROTHER    00:00:33    192.168.1.1     FastEthernet0/0
192.168.1.3       1   FULL/DR         00:00:30    192.168.1.3     FastEthernet0/0
```

Voisin OSPF de R3

```
r3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.2       1   FULL/BDR        00:00:35    192.168.1.2     FastEthernet0/0
192.168.1.1       1   FULL/DROTHER    00:00:37    192.168.1.1     FastEthernet0/0
```

Visualisation de la priorité OSPF

Commande : #show ip ospf interface fastEthernet 0/0

```
R1#show ip ospf interface fastEthernet 0/0

FastEthernet0/0 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 192.168.1.3, Interface address 192.168.1.3
  Backup Designated Router (ID) 192.168.1.2, Interface address 192.168.1.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 192.168.1.3  (Designated Router)
    Adjacent with neighbor 192.168.1.2  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
```

Modification de la priorité OSPF

Commande : #ip ospf priority 200

```
R1(config)#inter fa 0/0
R1(config-if)#ip ospf priority 200
```

Commande : #show ip ospf interface fastEthernet 0/0

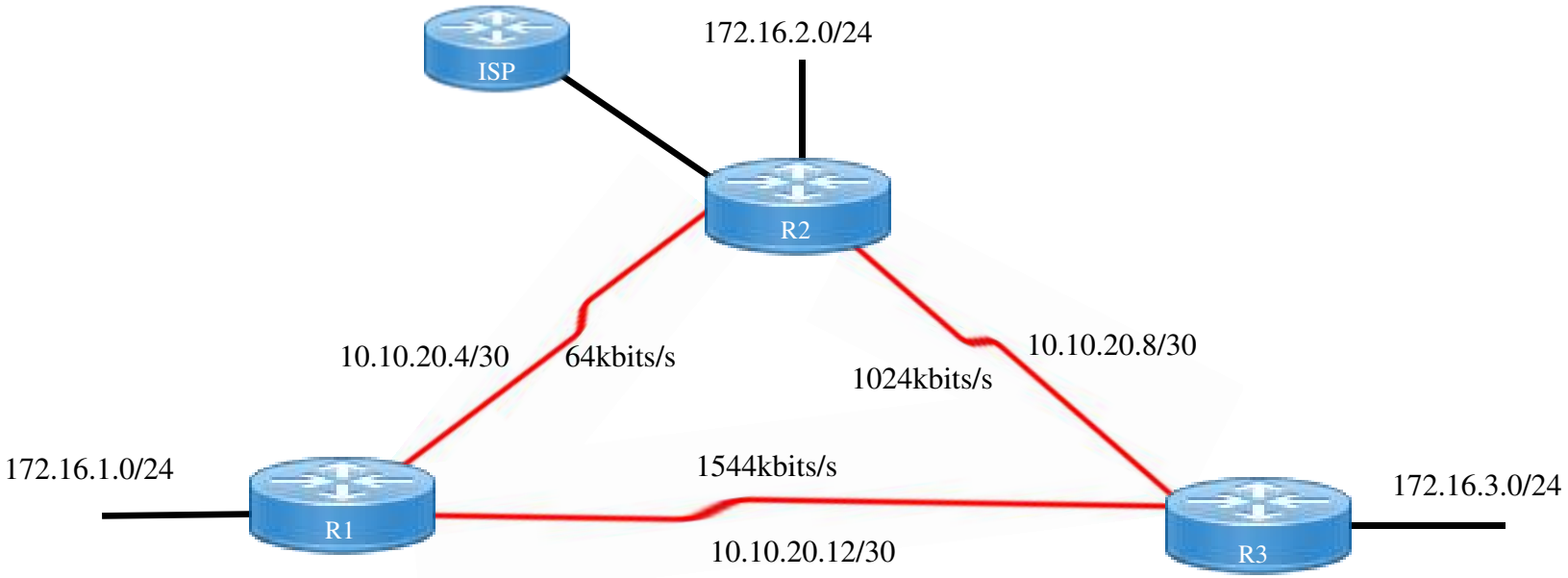
```
R1#show ip ospf interface fastEthernet 0/0

FastEthernet0/0 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 200
  Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
  Backup Designated Router (ID) 192.168.1.2, Interface address 192.168.1.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 192.168.1.3
    Adjacent with neighbor 192.168.1.2  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
```

Redistribution de la route par default

Commande : (config-router)#default-information originate

```
r2(config)#router ospf 1
r2(config-router)#default-information originate
```



Modification des timers OSPF

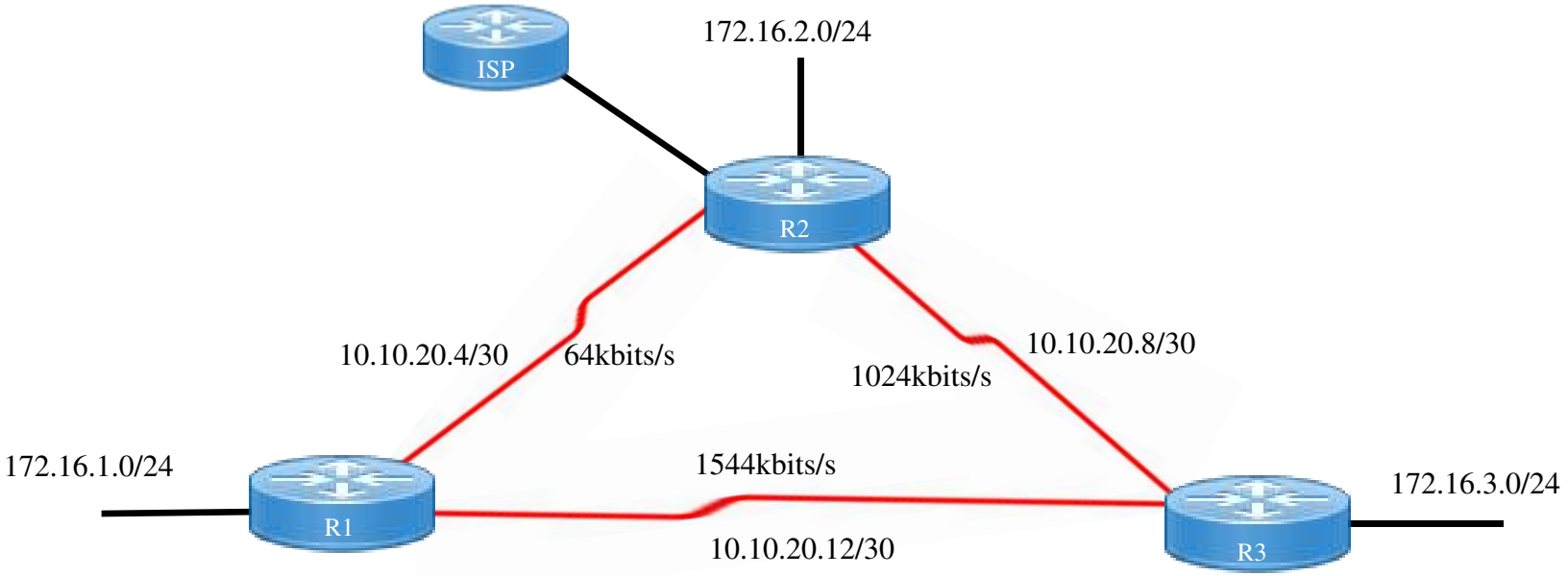
Commande : #show ip ospf neighbor

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:35    10.10.20.6      Serial0/1/0
10.3.3.3          0   FULL/  -        00:00:35    10.10.20.13     Serial0/1/1
```

Commande : (config-if)#ip ospf hello-interval 5 et ip ospf dead-interval 10

```
R1(config)#interface S0/1/0
R1(config-if)#ip ospf hello-interval 5
R1(config-if)#ip ospf dead-interval 10
```



Modification des timers OSPF

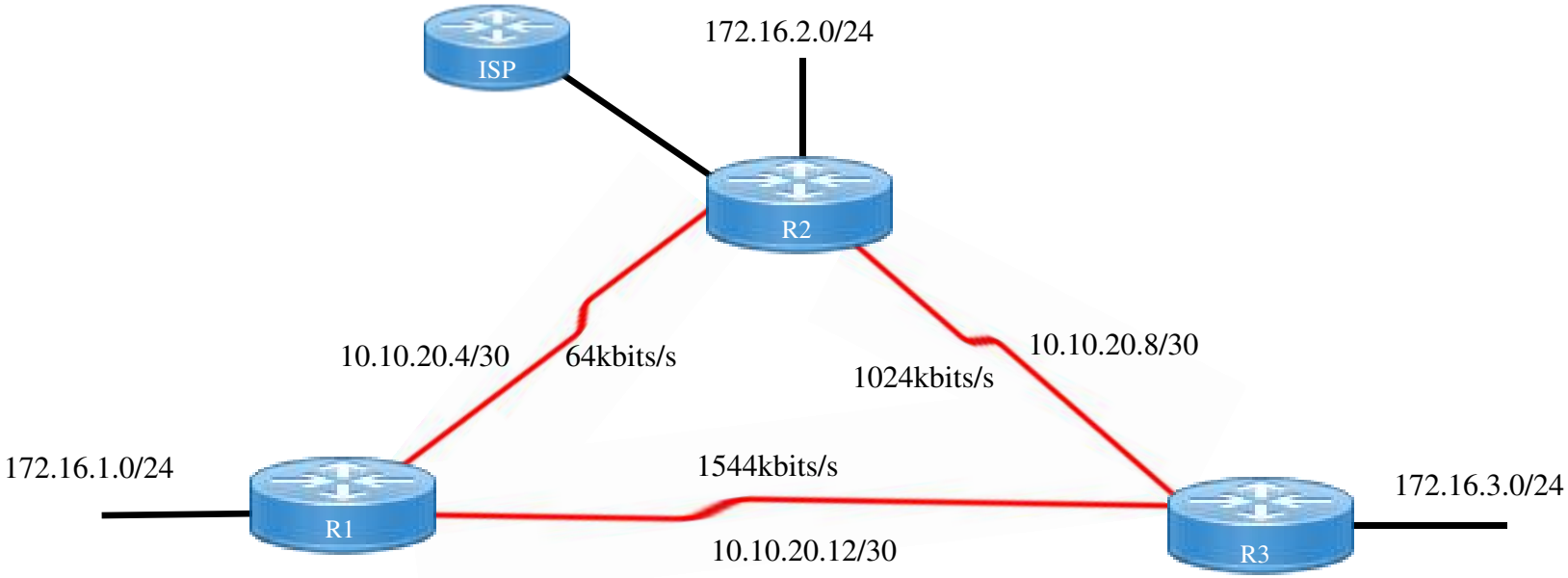
Commande : #show ip ospf neighbor

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.3.3.3          0   FULL/  -        00:00:32    10.10.20.13     Serial0/1/1
```

Commande : (config-if)#ip ospf hello-interval 5 et ip ospf dead-interval 10

```
R2(config)#interface S0/1/0
R2(config-if)#ip ospf hello-interval 5
R2(config-if)#ip ospf dead-interval 10
```

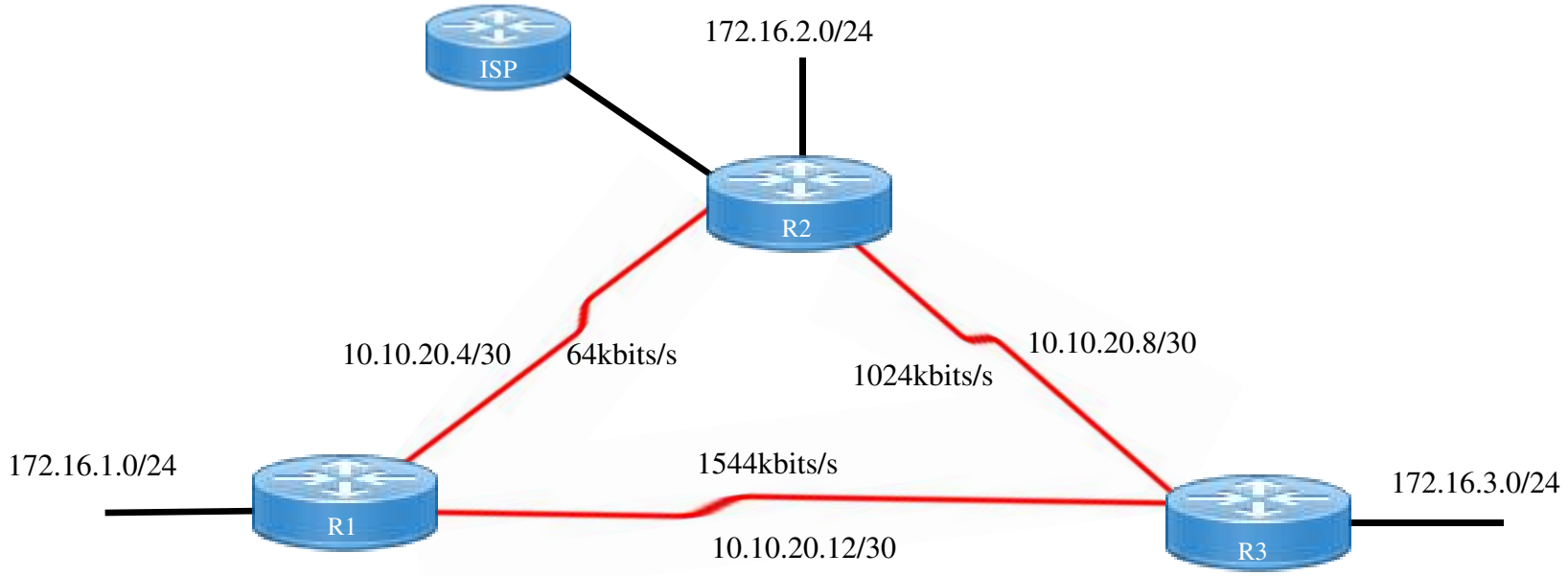


Modification des timers OSPF

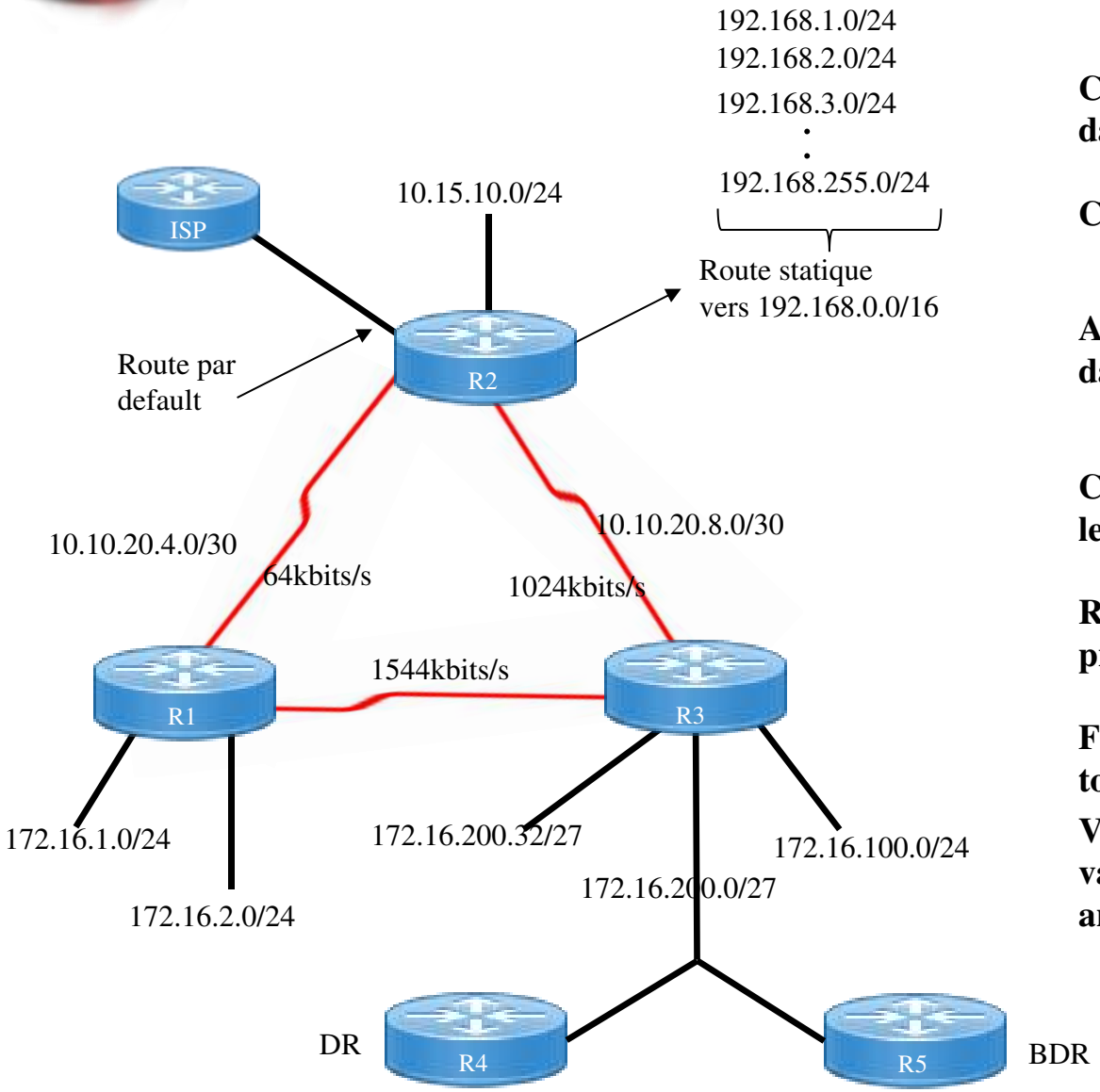
Commande : #show ip ospf neighbor

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:06    10.10.20.6      Serial0/1/0
10.3.3.3          0   FULL/  -        00:00:36    10.10.20.13     Serial0/1/1
```



TP : Routage OSPF



Configurez les interfaces comme indiqué dans le schéma

Configurer le protocole de routage OSPF

Ajouter la route statique comme indiqué dans la schéma

Configurer la route par default qui sort vers le routeur de du FAI

Redistribuer la route par default avec le protocole OSPf

Forcer la configuration des DR et BDR sur la topologie en accès multiple

Vérifier l'ensemble de vos configuration et valider le bon fonctionnement de votre architecture



Yann Guittet



Merci de votre attention

